Algebra as a structural toolkit

Foundations, splitting obstructions, integral reconstruction, and proof comparisons

Research notes for returning to algebra

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Returning to algebra: a compact working guide

For the continuation, use the reconstructive map after these foundational exercises. It connects invariant-complement obstructions, averaging, integral spectral reconstruction, and common sign lifts, with explicit hypotheses and stopping tests. The two expanded proof comparisons identify what can actually be shortened in the inspected parts of Xi, Hungerford, and Grillet.

The essential memory spine

Retain the following statements with their hypotheses and the type of every map. The rest of the toolkit should be reconstructible from them.

1. A map $V/U \to X$ is a map $V \to X$ that kills U.
2. Dualization is precomposition; evaluation gives $V \cong V^{**}$ in finite dimension. An identification with V^* needs extra data.
3. A direct sum is encoded by projections, and $P^2 = P$ gives $V = \operatorname{im} P \oplus \ker P$.
4. A split exact sequence has a section. Sections differ by a map from the quotient to the kernel; existence does not imply a preferred choice.
5. An operator is a $k[t]$-module. The PID structure theorem classifies finite-dimensional operators through invariant factors; Jordan form needs splitting of the minimal polynomial.
6. Coprime annihilator factors produce polynomial idempotents by Bézout.
7. Tensor products represent bilinear or balanced maps. A formula on pure tensors must respect the defining relations.
8. $W \otimes V^*$ identifies canonically with the finite-rank maps $V \to W$, hence with all of $\operatorname{Hom}(V, W)$ if either V or W is finite-dimensional. Trace is evaluation in finite dimension.
9. Tensor, symmetric and exterior algebras are free constructions with specified relations. Λ^r transports alternating data; the top exterior power gives a determinant line.
10. Forms transform by congruence, operators by similarity. A positive metric supplies adjoints, orthogonal projections, and the normal spectral theorem.
11. Exactness is image equals kernel. Over a field short exact sequences split and positive Ext/Tor vanish; homology of an arbitrary complex need not.
12. A representation is a module over a group algebra. Maschke needs invertible group order; Schur's scalar conclusion needs finite dimension and an algebraically closed field.

For Hadamard/Williamson add five facts: Gram orthogonality is an integral identity; tensors multiply valid orders; cyclic Fourier coordinates are CRT coordinates; compression loses lifting information; and equality of Boolean phases must be distinguished from independence of their signs.

A selective route, with stopping tests

First return: regain the language. Read the quotient/duality/splitting chapter, with Xi II §§1.4–1.6 and 2.4 beside it. Stop when you can construct the quotient map and its dual, describe the affine space of sections, and explain why an invariant complement is an additional demand. Xi I can be a reference for matrix maps and determinant characterizations rather than a sequential restart.

Second return: recover operator theory from one theorem. Choose Grillet VIII.6–7 or Hungerford IV.6 then VII.4–5 as the principal module route. Read Xi II §2.5's actual Bézout and quotient-induction proofs as a comparison, not a second complete linear algebra course. Stop when you can obtain primary projections, invariant factors, Jordan blocks under the field hypothesis, and the block counts from kernel growth.

Third return: turn multilinearity into maps. Read Xi II §6.2, especially Theorem 6.15 and the Hom/scalar-extension examples. Then use Grillet XI.5–6 and XIII.2–4 for balanced products, quotient constructions, and characteristic issues. Read Mac Lane III.1–2 and IV.1 when the universal examples are already familiar. Stop when you can define tensor-induced maps without a basis and derive Cauchy–Binet from $\Lambda^r(gf)$.

Keep a metric branch active. Read Xi II §§1.7, 3.3–3.5 and 3.8 as needed for forms, adjoints, normal operators, and projections. This is central for Hadamard Gram matrices. Defer the rest of affine/projective geometry only according to current goals, without concluding that the content was replaced.

Introduce homology at its first real obstruction. Read the exactness chapter here, then Hilton–Stammbach III.1–4 and IV.1–3 if you want the general construction. Grillet X.1, X.3 and XII are an alternative. Stop the first pass after computing the $k[t]/(t^2)$ extension and explaining nonzero field-complex homology. Spectral sequences and derived categories are not prerequisites for the current Hadamard bridge.

For the confirmed problem direction. Read the Hadamard bridge in order: Gram matrices and tensor closure; Xi III §5.6 finite fields; the character-sum proof and Paley blocks; cyclic group algebra and Fourier CRT; then the pinned third-round proofs with their color and Boolean equations. Use Xi III Chapter 3 or Grillet IX.7 for group algebras and representations, keeping Grillet's initial indecomposable/irreducible convention visible. Read the prior limitations before proposing a larger search. The essential stopping test is being able to say exactly which implication is necessary, sufficient, or only a relaxation.

Hall's further Lie theory, Lang's local arithmetic, Hartshorne's scheme/sheaf theory, and Hirsch's transversality/bundle development can then be pursued as separate branches. Their prerequisite bridges are recorded, not placed in front of every use of linear algebra.

Reconstruction exercises

These are deliberately small. Their purpose is to test whether a tool can be rebuilt, not to begin the later difficult problem-set campaign.

1. **Quotient versus complement.** For $U \subseteq V$, construct $(V/U)^* \cong U^\circ$ and show why no complement is needed. Then describe all sections of $V \to V/U$ once one has been chosen.
2. **One invariant line.** In $k[t]/(t^2)$, list the subspace killed by t. Prove it has no t-invariant complement, and identify the associated nonzero Ext class.
3. **Recover a projection.** Assuming $T^2(T-I)^2 = 0$, derive the projector $3T^2 - 2T^3$ from polynomial congruences. Check the argument in characteristics 2 and 3 without dividing by those integers.
4. **Recover Jordan data.** Let $T : V \to V$ with $\dim_k V = 10$ and $\lambda \in k$. The dimensions of $\ker(T-\lambda 1_V)^j$ for $j = 0, 1, \dots, 5$ are $0, 4, 7, 9, 10, 10$. Find all its block sizes, minimal polynomial, and the dimension of its commutant.
5. **Nonsplitting field.** Classify multiplication by t on $\mathbb{R}[t]/(t^2+1)$ as a real module and after complexification. Explain which use of "semisimple" is valid before extension.
6. **Tensor universality.** Construct the currying bijection and the canonical map $W \otimes V^* \to \mathrm{Hom}(V, W)$. Identify the image of the latter and prove injectivity. Explain why finite dimension of either V or W suffices for surjectivity, and give a failure when both are infinite-dimensional.
7. **Determinants from wedges.** Derive the minors formula for $\Lambda^2 f$ and the 2×2 Cauchy–Binet identity for a 2×3 matrix times a 3×2 matrix.

8. **Forms versus operators.** Give two congruent real symmetric matrices that are not similar. Derive $H^{-1}A^*H$ from the defining adjoint identity, stating the Hermitian linearity convention.
9. **Homology survives.** For $d : k^3 \to k^2$, $d(x, y, z) = (x, y)$, compute the two homology groups of the two-term complex, and write an explicit contractible/homology decomposition.
10. **The right group action.** Verify the inverse in the dual action, and differentiate it to obtain the minus sign in the Lie algebra dual action.
11. **Character sum.** Reproduce the count of $y^2 = x(x + a)$ over an odd finite field. Identify every use of $a \neq 0$ and $2 \neq 0$, then check one Paley block multiplication.
12. **Compression and phase dependence.** Check the core-nine compressed autocorrelation example. Explain why failure of its chosen lifts alone cannot prove absence of all lifts. Separately give a numerical assignment to three formally independent signs that violates $\xi_{p+q} = \xi_p\xi_q$.

Brief answer checks

For Exercise 1, pull back along the quotient, and use $s' = s + ih$. For 2, the killed subspace is kt and the class is $1 \in k$ under $\mathrm{Ext}^1_{k[t]}(k, k) \cong k$. For 3, require a polynomial to be zero modulo t^2 and one modulo $(t - 1)^2$; the integer identity remains valid after any reduction of characteristic.

For 4, the differences are $4, 3, 2, 1, 0$, so the sizes are $4, 3, 2, 1$. Since the kernels reach $\dim_k V = 10$, there is no other primary summand. The minimal polynomial is $(t-\lambda)^4$ and the commutant dimension is $\sum_{i,j} \min(i, j) = 30$ for $i, j = 1, 2, 3, 4$. For 5, the real module is simple; after complexification it is the direct sum of the i and $-i$ eigenspaces. It is semisimple over $\mathbb{R}[t]$ but not real diagonalizable.

For 6, currying needs no finite dimension. The canonical Hom/tensor map is always injective, with image exactly the finite-rank maps. Thus it is onto precisely when every map $V \to W$ has finite rank; finite dimension of either space suffices. For $V = W$ infinite-dimensional, the identity is outside its image. For 7, sum the three intermediate 2-subsets. For 8, I and $2I$ suffice, and $h(x, y) = \bar{x}^{\mathsf{T}}Hy$ is the convention used here. For 9, H_1 is the z-axis and $H_0 = 0$; the (x, y) coordinates map isomorphically to k^2. For 10, transpose reverses products and differentiation of an inverse at the identity contributes a minus sign. For 11, the count is $q-1$. For 12, choose all three phase signs negative: the product of the first two is positive, contradicting the third. These checks indicate the target, while the chapters provide the proofs to reconstruct.

What can be deferred

Defer a formula once its map or universal property can be reconstructed. Defer a subject when it has no role in the current destination. Do not defer the hypotheses, the equivalence relation, or a known failure case. For this collection, an effective memory sheet contains definitions and four or five counterexamples, with links back to proofs, rather than a long inventory of named theorems.

A reconstructive map: from decompositions to their obstructions

The basic study explains quotients, duality, operators as modules, tensor universality, exterior algebra, forms, exactness, and representations. The continuation asks a sharper question: **which extra structure survives a decomposition, and how can its failure be computed?** Four concrete outputs organize the new material:

- an invariant complement, or a reverse-intertwiner witness that it cannot exist;
- an equivariant section obtained by a justified averaging operation;
- an integral coefficient vector reconstructed from cyclotomic data, or a congruence obstruction;

- a common sign lift of compressed rows, or a residual correlation defect.

These outputs require different hypotheses. A vector-space splitting is not automatically invariant; a rational splitting is not automatically integral; an integral vector is not automatically a sign vector; four individually possible lifts need not satisfy their equations together.

Dependencies and what each step adds

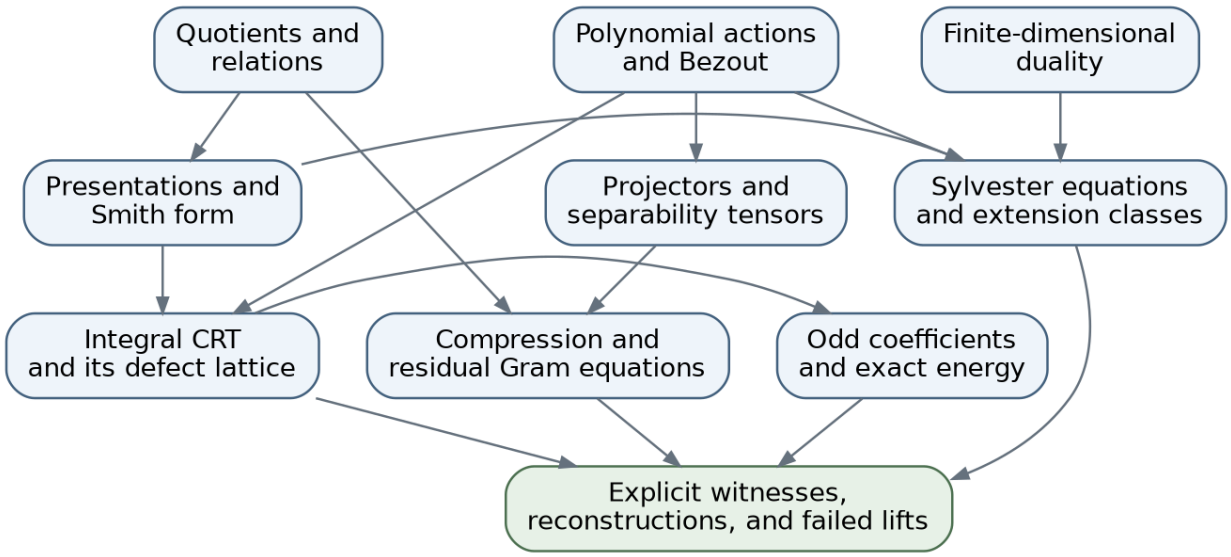


Figure 1: Dependencies for reconstruction. Arrows indicate prerequisite routes; each theorem retains the hypotheses stated below.

Begin with quotients and the factorization rule: a map descends precisely when it kills the relations. This one rule constructs tensor products, module presentations, quotient operators, and cyclic group-algebra compression. It tells us how to form the new object, not whether a section exists.

Next add polynomial evaluation and Bézout. Coprime annihilators give idempotent projectors. In the operator setting, the same identity gives the inverse of a Sylvester operator. With an invertible derivative it gives a separability tensor and a universal averaging operation. Over the integers, the denominators that appear are arithmetic information: they lead to resultants, finite cokernels, and Smith divisibility.

Add finite-dimensional duality to turn a failed linear equation into a witness. The trace pairing identifies the annihilator of the Sylvester image with reverse intertwiners. Add module presentations to compute the same obstruction as an extension class, and Smith form to determine how that class changes the middle module. The witness and the module type are different outputs.

Finally add a coefficient lattice, an involution, and real positivity. Cyclotomic evaluation preserves multiplication and involution and identifies the integral coefficient lattice with a finite-index sublattice of the product of cyclotomic integer rings. Allowing arbitrary tuples in that larger product forgets the integral compatibility. Trace reconstruction describes its exact image. Membership in the affine lattice of odd coefficient vectors, together with the exact total energy, forces signs. Compression discards a complementary ideal, so it retains only part of the Gram equation; the explicit core-nine defect lies in the discarded part.

No step requires importing schemes or differential topology. Homological language is useful here because its first extension group has an explicit matrix presentation; the Hadamard calculations themselves do not depend on a general course in derived functors.

Eight principles to reconstruct rather than memorize

1. Quotients encode relations

Minimum data. A module F, a submodule of relations N, and a module map $f : F \to Q$ with $N \subseteq \ker f$.

Construction. Define $\bar{f}(x + N) = f(x)$. The kernel condition is exactly what makes the answer independent of the representative, and surjectivity of $F \to F/N$ proves uniqueness.

Families recovered. Balanced tensor maps, cokernel maps, cyclic-module presentations, induced quotient operators, and the map $\mathbb{Z}[C_m] \to \mathbb{Z}[C_d]$.

Failure to detect. Factorization says nothing about a section. The quotient $\mathbb{Z} \to \mathbb{Z}/2$ has no additive section; an invariant quotient of a Jordan block need not have an invariant complement.

2. Bézout gives projectors and inverses

Minimum data. An endomorphism T of a module over a commutative ring, with $a(T)b(T) = 0$ and $ua + vb = 1$ in that ring's polynomial algebra.

Construction. The operators $v(T)b(T)$ and $u(T)a(T)$ add to the identity. The first acts as the identity on $\ker a(T)$ and vanishes on $\ker b(T)$; the second does the reverse. Their images lie in those respective kernels, and their cross product is zero, so they are complementary idempotents.

Families recovered. Primary decomposition, CRT coordinates, polynomial spectral projectors, and the divided-difference inverse of $h \mapsto Ah - hB$ when $a(A) = b(B) = 0$.

Failure to detect. Coprimality over $\mathbb{Q}[t]$ does not imply an integral Bézout identity. The factors t and $t - 2$ produce a rational graph with slope $1/2$ and an integral obstruction of order two.

3. A dual kernel certifies a missing image

Minimum data. A linear map $D : H \to H'$ of finite-dimensional vector spaces and perfect pairings with their dual spaces.

Construction. $\operatorname{im} D = (\ker D^*)^\perp$. One direction is evaluation; the converse extends a basis of the image to a basis of the codomain and constructs a separating linear functional.

Families recovered. Solvability certificates, annihilator identities, and the trace criterion $C \in \operatorname{im}(Ah - hB)$ exactly when $\operatorname{tr}(gC) = 0$ for every $gA = Bg$.

Failure to detect. A field-valued witness cannot see a torsion cokernel that disappears over that field. Integral divisibility requires more than rational trace equations.

4. A presentation records both a module and how it is attached

Minimum data. Relations $fu = 0$, $ge = cu$ over a PID R, with $f, g \neq 0$ and specified endpoint identifications with $R/(f), R/(g)$.

Construction. The relation matrix is $\left(\begin{smallmatrix} f & -c \\ 0 & g \end{smallmatrix}\right)$. Changing the chosen lift changes c by gh modulo f. The extension class is $c \bmod (f, g)$; the middle module has Smith factors $\gcd(f, g, c)$ and $fg/\gcd(f, g, c)$.

Families recovered. Cyclic Ext computations, Jordan-block gluing, integer-lattice quotients, and the difference between classification with fixed endpoint maps and classification after forgetting them.

Failure to detect. Equal middle-module types need not mean equivalent extensions. The residues 1 and $1+t$ modulo t^2 give distinct endpoint-preserving classes but the same middle module $k[t]/(t^4)$.

5. A separability tensor repairs maps

Minimum data. An associative unital algebra S over a field k, with a tensor $E = \sum_i a_i \otimes_k b_i$ satisfying $\sum_i a_i b_i = 1$ and $\sum_i s a_i \otimes b_i = \sum_i a_i \otimes b_i s$.

Construction. For left S-modules, send a k-linear map f to $v \mapsto \sum_i a_i f(b_i v)$. The two tensor identities prove S-linearity and show that this operation fixes every S-linear map.

Families recovered. Maschke averaging and polynomial spectral averaging. Sections and retractions are repaired when the surrounding maps are already module maps.

Failure to detect. Averaging an arbitrary idempotent map need not preserve idempotence. Endpoints annihilated by m do not imply that the middle module of an extension is annihilated by m. Over an imperfect field, a field extension can be semisimple as a ring without being separable over the ground field.

6. An injective rational decomposition can have an integral defect

Minimum data. Monic $f, g \in \mathbb{Z}[t]$ coprime over $\mathbb{Q}[t]$.

Construction. The integral CRT map has cokernel $\mathbb{Z}[t]/(f, g)$ as an additive group, of order $|\operatorname{Res}(f, g)|$. Smith form refines its order into exact congruences. In a product of cyclotomic factors, trace inversion tests the whole coefficient lattice.

Families recovered. Prime gluing, prime-power overlap jets, the cyclic integral Fourier lattice, and the resultant formula for the determinant of a Sylvester operator.

Failure to detect. Pairwise congruences among several factors can miss a congruence against the whole previously glued block. At order four, the tuple $(0, 0, 1 + i)$ passes every pairwise overlap but reconstructs half-integer coefficients.

7. Parity plus a sharp energy bound forces signs

Minimum data. A list $x_1, \ldots, x_N$ of odd integers with $\sum_j x_j^2 = N$.

Construction. Each squared coefficient is at least one. Equality in the sum forces every square to be one.

Families recovered. The all-order affine-lattice sign certificate: under the exact four-row spectral norm equations, membership of each tuple in $J + 2L$ forces every reconstructed coefficient to be a sign. Here L is the integral Fourier image and $J = (n, 0, \ldots, 0)$ is the image of $1 + t + \cdots + t^{n-1}$.

Failure to detect. Integral coefficients and the correct energy alone allow zeros and larger entries. At odd order, component parity can be combined with integral gluing because the CRT defect has no two-torsion. At even order this simplification fails; one must test the half-shifted tuple in L.

8. An idempotent splits equations but not discrete choices

Minimum data. A self-adjoint central idempotent e in a real algebra with involution.

Construction. Write $X = eX + (1 - e)X$. Cross products vanish, so a Gram equation splits into its two ideal components. For cyclic compression, e is averaging over the kernel subgroup.

Families recovered. An exact retained-energy calculation, the residual Gram equation, individual fiber lift counts, and the core-nine defect $8(S_p - S_k)$.

Failure to detect. A split real equation need not admit a solution in the prescribed finite fiber vertex sets. Exact relaxation is still relaxation when it forgets part of the domain.

A shorter route through the books

There are two coherent routes to canonical forms. A reader staying with undergraduate linear algebra can use Xi's Cayley–Hamilton proof, primary splitting, quotient-lift Jordan construction, and image–kernel chain algorithm, assuming the minimal polynomial splits over the working field. A reader who needs integral presentations and nonsplit operators can instead establish polynomial torsion by linear dependence among powers, prove one PID structure theorem, and derive rational and Jordan forms. Using Jordan as the sole proof of Cayley–Hamilton while also using Cayley–Hamilton to establish that Jordan route would be circular.

The source comparisons give precise inspected pages and stopping tests. They support compressing repeated theorem coverage, not discarding all alternate proofs. Xi's quotient-lift and filtration constructions teach different skills; Hungerford's bounded-primary splitting lemma has additional scope; Grillet's relative-basis proof is particularly useful for the integral chapters. Xi already proves tensor universality and two forms of Maschke's theorem. The graduate texts supply broader module settings, exactness arguments, and different dependencies.

For the present project, learn one proof of each common structural theorem, then reconstruct the new chapters in this order:

1. Extract C from an invariant-subspace problem, solve $Ah - hB = -C$, and give a trace witness for a nonsplit example.
2. Compute the same class through a cyclic presentation and determine its middle-module Smith factors. Check a characteristic-two example.
3. Construct the polynomial averaging tensor and the group averaging tensor, then explain precisely which sections each repairs.
4. Derive the prime integral CRT image and trace inverse. Find the missing congruence in the order-four pairwise counterexample.
5. Prove the affine-lattice sign theorem from odd coefficient energy and explain the odd/even distinction.
6. Count normalized symmetric lifts and reproduce the core-nine residual polynomial without enumerating Gram matrices by hand.

These are performance tests, not a page quota. Passing them means the reader can reconstruct a decomposition, name its domain of validity, and exhibit a concrete obstruction to the tempting stronger assertion.

Architecture: structures, maps, and obstructions

The organizing claim

Much ordinary linear algebra is the study of finite-dimensional vector spaces with additional structure. The most reusable move is to identify that structure, declare which maps preserve it, and then ask which familiar constructions remain available. An abstract tool earns its place when it preserves information or exposes an obstruction that a coordinate calculation conceals.

Four layers organize the collection:

1. **Linear structure.** Kernels, images, quotients, duals, direct sums, and tensor products describe how linear information is discarded, tested, assembled, and transported.
2. **An action or a multiplication.** An endomorphism makes a vector space a $k[t]$-module. A group action makes it a $k[G]$-module. A bilinear multiplication is a map $V \otimes V \to V$ satisfying specified identities.
3. **Decomposition and its obstructions.** Idempotents encode direct sums; coprime annihilators produce idempotents; semisimplicity guarantees invariant complements. Extensions measure ways in which complements fail to exist.
4. **Variation and locality.** Families of vector spaces introduce bundles and sheaves; changing coefficients introduces localization, integrality, and arithmetic. A theorem about each individual vector space need not control a family or an integral lattice.

These layers are connected, but none erases the others. A classification up to arbitrary linear isomorphism does not solve a problem about lengths, positive forms, an integral lattice, or smooth variation.

The prerequisite map

The following arrows mean a useful prerequisite for the stated destination, not a claim that every possible approach must take this route.

Foundation	Construction or theorem it supports	Destination and extra bridge
Equivalence relations, linear maps, subspaces	Quotients, kernels, first isomorphism theorem	Exact sequences; then modules and abelian categories if needed
Bases, linear functionals, quotients	Dual exact sequences, annihilators, evaluation	Adjoints after choosing a form; trace duality after separability
Direct sums, polynomial division, Bézout	Coprime primary decomposition	$k[t]$-modules, rational/Jordan forms, commuting operators
PID ideals and presentations	Smith normal form, cyclic module decomposition	Integral abelian groups and operator classification; higher-dimensional polynomial rings need different tools
Bilinear maps and quotients	Tensor products and Hom–tensor adjunction	Scalar extension, representations, multilinear algebra
Tensor algebra and relations	Symmetric and exterior algebras	Polynomial coordinate rings, determinants, differential forms after smoothness
Splitting and projective presentations	Ext^1, Tor_1	Extension obstructions; derived functors after complexes and homotopies

Foundation	Construction or theorem it supports	Destination and extra bridge
Groups, invariant subspaces, duals, tensors	Group representations, Maschke, Schur	Matrix Lie groups after topology and the exponential map; Lie algebra representations after differentiation
Finite field extensions and multiplication maps	Norm, trace, separable trace pairing	Lang: integral closure, Dedekind ideals, valuations, completions, ramification
Commutative rings, modules, localization	Affine coordinate rings, local rings, tangent spaces	Hartshorne: sheaves, schemes, locally free sheaves, cohomology
Real linear maps, inner products, multivariable calculus	Tangent spaces, normal spaces, quadratic forms	Hirsch: manifolds, function-space topology, approximation, transversality, bundles

An economical order for this study is

$$\text{maps and quotients} \longrightarrow \begin{cases} \text{duality and splitting,} \\ \text{polynomials and modules,} \\ \text{bilinear maps and tensors.} \end{cases}$$

These three paths rejoin at invariant decompositions, determinant lines, forms, and representations. Exactness becomes useful as soon as a construction stops preserving injections or an invariant complement fails. There is no need to put spectral sequences in front of that first obstruction.

What the collection actually contributes

Xi's 基础代数 I develops the concrete languages of equations, matrix maps, determinants, polynomial division, roots, symmetric polynomials, and resultants. Volume II develops vector spaces and operators, but also the metric and geometric meaning of those objects. Volume III adds transformation groups, representations, modules, Lie algebra examples, and Galois theory. This is a connected course, not merely three fragments awaiting replacement by a graduate book.

Grillet is particularly well aligned with the present structural route: VIII.6–7 connects PID modules to Jordan form, X handles splitting and projectivity, XI develops Hom and tensors, XII introduces Ext/Tor, and XIII builds free algebras and their quotients. Hungerford independently recommends PID torsion modules before his linear algebra chapter (printed p. xviii, PDF p. 15). This is strong evidence that the module viewpoint is a sound organizational choice, rather than a gratuitous modernization of the user's plan.

Mac Lane supplies a language for recognizing the same construction in different categories. His universal arrows and representability are useful after several concrete examples have been worked through. They do not by themselves provide Jordan bases, a spectral theorem, or a treatment of Euclidean geometry.

Hilton–Stammbach supplies the next layer when exactness fails: projective presentations, extension classes, complexes, and derived functors. Its route through extensions before general derived functors is especially compatible with a reader coming from linear algebra. Hall turns familiar matrix groups and the exponential into systematic Lie theory and develops representation theory beyond the initial $\mathfrak{sl}_2$ example.

Hartshorne and Lang are destinations requiring new bridges. Neither should be mistaken for an interchangeable abstract-algebra text. Hirsch is a different destination again: his preface (printed p. vii, PDF p. 8) deliberately avoids tensors and differential forms, and largely avoids algebraic topology. The relevant foundations for his approach are geometry, analysis, and topology as well as linear algebra.

The memory hierarchy

There are three different things to retain.

Definitions that reconstruct maps. Remember the quotient factorization property, tensor universality, evaluation, and the equations defining a split sequence or an idempotent. Each tells you how a new map must be defined and what must be checked for it to exist.

Theorems whose hypotheses control classification. Remember finite generation and the PID hypothesis in the module structure theorem; splitting of the minimal polynomial for Jordan form; positivity and the base field in the spectral theorem; invertibility of $|G|$ in Maschke's theorem; separability in trace duality. The missing hypothesis often explains the entire difficulty.

Geometric or computational interpretations. A quotient is not a preferred complement. A dual is not an inner-product identification. A tensor is not usually a pure tensor. A canonical form is not a canonical basis. A fiber is not a smooth bundle, and a finite-dimensional vector space over a number field is not its integral lattice. These distinctions protect the structural tools from misuse.

What this architecture does not promise

Two commuting operators give a module over $k[s,t]$, which is generally not a PID. The one-operator classification therefore does not automatically extend to pairs. Likewise, finite-dimensionality over k does not make a $k[G]$-module semisimple, and a bundle of rank n does not automatically possess a global frame. These are productive boundaries, not reasons to abandon abstraction.

The existing algebra direction has been confirmed as the Hadamard/Williamson study. Its dedicated bridge locates its Gram matrices, integral cyclic group algebras, finite-field characters, symmetry quotients, and Boolean constraints within this map. The other subject bridges retain their independent value; they are not presented as prerequisites for that particular problem.

1 Quotients, duality, and splitting

1.1 Conventions and the type of a map

Throughout the core, k is a field and vector spaces are finite-dimensional unless a paragraph explicitly removes that hypothesis. Rings are associative with identity, modules are unital, and modules are left modules unless sides are specified. A commutative ring is stated when tensor products are themselves used as modules over that ring. The zero vector space is allowed.

For a linear map $f : V \to W$, f is the map and $f(v)$ is its value at $v \in V$. Write $\operatorname{im} f \subseteq W$ and $\ker f \subseteq V$. Composition $g \circ f$ first applies f. An equality between maps includes their domains and codomains; when a subspace is identified with its image, the inclusion will be stated. This matters especially when comparing an operator on a quotient with a restriction to a complement.

1.2 Quotients: impose relations once

Let $U \subseteq V$ be a subspace. Define $v \sim w$ if $v - w \in U$ and $V/U = \{v + U : v \in V\}$. Addition and scalar multiplication of cosets are well defined because replacing representatives changes the result by an element of U. Let $q : V \to V/U$, $q(v) = v + U$.

Quotient principle. For any vector space X and linear map $f : V \to X$, there is a unique linear map $\bar{f} : V/U \to X$ with $\bar{f} \circ q = f$ if and only if $U \subseteq \ker f$.

Proof. Necessity follows from $q(u) = 0$ for $u \in U$. Conversely set $\bar{f}(v + U) = f(v)$. If $v - w \in U$, then $f(v) = f(w)$; hence this is well defined. It is linear by the coset operations. Surjectivity of q forces uniqueness. □

The construction has a concrete description, but its reusable content is the factorization property. When constructing a map from a quotient, specify its value on a representative and prove it annihilates the relations. When proving two maps from the quotient equal, precompose with q.

First isomorphism theorem. If $f : V \to W$ is linear, then

$$V/\ker f \xrightarrow{\tilde{f}} \operatorname{im} f, \qquad \tilde{f}(v + \ker f) = f(v)$$

is a canonical isomorphism. It is surjective by definition of the image; its kernel is zero by definition of $\ker f$. Thus rank–nullity follows from dimension of a quotient, rather than being an isolated formula.

Why universal objects are unique. If (Q, q) and (Q', q') have the same quotient property for $U \subseteq V$, there are unique maps $a : Q \to Q'$ and $b : Q' \to Q$ satisfying $aq = q'$ and $bq' = q$. Then $baq = q$, so uniqueness gives $ba = 1_Q$; similarly $ab = 1_{Q'}$. Uniqueness means a unique isomorphism preserving the specified structure, not a unique bare vector-space isomorphism.

Two useful reconstructions are

$$U/(U \cap W) \cong (U + W)/W, \qquad (V/U)/(W/U) \cong V/W \quad (U \subseteq W \subseteq V).$$

For the first, restrict the quotient map $V \to V/W$ to U and compute its kernel. For the second, send $(v + U) + (W/U)$ to $v + W$ and apply the quotient principle twice. These proofs explain the maps as well as the dimensions.

Recognition signal. Whenever a problem says "modulo," identifies choices, or asserts that a quantity depends only on an equivalence class, a quotient factorization is present. If an operation is not compatible with the relation, the quotient does not inherit that operation. For example, $T : V \to V$ induces $\bar{T} : V/U \to V/U$ precisely when $T(U) \subseteq U$.

1.3 Duality: test vectors by scalar-valued maps

Define $V^* = \operatorname{Hom}_k(V, k)$. For $f : V \to W$ define

$$f^* : W^* \to V^*, \qquad f^*(\lambda) = \lambda \circ f.$$

Then $(g \circ f)^* = f^* \circ g^*$ and $1_V^* = 1_{V^*}$. The reversal is forced by the domains. In dual bases the matrix of f^* is the transpose of the matrix of f; complex conjugation is absent here.

The evaluation map is

$$\epsilon_V : V \to V^{**}, \qquad \epsilon_V(v)(\lambda) = \lambda(v).$$

If $e_1, \dots, e_n$ is a basis with dual basis $e^1, \dots, e^n$, then $\epsilon_V(e_i)$ is the corresponding dual basis of V^{**}. Thus ϵ_V is an isomorphism. It is natural: for $f : V \to W$,

$$f^{**} \circ \epsilon_V = \epsilon_W \circ f.$$

Indeed, evaluate either side at $v \in V$ and $\lambda \in W^*$; both give $\lambda(f(v))$. The proof uses a basis to establish invertibility, but the map itself uses no chosen basis.

There is generally no such preferred identification $V \cong V^*$. A basis or a nondegenerate bilinear form provides one, and different choices provide different maps. Naturality of the double dual should not be confused with a metric on V.

For $U \subseteq V$ define its annihilator

$$U^\circ = \{\lambda \in V^* : \lambda(u) = 0 \text{ for every } u \in U\}.$$

The quotient principle identifies $(V/U)^*$ canonically with U°: pull back along q. Restriction to U gives an exact sequence

$$0 \longrightarrow (V/U)^* \xrightarrow{q^*} V^* \xrightarrow{\text{res}} U^* \longrightarrow 0.$$

To prove surjectivity of restriction, extend a basis of U to one of V, and prescribe zero on the added vectors. The extension exists; it is usually not unique. The sequence is canonical even though a chosen splitting is not.

It follows that $\dim U^\circ = \dim V - \dim U$. Under $V \cong V^{**}$, $(U^\circ)^\circ = U$: inclusion is immediate, and the dimensions agree. For $f : V \to W$,

$$\ker f^* = (\operatorname{im} f)^\circ, \qquad \operatorname{im} f^* = (\ker f)^\circ.$$

The second identity follows by factoring a functional vanishing on $\ker f$ through $\operatorname{im} f$ and extending it to W. Consequently $\operatorname{rank} f^* = \operatorname{rank} f$ and row rank equals column rank. These are manifestations of testing a map by functionals.

Boundary. For infinite-dimensional algebraic vector spaces, dualization of short exact sequences is still exact when bases and their usual choice principles are available. But $V \to V^{**}$ need not be surjective and $V^* \otimes W \to \operatorname{Hom}(V, W)$ generally has only finite-rank maps in its image. Continuous duals in analysis form a different theory; the algebraic extension argument need not preserve continuity.

1.4 Direct sums and idempotents

The external direct sum $A \oplus C$ has inclusions i_A, i_C and projections p_A, p_C satisfying

$$p_A i_A = 1_A, \quad p_C i_C = 1_C, \quad p_A i_C = p_C i_A = 0, \quad i_A p_A + i_C p_C = 1_{A\oplus C}.$$

These equations are more useful than the ordered-pair notation: they allow block matrices and maps to be reconstructed without coordinates. A finite direct sum is both a product and a coproduct. For infinite families, the direct sum consists of finitely supported tuples, whereas the product includes all tuples; they are different constructions.

Idempotent principle. For an endomorphism $P : V \to V$, the equation $P^2 = P$ is equivalent to a decomposition

$$V = \operatorname{im} P \oplus \ker P$$

in which P is projection onto the first summand.

Proof. Write $v = P(v) + (v - P(v))$; the second term lies in $\ker P$. If $x = P(y)$ and $P(x) = 0$, then $x = P^2(y) = P(y) = 0$. Conversely projection in a direct sum plainly satisfies $P^2 = P$. □

This works for modules over any ring. If P commutes with an operator T, both summands are T-invariant. Conversely a decomposition into T-invariant summands gives a projection commuting with T. Thus finding a decomposition can be translated into finding idempotents in a commutant.

Failure of a tempting test. Pairwise zero intersection of three subspaces does not imply their sum is direct. In k^2, the lines spanned by e_1, e_2, and $e_1 + e_2$ have pairwise intersection zero, but satisfy a nontrivial three-term relation. The right test is that the addition map $\bigoplus_i U_i \to V$ be injective, or that $U_i \cap \sum_{j\neq i} U_j = 0$ for each i.

1.5 Splitting: existence versus choice

A short exact sequence

$$0 \longrightarrow A \xrightarrow{i} B \xrightarrow{p} C \longrightarrow 0$$

means that i is injective, p is surjective, and $\ker p = \operatorname{im} i$. It splits if there exists a linear section $s : C \to B$ with $p \circ s = 1_C$. In that case

$$F : A \oplus C \to B, \qquad F(a, c) = i(a) + s(c)$$

is an isomorphism preserving the inclusion and quotient maps.

Proof. If $F(a, c) = 0$, apply p to get $c = 0$, then injectivity of i gives $a = 0$. For $b \in B$, the element $b - s(p(b))$ lies in $\ker p$, so is $i(a)$ for a unique a. This also describes F^{-1}. A retraction $r : B \to A$, $ri = 1_A$, is equivalent: its kernel is a complement to $i(A)$ and restricts under p isomorphically to C. □

Over a field, lift a basis of C to B to obtain s. Over a general ring, the lifts may violate relations in C; that is the first source of extension obstructions.

The set of sections, if nonempty, is an affine space under $\mathrm{Hom}(C, A)$. For sections s, s', $p(s' - s) = 0$, so uniquely $s' - s = i \circ h$ for $h : C \to A$. Conversely $s + ih$ is a section for any h. This statement holds in module categories with the indicated Hom group. It precisely locates the ambiguity.

There is no natural complement in general. In the split model $B = A \oplus C$, the automorphism

$$u_h(a, c) = (a + h(c), c)$$

preserves i and p but sends the section $c \mapsto (0, c)$ to $c \mapsto (h(c), c)$. Whenever $h \neq 0$, no distinguished section can be invariant under all such structure-preserving automorphisms. A chosen positive inner product changes the problem: the orthogonal complement then becomes canonical relative to that extra data.

1.6 What to reconstruct and when

For an ordinary rank problem, first factor through the image. For constraints on functionals, take annihilators. For a proposed decomposition, construct its projections and check their equations. For an invariant subspace, ask explicitly whether the desired complement must preserve an operator or a group action. The underlying vector-space splitting proves less than an equivariant one.

The main source anchors are Xi II §§1.4–1.6 and 2.4 (printed pp. 14–24, 67–71; PDF pp. 20–30, 73–77), Grillet VIII.2–5 and X.1 (especially X.1.5, printed p. 395, PDF p. 407), and Mac Lane III.1–2 for the broader universal-property language. Proofs above are written out independently rather than inferred from those references.

2 Operators as modules and polynomial projections

2.1 Put the operator into the scalar ring

Let $T : V \to V$ be k-linear. Define an action of $R = k[t]$ on V by

$$p \cdot v = p(T)(v), \qquad p(T) = a_0 1_V + a_1 T + \cdots + a_d T^d \quad \text{if } p(t) = \sum_{j=0}^{d} a_j t^j.$$

Polynomial evaluation is a unital k-algebra homomorphism $k[t] \to \mathrm{End}_k(V)$, because the powers of a single operator commute. Denote the resulting module by V_T.

For another operator $S : W \to W$, a map $f : V \to W$ is $k[t]$-linear exactly when it is k-linear and

$$f \circ T = S \circ f.$$

The forward implication follows by applying f to $t \cdot v$; the reverse follows first for powers, then for polynomials. Thus operators and intertwining maps form the same category as $k[t]$-modules whose underlying k-spaces lie in the chosen dimension class. In particular, finite-dimensional operators correspond to modules finite-dimensional over k, a smaller class than all finitely generated $k[t]$-modules. Invariant subspaces are submodules, invariant quotient operators are quotient modules, and similarity means module isomorphism. This translation preserves the maps, not merely the list of isomorphism classes.

If $n = \dim_k V < \infty$, a k-basis generates V_T over R. Moreover, $1_V, T, \dots, T^{n^2}$ are linearly dependent in $\mathrm{End}_k(V)$, so some nonzero polynomial annihilates V_T. Its annihilator ideal has a unique monic generator m_T, the minimal polynomial. For $V = 0$, set $m_T = 1$. This argument avoids using Cayley–Hamilton before proving it.

2.2 The PID structure theorem and its mechanism

Structure theorem. If R is a principal ideal domain and M is a finitely generated R-module, then

$$M \cong R^r \oplus \bigoplus_{i=1}^{s} R/(d_i), \qquad d_1 \mid d_2 \mid \cdots \mid d_s,$$

where the d_i are nonzero nonunits. The free rank and the invariant-factor ideals (d_i) are unique. Over $k[t]$, choose the d_i monic. The isomorphism and its cyclic generators are generally not canonical.

Here is the construction in the Euclidean rings relevant to ordinary linear algebra, $k[t]$ and $\mathbb{Z}$. Write M as a cokernel of a matrix $A : R^a \to R^b$. Such a finite presentation exists: a submodule of a finite free module over a PID is finite free. Invertible row and column operations change the bases of the target and source and preserve its cokernel.

If $A \neq 0$, move a nonzero entry into the first position. Divide entries in its column and row by this pivot. A nonzero remainder can be moved to the pivot and has smaller Euclidean size. Repetition ends with a pivot dividing its entire row and column, which can then be cleared. If the pivot fails to divide an entry of the remaining block, add the corresponding row to the first row; a subsequent column division produces a smaller nonzero pivot. Thus this stage also terminates. The pivot now divides the remaining block. Repeat on that block to obtain a diagonal matrix with $d_i \mid d_{i+1}$, including units and zero entries as appropriate. Taking cokernels gives the theorem's existence statement in these Euclidean cases.

For uniqueness, let $I_j(A)$ be the ideal generated by all $j \times j$ minors. These ideals are unchanged by invertible row or column operations; multilinearity of minors gives containment in one direction and the inverse operation gives the other. For Smith diagonal form,

$$I_j(A) = (d_1 d_2 \cdots d_j)$$

for the nonzero diagonal positions, including unit positions. Thus the successive products, and hence the factors up to units, are determined. For arbitrary module presentations, invariance of the corresponding presentation ideals is a further theorem; alternatively uniqueness of the abstract decomposition follows from primary filtrations below. We are not identifying matrices of different presentation sizes without that extra argument.

For a general PID there need not be a Euclidean norm, so the specific descent algorithm above is not a proof in that generality. Grillet VIII.6.1–6.3 (printed pp. 336–342, PDF pp. 348–354) gives the general proof using principal ideals and a basis theorem. This is a cited extension of the construction just explained.

Why finite-dimensional operator modules have no free summand. The k-dimension of $k[t]$ is infinite, whereas $\dim_k V$ is finite. Hence

$$V_T \cong \bigoplus_{i=1}^{s} k[t]/(d_i), \qquad d_1 \mid \cdots \mid d_s.$$

Every block $k[t]/(d)$ has basis $1, t, \ldots, t^{d_0-1}$ if $d_0 = \deg d > 0$. Multiplication by t has companion matrix

$$C(d) = \begin{pmatrix} 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -a_{d_0-1} \end{pmatrix}, \quad d(t) = t^{d_0} + a_{d_0-1}t^{d_0-1} + \cdots + a_0.$$

The subdiagonal entries are all 1 and unspecified entries are 0; for $d_0 = 1$ this is the one-entry matrix $(-a_0)$. The defining relation $d(t) = 0$ explains the last column. Expansion of $\det(xI - C(d))$ along the last column gives $d(x)$; equivalently the cofactor of its jth last-column entry is the appropriate power of x. The invariant-factor blocks give the rational canonical form over any field. Consequently

$$\chi_T = \prod_i d_i, \qquad m_T = d_s \quad (V \neq 0).$$

In particular $m_T \mid \chi_T$, so $\chi_T(T) = 0$. This derives Cayley–Hamilton from the structure theorem, rather than assuming it in the proof that the module is torsion. For $V = 0$ the empty product is 1.

There is also a useful presentation that makes contact with matrices:

$$V_T \cong \operatorname{coker}(tI - A : R^n \to R^n),$$

where A is the matrix of T in a chosen basis. Send a vector polynomial $p(t) = \sum_j t^j v_j$ to $\sum_j A^j v_j$. The relation $tv \equiv Av$ reduces every class modulo $(tI - A)R^n$ to a constant. It also shows that a constant class maps to itself in k^n, so no nonzero constant is killed. Thus the induced map is bijective. Computing Smith form of $tI - A$ therefore computes similarity invariants of A; this is not the same as applying arbitrary left and right equivalence to A over k.

2.3 Bézout and Chinese remainder projections

Suppose $f_1, \ldots, f_r \in k[t]$ are pairwise coprime and $m = f_1 \cdots f_r$ annihilates T. Set $g_i = m/f_i$. Bézout gives $a_i g_i \equiv 1 \pmod{f_i}$. Let $e_i = a_i g_i$ in $k[t]/(m)$. Its residues are 1 modulo f_i and 0 modulo every other factor. The Chinese remainder theorem, or divisibility by each coprime factor, gives

$$e_i^2 = e_i, \qquad e_i e_j = 0 \ (i \neq j), \qquad \sum_i e_i = 1 \qquad (\text{mod } m).$$

Evaluate at T to obtain projections $P_i = e_i(T)$ satisfying these same identities in $\operatorname{End}_k(V)$. Their images are exactly $\ker f_i(T)$: first $f_i e_i$ is divisible by m, so the image lies there; conversely $e_i - 1$ is divisible by f_i, so $e_i(T)$ fixes that kernel. Hence

$$V = \bigoplus_i \ker f_i(T).$$

The polynomials a_i are not unique, but the projections are uniquely determined by the specified factors and the action. If m is the minimal polynomial and the factors are its primary powers, these are the canonical primary summands. Any operator commuting with T commutes with all the P_i, and thus preserves these summands.

Small example. If $T^2(T - 1_V) = 0$, take $P_1 = T^2$ and $P_0 = 1_V - T^2$. Indeed $T^3 = T^2$ implies $T^4 = T^2$, so $P_1^2 = P_1$, and

$$V = \ker T^2 \oplus \ker(T - 1_V).$$

With $T^2(T - 1_V)^2 = 0$ instead, one may take

$$P_1 = 3T^2 - 2T^3, \qquad P_0 = 1_V - 3T^2 + 2T^3.$$

These integer-coefficient formulas work in every characteristic: their defining congruences are polynomial identities over $\mathbb{Z}$. There is no division by 2 or 3. Only the distinctness of 0 and 1 is used.

What fails without coprimality. Take $R/(t^2)$. Its submodule $tR/(t^2)$ and quotient $R/(t)$ are both one-dimensional over k. They do not give an invariant direct sum: the only vectors killed by t are in $tR/(t^2)$. Dividing one repeated primary factor into smaller pieces does not create Bézout projectors.

2.4 Jordan form and what is actually canonical

Factor the invariant factors into monic irreducibles. Chinese remaindering gives blocks $k[t]/(p^a)$. This elementary-divisor decomposition exists over every field. If the minimal polynomial splits over k, each $p = t - \lambda$. Writing $N = T - \lambda 1_V$ on a block of size a, choose a cyclic vector v with $N^{a-1}v \neq 0$ and $N^a v = 0$. The ordered basis

$$N^{a-1}v, N^{a-2}v, \dots, Nv, v$$

gives the Jordan block $J_a(\lambda)$ with 1 on the superdiagonal. Thus Jordan form requires splitting of m_T over k, not necessarily that k itself be algebraically closed.

To recover block sizes without choosing chains, put $a_j = \dim \ker(T - \lambda 1_V)^j$ and $a_0 = 0$. Every λ-block of size b contributes $\min(j, b)$; other primary blocks contribute zero because $T - \lambda 1_V$ is invertible there. Therefore

$$\begin{aligned} a_j - a_{j-1} &= \#\{\lambda\text{-blocks of size at least } j\}, \\ 2a_j - a_{j-1} - a_{j+1} &= \#\{\lambda\text{-blocks of size exactly } j\}. \end{aligned}$$

For example, nullities $0, 3, 5, 6, 6$ give three λ-blocks, two of size at least 2, one of size at least 3, and none of size at least 4: the sizes are $3, 2, 1$. These are all the blocks if $\dim_k V = 6$; otherwise another primary summand can remain invisible to these kernels. Even saying that λ is the only eigenvalue in k does not exclude a nonsplitting summand. This gives uniqueness of the λ-block sizes

and a practical reconstruction procedure. More generally a p^b-block contributes $\deg(p)\min(j,b)$ to $\dim_k \ker p(T)^j$, so these filtrations recover the elementary divisors over a nonsplitting field as well.

The primary subspaces and block-size multiset are intrinsic. Individual cyclic summands, Jordan chains, their ordering, and a conjugating matrix are choices. Conflating a canonical list of invariants with a canonical basis leads to many false naturality claims.

Diagonalizability and semisimplicity. T is diagonalizable over k if and only if m_T is a product of distinct linear factors. The $k[t]$-module V_T is semisimple if and only if m_T is a product of distinct monic irreducible factors. These differ when the field does not split m_T. A real plane rotation through a nonzero angle other than π can be a simple $\mathbb{R}[t]$-module without having a real eigenvector.

2.5 Hom spaces and the commutant

A module map $R/(f) \to R/(g)$ is determined by the image y of 1, and its well-definedness condition is $fy = 0$ in $R/(g)$. For monic $d = \gcd(f,g)$ write $f = df_0$, $g = dg_0$ with $\gcd(f_0, g_0) = 1$. The condition $g \mid fy$ is equivalent to $g_0 \mid y$. Thus

$$\operatorname{Hom}_R(R/(f), R/(g)) \cong R/(d), \qquad h \bmod d \longmapsto (1 \longmapsto g_0 h \bmod g).$$

Cancellation here takes place in the domain $k[t]$. In particular

$$\dim_k \operatorname{Hom}_R(R/(f), R/(g)) = \deg \gcd(f,g).$$

The commutant of T is $\operatorname{End}_{k[t]}(V_T)$. If the elementary divisors are $(t-\lambda)^{b_{\lambda,i}}$, then

$$\dim_k \{S : ST = TS\} = \sum_{\lambda} \sum_{i,j} \min(b_{\lambda,i}, b_{\lambda,j}).$$

This computes the number of independent commuting maps without solving all matrix-entry equations. For a single cyclic block its commutant is $k[T]$. For multiple identical blocks there are extra intertwiners, so a commuting operator need not be a polynomial in T.

2.6 Fitting decomposition and the boundary of the method

For $N \geq \dim V$, both the kernels and images of powers of T have stabilized. Then

$$V = \ker T^N \oplus \operatorname{im} T^N.$$

If $v = T^N w$ lies in the intersection, then $T^{2N} w = 0$; stabilization gives $w \in \ker T^N$, so $v = 0$. Rank–nullity gives the direct sum. The restriction to the first summand is nilpotent and to the second is bijective. These subspaces are canonical, though a basis for either is not. This is the primary split into the factor t and the factors coprime to it, obtained without explicitly factoring the minimal polynomial.

For two commuting operators the ring is $k[s,t]$. It is not a PID: the ideal (s,t) is not principal. A simultaneous eigenbasis exists for a finite family of commuting diagonalizable operators over a common splitting field: decompose into eigenspaces of one operator and note that the others preserve them. Their restrictions remain diagonalizable because their minimal polynomials divide the

original square-free split minimal polynomials, so induction applies. Without diagonalizability, the one-variable module classification does not provide simultaneous Jordan form.

Source route. Xi II Theorems 2.51–2.57 and the existence/uniqueness proofs (printed pp. 72–77; PDF pp. 78–83) already contain Bézout decomposition, quotient induction, and intrinsic block counts. Grillet VIII.6–7 supplies the module derivation (printed pp. 336–346; PDF pp. 348–358); Hungerford IV.6 and VII.4–5 supplies another module-first route. The presentation, centralizer, and Fitting arguments above are included so the route is operational rather than a list of references.

3 Hom and tensor products

3.1 Hom records maps and variance

$\mathrm{Hom}_k(V, W)$ is a vector space under pointwise operations. Maps $a : V' \to V$ and $b : W \to W'$ induce

$$\mathrm{Hom}(V, W) \to \mathrm{Hom}(V', W'), \qquad f \longmapsto b \circ f \circ a.$$

The source variable reverses arrows and the target variable preserves them. For modules over a noncommutative ring, $\mathrm{Hom}_R(M, N)$ is always an abelian group but generally has no natural left R-module structure. Commutativity or specified bimodule actions must be supplied before treating it as one.

The kernel universal property gives left exactness. For a short exact sequence $0 \to A \xrightarrow{i} B \xrightarrow{p} C \to 0$ of R-modules,

$$0 \to \mathrm{Hom}_R(X, A) \xrightarrow{i\circ -} \mathrm{Hom}_R(X, B) \xrightarrow{p\circ -} \mathrm{Hom}_R(X, C)$$

is exact. A map $X \to B$ killed by p factors uniquely through i. There need not be surjectivity at the right: lifting the identity of C would be a section of p. Dually,

$$0 \to \mathrm{Hom}_R(C, X) \to \mathrm{Hom}_R(B, X) \to \mathrm{Hom}_R(A, X)$$

is exact. A map on the submodule A need not extend to B. Over a field both failures disappear, by choosing a splitting. The resulting exactness is canonical; its proof need not make a canonical choice.

3.2 Tensor universality and a construction

A tensor product of vector spaces V, W is a vector space $V \otimes_k W$ together with a bilinear map $\tau : V \times W \to V \otimes_k W$ such that every bilinear map $\beta : V \times W \to X$ factors uniquely as

$$\beta = \tilde{\beta} \circ \tau, \qquad \tilde{\beta} : V \otimes_k W \to X \text{ linear}.$$

Write $v \otimes w = \tau(v, w)$. The pair, not the bare vector space, is the universal object.

For existence, take the free vector space on symbols $[v, w]$ and quotient by the span of all bilinearity relations:

$$[v+v',w]-[v,w]-[v',w], \quad [v,w+w']-[v,w]-[v,w'],$$
$$[av,w]-a[v,w], \quad [v,aw]-a[v,w].$$

Every bilinear β extends linearly from the symbols and annihilates these relations, hence factors through the quotient. The symbols span, giving uniqueness. Universal uniqueness is proved by composing the two comparison maps, as for quotients.

If (e_i) and (f_j) are bases, then $(e_i \otimes f_j)$ is a basis. Spanning follows by expansion. For independence, the coefficient maps $\beta_{ij}(v,w) = e^i(v)f^j(w)$ induce linear functionals on the tensor product extracting each coefficient of a finite relation. This proof works for infinite algebraic bases too, because each vector and tensor uses only finite sums. It does not assert that dual bases span the whole algebraic dual in that case.

Recognition signal. A bilinear expression should usually be turned into a linear map on a tensor product before choosing coordinates. A formula on pure tensors defines a linear map only after its bilinearity or balancing has been checked. Pure tensors span; they are not generally all the elements.

For instance, $e_1 \otimes f_1 + e_2 \otimes f_2$ is not pure when the two displayed pairs are independent. Under coefficient matrices it has rank 2, whereas a pure tensor has entries $v_i w_j$ and rank at most 1.

3.3 Functorial maps, symmetry, and associativity

For $a : V \to V'$ and $b : W \to W'$, bilinearity gives the unique linear map

$$a \otimes b : V \otimes W \to V' \otimes W', \qquad (a \otimes b)(v \otimes w) = a(v) \otimes b(w).$$

Checking on pure tensors proves $(a' \otimes b') \circ (a \otimes b) = (a' \circ a) \otimes (b' \circ b)$. The same method gives canonical isomorphisms

$$V \otimes W \cong W \otimes V, \qquad (U \otimes V) \otimes W \cong U \otimes (V \otimes W),$$
$$k \otimes V \cong V, \qquad (U \oplus V) \otimes W \cong (U \otimes W) \oplus (V \otimes W).$$

These identify objects by specified maps; they are not literal equalities of set-theoretic constructions. Associativity is especially transparent by observing that both iterated tensor products represent trilinear maps.

In chosen product bases, the matrix of $a \otimes b$ is the Kronecker product. If A is $n \times n$ and B is $m \times m$,

$$\operatorname{tr}(A \otimes B) = \operatorname{tr}(A)\operatorname{tr}(B), \quad \det(A \otimes B) = \det(A)^m \det(B)^n.$$

For trace, sum the diagonal entries $a_{ii}b_{jj}$. For determinant, write $A \otimes B = (A \otimes I_m)(I_n \otimes B)$; the second factor is a direct sum of n copies of B, and the first is conjugate by the basis permutation $e_i \otimes f_j \mapsto f_j \otimes e_i$ to a direct sum of m copies of A. This proves the identity over every field and also over commutative rings. No diagonalization hypothesis is required.

3.4 Currying and the Hom–tensor adjunction

There is a canonical isomorphism, with no finite-dimensional hypothesis,

$$\mathrm{Hom}_k(U \otimes V, W) \cong \mathrm{Hom}_k(U, \mathrm{Hom}_k(V, W)).$$

It sends F to the map $u \mapsto (v \mapsto F(u \otimes v))$. Conversely a linear $g : U \to \mathrm{Hom}(V, W)$ yields the bilinear map $(u, v) \mapsto g(u)(v)$, hence a unique F. These operations are inverse, and pre/postcomposition proves naturality in all variables.

One can remember this as: "a linear map out of a tensor is a bilinear map, and a bilinear map is a linear family of linear maps." The adjunction is the precise version of that sentence, not an extra theorem to memorize separately.

For modules over a commutative ring R the same formulas work. Over a noncommutative ring one uses a right module and a left module and tracks the remaining bimodule actions. A correct underlying abelian-group version is

$$\mathrm{Hom}_{\mathbb{Z}}(M \otimes_R N, A) \cong \mathrm{Hom}_{R^{\mathrm{op}}}(M, \mathrm{Hom}_{\mathbb{Z}}(N, A)),$$

where M is right R, N is left R, and $(\phi \cdot r)(n) = \phi(rn)$ gives the Hom group its right action. The tensor represents biadditive maps satisfying $\beta(mr, n) = \beta(m, rn)$, called balanced maps. Two arbitrary left modules over a noncommutative ring cannot simply be tensored over R using the commutative formulas.

3.5 Finite-dimensional Hom as a tensor and trace as evaluation

Define

$$\Theta : W \otimes V^* \to \mathrm{Hom}(V, W), \qquad \Theta(w \otimes \lambda)(v) = \lambda(v)w.$$

If V is finite-dimensional, its inverse sends f to $\sum_i f(e_i) \otimes e^i$. Indeed evaluation at $v = \sum_i e^i(v)e_i$ recovers $f(v)$. Conversely applying the inverse to a pure tensor recovers it. The inverse expression is basis-independent because it is the inverse of a specified canonical map. W need not be finite-dimensional for this result.

For $W = V$, compose Θ^{-1} with the evaluation contraction $v \otimes \lambda \mapsto \lambda(v)$. This defines trace invariantly, and the basis formula is $\mathrm{tr}\, T = \sum_i e^i(T(e_i))$. The tensor corresponding to 1_V is $\sum_i e_i \otimes e^i$; it is basis-independent even though each summand is not.

Composition of rank-one maps is

$$\Theta(v \otimes \lambda) \circ \Theta(w \otimes \mu) = \lambda(w)\Theta(v \otimes \mu).$$

Taking traces gives $\lambda(w)\mu(v)$ in either order. Rank-one maps span all endomorphisms, so $\mathrm{tr}(ST) = \mathrm{tr}(TS)$. This is an evaluation identity, which explains why trace is invariant under similarity and why every commutator has trace zero. The converse is not needed here, and in positive characteristic trace of the identity can be zero without the identity being zero.

For infinite-dimensional V, every tensor in $W \otimes V^*$ still maps to a finite-rank operator, and every finite-rank operator arises this way. The canonical map is injective: express a tensor with linearly independent W-components and use coordinate functionals to show its coefficients vanish if the operator vanishes. Thus its image is exactly the finite-rank maps. In particular, finite-dimensional W also makes Θ surjective, even for infinite-dimensional V: for a basis w_i of W with coordinate functionals w^i, write $f(v) = \sum_i (w^i \circ f)(v) w_i$. The exact surjectivity criterion is that every linear map $V \to W$ have finite rank; finite dimension of either space is sufficient. When $V = W$ is infinite-dimensional the identity is a concrete obstruction to surjectivity.

This also exposes the issue in Xi II's remark after Theorem 6.19, printed p. 278 (PDF p. 284). The preceding definition uses **all** multilinear forms on $V^p \times (V^*)^q$. For $p = q = 1$, evaluation $(v, \lambda) \mapsto \lambda(v)$ is such a form, but it cannot be a finite sum $\sum_i \alpha_i(v)\lambda(w_i)$: that would represent 1_V as a finite-rank map. The finite-dimensional theorem is sound; its unqualified infinite-dimensional extension needs to be withdrawn or the tensor space restricted. The printed remark and definition were both checked as page images, so this is a mathematical correction, not an OCR inference.

3.6 Exactness, quotients, and change of scalars

Over a field, tensoring is exact. One proof chooses a splitting of a short exact sequence and uses distribution over direct sums. Alternatively a basis of W identifies $- \otimes W$ with a direct sum of copies of the original map. These are proofs of a canonical exactness statement, not canonical splittings.

Over a commutative ring, tensoring is always right exact:

$$A \xrightarrow{f} B \longrightarrow C \longrightarrow 0 \quad \Longrightarrow \quad A \otimes N \xrightarrow{f \otimes 1} B \otimes N \longrightarrow C \otimes N \longrightarrow 0.$$

To see the kernel, take the quotient of $B \otimes N$ by the image of $A \otimes N$. Bilinear maps from $B \times N$ vanishing on that image are precisely the ones factoring through $C \times N$. The quotient and $C \otimes N$ therefore satisfy the same universal property. In particular

$$(M/U) \otimes_R N \cong (M \otimes_R N)/\operatorname{im}(U \otimes_R N).$$

The word "image" matters: the induced map $U \otimes_R N \to M \otimes_R N$ need not be injective. For $0 \to \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \to \mathbb{Z}/2 \to 0$, tensoring with $\mathbb{Z}/2$ turns the first displayed map into zero. Thus a tensor product does not always preserve subobjects. A module for which tensoring preserves all short exact sequences is called flat.

If L/k is a field extension, $L \otimes_k V$ has an L-action $a(b \otimes v) = ab \otimes v$. A k-basis (e_i) gives an L-basis $(1 \otimes e_i)$, so dimension over the **new** field remains $\dim_k V$. The map $v \mapsto 1 \otimes v$ is injective, and scalar extension is exact. A real operator may acquire eigenvectors after extending to $\mathbb{C}$; its complexified invariant decomposition need not descend to real one-dimensional subspaces. The coefficient field is part of every classification statement.

Sources. Xi II §6.2, especially Definition 6.11, Theorem 6.15, and Examples 6.13–6.14 (printed pp. 273–278; PDF pp. 279–284), contains actual tensor universality and its applications. Grillet XI.5–8 (printed pp. 434–456; PDF pp. 446–468) supplies balanced products, general exactness, dual modules, and flatness. The concrete calculations here show why that extra generality matters.

4 Free algebras, exterior powers, determinants, and minors

4.1 Free constructions remember their extension property

The tensor algebra of a vector space is

$$T(V) = \bigoplus_{r \geq 0} V^{\otimes r}, \qquad V^{\otimes 0} = k,$$

with multiplication by concatenation of tensors. Each element has finitely many nonzero degree components. The inclusions of k and V are part of the construction. For any unital associative k-algebra A, a linear map $f : V \to A$ extends uniquely to an algebra homomorphism

$$\hat{f} : T(V) \to A, \qquad \hat{f}(v_1 \otimes \cdots \otimes v_r) = f(v_1) \cdots f(v_r).$$

The product on the right is multilinear, giving a map on each tensor power; concatenation proves multiplicativity, and generation by V proves uniqueness. An algebra generated by a specified copy or image of V is therefore a quotient of $T(V)$ by its relations. The quotient has to impose exactly the intended identities; it cannot be guessed from a superficial resemblance of formulas.

The symmetric algebra is

$$\mathrm{Sym}(V) = T(V)/\langle v \otimes w - w \otimes v : v, w \in V \rangle.$$

It is the free **commutative** unital algebra on V. The extension above factors precisely when the images $f(v)$ commute. With a basis $e_1, \ldots, e_n$, sending e_i to a variable x_i identifies it with $k[x_1, \ldots, x_n]$. The variables and monomial basis depend on the chosen basis, while the free property does not.

The homogeneous degree-r part $\mathrm{Sym}^r(V)$ represents symmetric r-linear maps from V^r. Notice the variance: polynomial expressions on a space V belong to $\mathrm{Sym}(V^*)$, not $\mathrm{Sym}(V)$, once its vectors are regarded as points. Over a finite field, formal polynomials are not determined by their values on all points: $x^q - x$ vanishes on $\mathbb{F}_q$ but is a nonzero polynomial. Thus a coordinate ring should not silently be replaced by the ring of set functions on its rational points.

4.2 Exterior algebra: alternation is the relation

Define

$$\Lambda(V) = T(V)/\langle v \otimes v : v \in V \rangle.$$

The product is denoted $\wedge$. Expanding $(v + w) \wedge (v + w) = 0$ gives $v \wedge w = -w \wedge v$, and exchanging blocks gives $\alpha \wedge \beta = (-1)^{rs} \beta \wedge \alpha$ for homogeneous degrees r, s. The defining equation $v \wedge v = 0$ remains necessary in characteristic 2: a sign rule alone does not imply it there.

The degree-r piece $\Lambda^r(V)$ represents alternating r-linear maps: an alternating map vanishes whenever two arguments are equal. Its unique factorization through $V^{\otimes r}$ kills the repeated-vector relations; conversely the wedge map is alternating. This construction works over any commutative ring, not only in characteristic zero.

With an ordered basis $e_1, \ldots, e_n$, the wedges

$$e_{i_1} \wedge \cdots \wedge e_{i_r}, \qquad 1 \le i_1 < \cdots < i_r \le n$$

form a basis. Reordering and removing repetitions proves spanning. For independence, for each ordered subset $I = (i_1, \dots, i_r)$ define on vectors with coordinates v_{ab} the alternating coefficient form

$$\beta_I(v_1, \dots, v_r) = \sum_{\sigma \in S_r} \operatorname{sgn}(\sigma) \prod_{a=1}^{r} v_{i_a, \sigma(a)}.$$

When two columns coincide, pair the permutations exchanging those columns; their terms sum to zero, in characteristic 2 as well. Thus β_I factors through $\Lambda^r(V)$ and takes value 1 on the I basis wedge and 0 on the other ascending ones. This proves independence without dividing by $r!$. Therefore

$$\dim \Lambda^r(V) = \binom{n}{r}, \qquad \Lambda^r(V) = 0 \ (r > n).$$

For $r = 0$, $\Lambda^0(V) = k$. The zero-dimensional top exterior power is therefore a line too.

4.3 Antisymmetrizing tensors is a model with extra hypotheses

If $r!$ is invertible in k, the operator

$$\operatorname{Alt}_r = \frac{1}{r!} \sum_{\sigma \in S_r} \operatorname{sgn}(\sigma)\sigma : V^{\otimes r} \to V^{\otimes r}$$

is idempotent. Its image identifies with $\Lambda^r(V)$: the quotient map restricted to that image is inverse to the map induced by Alt_r. With this normalization, exterior multiplication in the image is $\alpha \wedge \beta = \operatorname{Alt}_{r+s}(\alpha \otimes \beta)$. Other common conventions place factorial factors in the identification, so one must check which model a formula uses.

The analogous averaging operator without signs models symmetric powers when $r!$ is invertible. In small characteristic the quotient definitions remain valid but these normalized projectors do not. Symmetric powers (coinvariants) and invariant symmetric tensors must not be identified using a nonexistent $1/r!$. Polarization formulas with factorial denominators have the same limitation. Xi II §6.4 explicitly assumes characteristic zero for its antisymmetric-tensor model (printed p. 292, PDF p. 298); Grillet XIII.4 uses the quotient construction over a commutative ring (printed pp. 523–527, PDF pp. 535–539). These are related constructions with different stated scope.

4.4 Functoriality, rank, and determinants

For $f : V \to W$, the alternating map $(v_1, \dots, v_r) \mapsto f(v_1) \wedge \cdots \wedge f(v_r)$ induces

$$\Lambda^r f : \Lambda^r V \to \Lambda^r W.$$

Checking decomposable wedges gives $\Lambda^r(g \circ f) = \Lambda^r g \circ \Lambda^r f$ and $\Lambda^r 1_V = 1_{\Lambda^r V}$. For $r > 0$,

$$\Lambda^r f \neq 0 \quad \iff \quad \operatorname{rank} f \ge r.$$

If the image has dimension less than r, every image wedge is zero. If the rank is at least r, choose r vectors whose images are independent and extend those images to a basis of W; their wedge is nonzero by the basis theorem. This is the intrinsic meaning of testing rank by minors.

For $n = \dim V$ define the determinant line $\det V = \Lambda^n V$. It is one-dimensional. An endomorphism $T : V \to V$ therefore acts on it by a unique scalar:

$$\Lambda^n T = (\det T) 1_{\det V}.$$

This scalar is the determinant. Functoriality immediately gives $\det(ST) = \det S \det T$, and $\det T \neq 0$ precisely when T is invertible. For $V = 0$ the determinant is 1 by the degree-zero convention. Expanding $T(e_1) \wedge \cdots \wedge T(e_n)$ gives the usual signed permutation formula. Thus elimination formulas can be reconstructed from alternation and the chosen volume basis.

If $f : V \to W$ maps two different n-dimensional spaces, it determines a map $\det V \to \det W$. There is no preferred scalar determinant until trivializations of both lines are chosen. This distinction is useful in coordinate changes, orientation, and families of vector spaces.

4.5 Minors and Cauchy–Binet from composition

Let A be the matrix of $f : k^n \to k^m$. In ascending wedge bases, the entry of $\Lambda^r f$ in row subset I and column subset J is the minor $\det A_{I,J}$. This follows by expanding the image of $e_{j_1} \wedge \cdots \wedge e_{j_r}$ and collecting the I coefficient.

For a second matrix $B : k^m \to k^\ell$, functoriality gives

$$\det(BA)_{I,J} = \sum_{\substack{K \subseteq \{1,\dots,m\} \\ |K| = r}} \det B_{I,K} \det A_{K,J}.$$

This is the full minors version of Cauchy–Binet. The familiar determinant formula for a rectangular product is the case $r = n = \ell$. The same proof works over commutative rings using finite free modules and the exterior basis theorem. A rank interpretation over a field should not be carried unchanged to rings with zero divisors.

Expanding $\det(xI - T)$ by the columns in which $-T$ is selected gives

$$\chi_T(x) = \sum_{r=0}^{n} (-1)^r \operatorname{tr}(\Lambda^r T) x^{n-r}.$$

The trace of $\Lambda^r T$ is the sum of principal r-minors. In particular $\operatorname{tr} T$ and $\det T$ are the first and last nontrivial elementary symmetric coefficients. This works without eigenvalues in k. In positive characteristic, recovering these coefficients from power traces via Newton identities can require division by a nonunit integer; the exterior-power formula itself does not.

4.6 Determinant lines of exact sequences

For a short exact sequence of finite-dimensional vector spaces $0 \to A \xrightarrow{i} B \xrightarrow{p} C \to 0$, put $a = \dim A$ and $c = \dim C$. There is a canonical isomorphism

$$\det A \otimes \det C \xrightarrow{\sim} \det B.$$

Send a decomposable tensor to

$$(a_1 \wedge \cdots \wedge a_a) \otimes (c_1 \wedge \cdots \wedge c_c) \longmapsto i(a_1) \wedge \cdots \wedge i(a_a) \wedge \tilde{c}_1 \wedge \cdots \wedge \tilde{c}_c,$$

where $p(\tilde{c}_j) = c_j$. Changing a lift adds a vector of $i(A)$, whose extra occurrence makes that term zero. Multilinearity and alternation give a map on the determinant lines, and lifting a basis of C gives a basis of B, so it is an isomorphism. It is independent of the chosen lifts, although the splitting of the original sequence is not. The order "A then C" is part of the convention; reversing it contributes $(-1)^{ac}$.

An operator preserving A therefore satisfies $\det T_B = \det T_A \det T_C$. Applied to $xI - T$ over $k(x)$, or directly to its block triangular matrix, this gives $\chi_{T_B} = \chi_{T_A}\chi_{T_C}$ even without an invariant complement. Multiplication of invariants across an exact sequence does not imply splitting.

4.7 Contraction and reconstruction

For $\lambda \in V^*$ and $r \geq 1$ define contraction $\iota_\lambda : \Lambda^r V \to \Lambda^{r-1} V$ by

$$\iota_\lambda(v_1 \wedge \cdots \wedge v_r) = \sum_{j=1}^{r} (-1)^{j-1} \lambda(v_j) v_1 \wedge \cdots \hat{v}_j \cdots \wedge v_r.$$

The right side is alternating and multilinear in the v_j, so the universal property makes it well defined. Set contraction on $\Lambda^0 V = k$ to zero; together these maps define a degree -1 operator on the exterior algebra. Splitting the sum into the positions of α and β gives

$$\iota_\lambda(\alpha \wedge \beta) = (\iota_\lambda \alpha) \wedge \beta + (-1)^r \alpha \wedge \iota_\lambda \beta \quad (\deg \alpha = r).$$

This supplies a controlled introduction to signs in graded constructions. Exterior powers, contractions, and determinant lines are useful now; differential forms additionally require a manifold and smooth variation. They should not be identified merely because the same wedge symbol appears.

The core memory is short: free algebra extends linear maps, quotient imposes relations, exterior power transports alternating data, and the top exterior power is a line. Minors, rank tests, determinant multiplication, and Cauchy–Binet can be rebuilt from those principles.

5 Forms, adjoints, and the correct equivalence relation

5.1 Similarity, equivalence, and congruence classify different data

For an endomorphism $T : V \to V$, changing the basis by columns S changes its matrix from A to $S^{-1}AS$. The source and target are the same space and use the same changed basis: this is similarity.

For a map $f : V \to W$ with independently changed source and target bases, the matrix becomes $Q^{-1}AP$. This is matrix equivalence; over a field its only invariant is rank.

For a bilinear form $b : V \times V \to k$ with matrix B, $b(x, y) = x^{\mathsf{T}} B y$. The same change of vector basis in both arguments gives $B \mapsto S^{\mathsf{T}} B S$, called congruence. It is not similarity. The associated map

$$b^\flat : V \to V^*, \qquad b^\flat(v)(w) = b(v, w)$$

has a dual space as target; the two basis changes are linked by duality. A nondegenerate form is exactly one for which this map is an isomorphism. This identification depends on b.

For a Hermitian form over $\mathbb{C}$, this study uses the convention

$$h(x, y) = \overline{x}^{\mathsf{T}} H y,$$

conjugate-linear in the first argument and linear in the second, with $H = \overline{H}^{\mathsf{T}}$. The matrix changes by $H \mapsto S^* H S$ where $S^* = \overline{S}^{\mathsf{T}}$. The map $v \mapsto h(v, -)$ into the algebraic dual is conjugate-linear. Xi's inner-product convention is linear in the first argument instead; translate the convention rather than copying an adjoint formula unchanged.

A decisive example. Over $\mathbb{R}$, I_2 and $2I_2$ are congruent via $S = \sqrt{2}I_2$, but are not similar because they have different eigenvalues. Similarly the real symmetric matrices $\operatorname{diag}(1, -1)$ and $\operatorname{diag}(2, -2)$ have the same inertia but different similarity classes. A theorem about Jordan form cannot replace a theorem about forms.

5.2 Symmetric forms and quadratic forms

Assume $\operatorname{char} k \neq 2$. A symmetric bilinear form determines $q(v) = b(v, v)$, and

$$b(v, w) = \frac{q(v + w) - q(v) - q(w)}{2}.$$

Thus symmetric forms and quadratic forms encode the same data in this characteristic. In characteristic 2, $b(v, v)$ can be identically zero for a nonzero alternating form, and a quadratic form needs its own definition and polarization theory. Dividing by 2 conceals an essential hypothesis.

Every symmetric bilinear form over a field of characteristic different from 2 has a diagonal matrix in some basis. If $b \neq 0$, polarization shows there is v with $b(v, v) \neq 0$. Then for any w,

$$w = \frac{b(v, w)}{b(v, v)} v + \left(w - \frac{b(v, w)}{b(v, v)} v \right)$$

is a decomposition into kv and its b-orthogonal complement. The sum is direct since $b(v, v) \neq 0$. Induct on dimension. If the remaining restriction is zero, it contributes a radical summand. This proof explains both diagonalization and its field dependence: nonzero diagonal entries can be rescaled only by squares available in k.

Over $\mathbb{R}$, rescale the nonzero coefficients to $+1$ or -1. The numbers (p, q, r) of positive, negative, and zero coefficients are unique (Sylvester's law of inertia). To see uniqueness of p, the diagonal form has a positive-definite subspace of dimension p. Any subspace of dimension greater than p meets the span of the negative and zero coordinate vectors nontrivially by the dimension inequality, and hence cannot be positive definite. Thus p is the maximal dimension of a positive-definite subspace, an intrinsic number. The same argument gives q, and r is the dimension of the radical.

Over an algebraically closed field of characteristic different from 2, every nonzero coefficient is a square, so rank classifies symmetric forms up to congruence. Over $\mathbb{Q}$ or a number field, rank alone does not: square classes and further arithmetic invariants matter. "Diagonal" is not the same as "classified."

5.3 Alternating forms and symplectic pairs

A bilinear form is alternating if $b(v, v) = 0$ for all v. This implies $b(v, w) = -b(w, v)$ in every characteristic. Conversely skew-symmetry implies alternation only when $2 \neq 0$.

If $b \neq 0$, choose u, v with $b(u, v) \neq 0$ and rescale to $b(u, v) = 1$. The span $P = \langle u, v \rangle$ has matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and is nondegenerate. For any w define

$$w' = w + b(v, w)u - b(u, w)v.$$

Then $b(u, w') = b(v, w') = 0$, so $V = P \oplus P^{\perp}$. Induction decomposes an alternating form into symplectic pairs and a zero radical. Its rank is even, and a nondegenerate alternating space has even dimension. This argument works in characteristic 2 with the definition of alternation retained.

The essential memory is the nondegenerate two-dimensional pair and the orthogonal-complement induction, not a special sequence of row operations.

5.4 Adjoint operators require a chosen form

Let h be a nondegenerate Hermitian form on a finite-dimensional complex space. There is a unique $T^{\dagger}$ such that

$$h(Tv, w) = h(v, T^{\dagger}w) \qquad (v, w \in V).$$

In a basis with Gram matrix H, its matrix is

$$[T^{\dagger}] = H^{-1}[T]^{*}H.$$

This proves existence and uniqueness and shows explicitly why the adjoint is ordinary conjugate transpose only in an orthonormal basis. It also gives

$$(ST)^{\dagger} = T^{\dagger}S^{\dagger}, \qquad (aS + bT)^{\dagger} = \bar{a}S^{\dagger} + \bar{b}T^{\dagger},$$
$$(T^{\dagger})^{\dagger} = T.$$

For a nondegenerate real symmetric form the analogous formula is $H^{-1}A^{\mathsf{T}}H$. The algebraic dual map T^{*} is defined without a form and has a different type; the adjoint uses the form to return to V.

If h is positive definite, every subspace has a canonical orthogonal complement relative to h. Moreover

$$U \text{ is } T\text{-invariant} \iff U^{\perp} \text{ is } T^{\dagger}\text{-invariant.}$$

For example, if $u \in U$ and $w \in U^{\perp}$, $h(u, T^{\dagger}w) = h(Tu, w) = 0$. This is the mechanism behind invariant orthogonal decompositions. For an indefinite nondegenerate form a subspace can intersect

its own orthogonal complement: the isotropic line $\mathbb{R}(1,1)$ for $\mathrm{diag}(1,-1)$ does. Nondegeneracy of the ambient form alone does not give a direct sum with every orthogonal complement.

5.5 The finite-dimensional spectral theorem, with the hypotheses visible

Theorem. On a finite-dimensional positive-definite complex inner-product space, T has an orthonormal eigenbasis if and only if $TT^\dagger = T^\dagger T$.

Proof. One direction follows from a diagonal matrix and its conjugate transpose. For the other, call T normal. Any $N = T - \lambda 1_V$ is normal and

$$\|Nv\|^2 = h(v, N^\dagger N v) = h(v, N N^\dagger v) = \|N^\dagger v\|^2.$$

Hence $Tv = \lambda v$ implies $T^\dagger v = \bar{\lambda} v$. The complex characteristic polynomial has a root, so choose a unit eigenvector v. The line kv is invariant under both T and $T^\dagger$; its orthogonal complement is therefore also invariant under both. The restrictions commute and are adjoints there, so induction gives the orthonormal eigenbasis. □

A self-adjoint operator has real eigenvalues: if $Tv = \lambda v$, then $h(v, Tv) = \lambda \|v\|^2$ is real. A unitary operator has eigenvalues of modulus 1 by norm preservation. Real self-adjoint operators admit real orthonormal eigenbases, for instance by complexifying and observing that real eigenvalues yield real eigenspaces of the required dimensions. A real normal operator need not have real eigenvectors: a planar rotation gives a 2×2 real block instead. The field and the positive metric are doing actual work.

The nonzero nilpotent matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has a Jordan basis, but no orthonormal eigenbasis. It is not normal under the standard positive inner product. A purely module-theoretic classification does not replace the spectral theorem's metric conclusion.

5.6 Gram matrices, least squares, and determinant bounds

For vectors $v_1, \ldots, v_r$ in a positive inner-product space, their Gram matrix is $G = (h(v_i, v_j))$. If X is the matrix with these vectors as columns in an orthonormal basis, then $G = X^* X$. Thus

$$c^* G c = \left\| \sum_j c_j v_j \right\|^2 \geq 0,$$

and G is positive definite exactly when the vectors are independent. This is the correct reason the normal-equations matrix $X^* X$ is invertible for a full-column-rank least-squares problem. If X is rank deficient, the minimizer in coefficient space need not be unique, though the orthogonal projection of the target onto $\mathrm{im}\, X$ is unique.

Gram–Schmidt writes independent vectors as $X = QR$ with orthonormal columns of Q and upper triangular R. Its diagonal entries are the lengths of successive orthogonal residuals, so

$$\det G = \prod_j |r_{jj}|^2 \leq \prod_j \|v_j\|^2.$$

Dependent vectors give determinant zero, so the inequality holds in general. Equality for nonzero vectors occurs precisely when all projections onto earlier vectors vanish, equivalently the vectors are pairwise orthogonal. For a square real matrix this yields Hadamard's determinant inequality $|\det X| \le \prod_j \|v_j\|$. A sign matrix with orthogonal rows attains the bound, linking metric geometry directly to the intended Hadamard problem without using an unnecessary abstraction.

5.7 Affine and projective information that must be retained

For an affine quadratic polynomial $q(x) = b(x,x) + \ell(x) + c$ in characteristic different from 2, a translation $x = y + a$ changes its linear part to $\ell + 2b(a,-)$. If b is nondegenerate one can choose a unique a to cancel that part. If b is degenerate, the component of ℓ on the radical cannot be removed this way and controls further geometric types. The constant and the permitted transformations also matter. Similarity invariants of a coefficient matrix do not classify affine quadrics.

Projectivizing identifies nonzero scalar multiples of a vector, and homogeneous quadratic equations are themselves defined up to nonzero scale. Real points, signature, isotropic directions, and the distinction between affine and projective transformations are information a general algebra text may not teach in the same way as Xi II Chapters 4–5. Those chapters are selectively deferable for the operator route; their geometric content is not thereby proved replaceable.

Source anchors. Xi II §1.7 (printed pp. 25–43), Chapter 3 (pp. 85–160), and Chapters 4–5 (pp. 161–262) supply forms, adjoints, metric examples, orthogonal projection, and geometry. The normal spectral theorem is Theorem 3.62 (printed pp. 126–128; PDF pp. 132–134). The book comparison distinguishes these substantial results from the graduate books' module and category-theoretic strengths.

6 Exact sequences, extensions, and calibrated homological algebra

6.1 Exactness records how pieces fit

A sequence $A \xrightarrow{f} B \xrightarrow{g} C$ is exact at B if $\operatorname{im} f = \ker g$. The weaker condition $g \circ f = 0$ only gives containment. The quotient $\ker g / \operatorname{im} f$ measures the difference. For a short exact sequence of finite-dimensional vector spaces, $\dim B = \dim A + \dim C$; the dimension identity alone does not specify the maps or prove exactness.

An exact sequence of modules splits when its quotient map has a module-linear section. A module P is projective if every map $P \to C$ lifts across every surjection $B \to C$. Equivalently every short exact sequence ending in P splits: apply the lifting property to 1_P, or form the pullback of a proposed map along the surjection. A module is projective if and only if it is a direct summand of a free module. For the nontrivial direction, choose a free surjection $F \to P$ and split it; for the converse, lift basis vectors of a free module and restrict the lift to the summand.

All vector spaces over k are free, hence projective. They are also injective: a map defined on a subspace extends by a basis extension. Consequently

$$\operatorname{Ext}^j_k(V,W) = 0, \qquad \operatorname{Tor}^k_j(V,W) = 0 \qquad (j > 0).$$

One may use the projective resolution concentrated in degree 0 to see both vanishings. They are statements about the coefficient ring k and its module category. They do not imply that every complex of vector spaces is exact, nor that $\operatorname{Ext}_{k[t]}$ or $\operatorname{Ext}_{k[G]}$ vanishes.

6.2 The smallest equivariant nonsplitting example

Let $R = k[t]$, $M = R/(t^2)$, and $K = R/(t)$. Define

$$0 \longrightarrow K \xrightarrow{i} M \xrightarrow{p} K \longrightarrow 0, \qquad i(a \bmod t) = at \bmod t^2, \quad p(f \bmod t^2) = f \bmod t.$$

The maps are well defined and R-linear; the image of i is exactly the kernel of p. As vector spaces, the sequence splits via $s(a) = a \bmod t^2$. As modules, it does not split. Any lift of $1 \in K$ has the form $1 + ct$ in M, and

$$t(1 + ct) = t \neq 0,$$

whereas $t \cdot 1 = 0$ in K. Thus no module-linear section exists. In the vector basis $(t, 1)$, multiplication by t has matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Its invariant line kt has no invariant complement. This is the exact same obstruction expressed as a module extension, a failure of equivariant splitting, or a Jordan block.

The underlying vector-space splitting is useful, but forgetting the action has discarded the information responsible for the obstruction.

6.3 Extensions and the first Ext group, with an explicit computation

An extension of M by N is a short exact sequence $0 \to N \to E \to M \to 0$. Two extensions are equivalent when an isomorphism of their middle terms commutes with the specified inclusion and quotient maps and is the identity on N and M. Merely having isomorphic middle modules is a weaker relation. The equivalence classes form $\operatorname{Ext}^1_R(M, N)$ under Baer sum, with the split extension as zero. The general equivalence with the derived-functor definition is cited from Hilton–Stammbach III.1–2; we can construct the important cyclic case directly.

Let R be a commutative domain, $0 \neq f \in R$, and N an R-module. The free resolution

$$0 \longrightarrow R \xrightarrow{\times f} R \longrightarrow R/(f) \longrightarrow 0$$

is exact. Apply $\operatorname{Hom}_R(-, N)$. Evaluation at 1 identifies the two Hom modules with N, and the induced map is multiplication by f. Hence

$$\operatorname{Hom}_R(R/(f), N) = \ker(f : N \to N), \qquad \operatorname{Ext}^1_R(R/(f), N) \cong N/fN,$$

and the positive Ext groups of degree greater than 1 vanish for this first argument. The injectivity of $\times f$ uses both the domain and $f \neq 0$ hypotheses; over a ring with zero divisors this two-term complex may not be a resolution.

The extension meaning of N/fN can be seen without the derived-functor terminology. For an extension, choose $e \in E$ over 1 mod f. Then $fe = i(n)$ for a unique $n \in N$. Replacing e by $e + i(n_0)$ changes n by fn_0. Conversely, for $n \in N$ define

$$E_n = (N \oplus Re)/R(-n, fe),$$

meaning the relation $fe = n$. The copy of N injects because $rf = 0$ forces $r = 0$. The quotient is $R/(f)$; every extension has this form by expressing its elements using N and a lift of 1. Changes of the lift give precisely the equivalences $n \sim n + fn_0$. The zero class permits a lift killed by f, and therefore a section. Pullback followed by addition of the two copies of N sends classes $[n], [n']$ to $[n + n']$, explaining the addition law.

For $R = k[t]$ and $N = K = R/(t)$ this gives

$$\operatorname{Ext}^1_{k[t]}(K, K) \cong k.$$

The class $a \in k$ has generators u, e and relations $tu = 0$, $te = au$. If $a = 0$ it is split. For every $a \neq 0$, the middle module is isomorphic to $R/(t^2)$, but the extension classes for different a remain different with the endpoints fixed. Thus an extension invariant contains more information than an isomorphism class of a single operator.

6.4 Tor detects lost injections

Tensor the same free resolution with N. Its positive homology is

$$\operatorname{Tor}^R_1(R/(f), N) = \ker(f : N \to N), \qquad R/(f) \otimes_R N \cong N/fN.$$

The groups of degree greater than 1 vanish for this first argument. For $R = \mathbb{Z}$, $f = m \neq 0$, and $N = \mathbb{Z}/n$,

$$\mathbb{Z}/m \otimes \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m, n), \qquad \operatorname{Tor}^{\mathbb{Z}}_1(\mathbb{Z}/m, \mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n).$$

For the kernel, multiplication by m on $\mathbb{Z}/n$ kills the subgroup generated by $n/\gcd(m, n)$, which has that order. This supplies a simple nonzero Tor example and explains the injection failure after tensoring the multiplication-by-m sequence.

An obstruction in Ext asks whether maps can lift or extend; an obstruction in Tor asks what tensoring has collapsed. They are useful descriptions only when the coefficient ring and the relevant sequence are retained.

6.5 Complexes and homology over a field

A chain complex is a sequence of maps $d_n : C_n \to C_{n-1}$ satisfying $d_{n-1} \circ d_n = 0$. Define

$$Z_n = \ker d_n, \qquad B_n = \operatorname{im} d_{n+1}, \qquad H_n(C) = Z_n/B_n.$$

A chain map $f : C \to D$ satisfies $d^D_n \circ f_n = f_{n-1} \circ d^C_n$. It carries cycles to cycles and boundaries to boundaries, hence induces $H_n(f)$. The quotient property constructs that map, and composition makes homology a functor.

Why the field does not force zero homology. The complex with a single nonzero term $C_0 = k$ has $H_0 = k$. In a two-term complex $0 \to k^2 \xrightarrow{d} k^2 \to 0$ with $d(x, y) = (x, 0)$, both H_1 and H_0 have dimension 1. All vector-space short exact sequences involved split; the map d still need not have maximal rank or have image equal to a prescribed cycle space.

In fact splitting gives a normal description of a complex. Choose complements

$$Z_n = B_n \oplus \mathcal{H}_n, \qquad C_n = Z_n \oplus L_n.$$

Then $\mathcal{H}_n \cong H_n(C)$ and $d_n : L_n \to B_{n-1}$ is an isomorphism: it is injective because $L_n \cap \ker d_n = 0$, and its image is all B_{n-1}. Thus

$$C_n = B_n \oplus \mathcal{H}_n \oplus L_n,$$

with differential zero on $B_n \oplus \mathcal{H}_n$ and an isomorphism from L_n onto B_{n-1}. A complex is therefore a direct sum of its homology with zero differential and two-term contractible pieces, after choices. The decomposition does not say the homology summands are zero.

More explicitly let $j : \mathcal{H} \to C$ and $q : C \to \mathcal{H}$ be inclusion and projection. Define $s_n : C_n \to C_{n+1}$ as the inverse of $d_{n+1} : L_{n+1} \to B_n$ on B_n, and zero on $\mathcal{H}_n \oplus L_n$. Then

$$d_{n+1}s_n + s_{n-1}d_n = 1_{C_n} - j_n q_n.$$

This is a chain homotopy from the identity to the projection onto chosen homology representatives. In particular an acyclic complex of vector spaces is contractible. Neither the contraction nor representatives are generally natural. Over $k[t]$, the required invariant complements can fail, so the same decomposition need not exist in that module category.

6.6 A connecting homomorphism is a controlled lifting construction

Let $0 \to A \xrightarrow{i} B \xrightarrow{p} C \to 0$ be a short exact sequence of chain complexes, exact in every degree. If $c \in C_n$ is a cycle, lift it to $b \in B_n$. Then $p(db) = dc = 0$, so uniquely $db = i(a)$ for $a \in A_{n-1}$. Since $i(da) = d^2b = 0$, injectivity gives $da = 0$. Define

$$\delta : H_n(C) \to H_{n-1}(A), \qquad \delta([c]) = [a].$$

A different lift changes b by $i(a_0)$ and therefore changes a by da_0, a boundary. Changing c by a boundary and lifting its primitive changes b by a boundary in B, which does not alter a except for the preceding lift ambiguity. Thus δ is well defined and linear.

The resulting sequence

$$\cdots \to H_n(A) \to H_n(B) \to H_n(C) \xrightarrow{\delta} H_{n-1}(A) \to H_{n-1}(B) \to \cdots$$

is exact. For example, $\delta([c]) = 0$ means $a = da_0$; replacing b by $b - i(a_0)$ gives a cycle lifting c, so $[c]$ comes from $H_n(B)$. Conversely a cycle lift plainly gives $\delta([c]) = 0$. At $H_{n-1}(A)$, a cycle a dies in B precisely when $i(a) = db$; then $p(b)$ is a cycle and maps under δ to $[a]$. At $H_n(B)$, if $p(b) = dc'$ lift c' to b' and observe that $b - db'$ lies in $i(A)$ and is a cycle. These checks prove exactness at all three kinds of position.

This explains why exact sequences are not decorative notation: the connecting map measures the obstruction to lifting a cycle as a cycle.

6.7 A sensible stopping point for the homological toolkit

For ordinary finite-dimensional linear algebra, retain exactness, splitting, the explicit $k[t]/(t^2)$ example, complexes, and the one-step resolution calculations. Derived functors become worthwhile when a concrete functor fails to be exact or an equivariant extension remains nonsplit. General spectral sequences, derived categories, and long resolutions can be deferred until a specific problem needs them.

For the Hadamard/Williamson direction, a common-lift obstruction is not automatically an Ext class: sign constraints define a nonlinear finite set, not a module extension with an identified coefficient category. Calling it "homological" without constructing that category and obstruction would add a label without mathematical content.

Sources. Grillet X.1, X.3, XI.2, XI.6, and XII.1–5 give the general framework. Hilton–Stammbach III.1–4 (printed pp. 84–99; PDF pp. 95–110) is especially useful for extension equivalence and computations; IV.1–4 (printed pp. 117–130; PDF pp. 128–141) supplies complexes, connecting maps, homotopy, and resolutions. The cyclic extension construction and field-complex decomposition above are fully derived so the reader can use the ideas before reading a whole homological algebra text.

7 Representations and semisimplicity

7.1 An action becomes a module

A representation of a group G on a k-vector space is a homomorphism $\rho : G \to \mathrm{GL}(V)$. Extend linearly to the group algebra $k[G]$, whose elements are finite sums $\sum_g a_g g$ and whose product extends the group law bilinearly. Then

$$\left(\sum_g a_g g\right) \cdot v = \sum_g a_g \rho(g)(v)$$

makes V a $k[G]$-module. Conversely such a module restricts to a group action because g^{-1} supplies the inverse operator. Intertwining maps are exactly module maps, invariant subspaces are submodules, and irreducible representations are nonzero simple modules.

For the infinite cyclic group, $k[G] \cong k[t, t^{-1}]$; this describes an invertible operator. For a cyclic group of order m, $k[G] \cong k[t]/(t^m - 1)$; its representations are operators satisfying $T^m = 1_V$. If $\operatorname{char} k$ does not divide m, the polynomial $t^m - 1$ is square-free, since its derivative is mt^{m-1}. Over a field containing its roots, such operators are diagonalizable. Both separability and the splitting field enter this elementary representation example.

7.2 Maschke's averaging argument

Theorem. If G is finite and $|G|$ is invertible in k, every invariant subspace of a finite-dimensional representation has an invariant complement. Hence every finite-dimensional representation is a direct sum of irreducibles.

Proof. Let $U \subseteq V$ be invariant. Choose an ordinary linear projection $P : V \to V$ with image U and $P(u) = u$ for every $u \in U$. Define the endomorphism

$$\overline{P} = \frac{1}{|G|} \sum_{g \in G} \rho(g) P \rho(g)^{-1}.$$

Each summand has image in U and is the identity on U, so the average does too; in particular $\overline{P}^2 = \overline{P}$. Reindexing the sum proves that $\overline{P}$ commutes with every $\rho(h)$. Therefore $\ker \overline{P}$ is an invariant complement. If V is not irreducible, split it and repeat, decreasing dimensions. □

The averaged projection depends on the original projection; the theorem asserts existence, not a preferred decomposition. Over $\mathbb{R}$ or $\mathbb{C}$, one can instead average a positive inner product. It remains positive because a nonzero vector has a positive contribution in each summand. Its invariant orthogonal complements prove complete reducibility with a geometric meaning.

Characteristic failure. In characteristic p, let a generator of C_p act on k^2 by $1 + N$, where $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Since $N^2 = 0$, $(1 + N)^p = 1$ in characteristic p (also when $p = 2$). This is a representation with invariant line ke_1 and no invariant complement, exactly as in the $k[t]/(t^2)$ example. Averaging would require $1/p$.

Finite-dimensional representations of infinite groups need not be semisimple even over $\mathbb{C}$: send $n \in \mathbb{Z}$ to $1 + nN$. The same invariant line fails to split. Finite-dimensionality alone does not remove extensions.

7.3 Schur's lemma and the role of the field

If S, T are irreducible representations, any intertwiner $f : S \to T$ is either zero or an isomorphism. Its kernel and image are invariant, and irreducibility forces the result. Thus $\mathrm{End}_G(S)$ is a division algebra over k.

If k is algebraically closed and S is finite-dimensional, every endomorphism f has an eigenvalue λ. The intertwiner $f - \lambda 1_S$ has nonzero kernel, so it is zero. Therefore $\mathrm{End}_G(S) = k1_S$. Neither the algebraic-closure nor finite-dimensional hypothesis should be suppressed.

Over $\mathbb{R}$, let C_4 act on the plane by a quarter-turn $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. There is no real invariant line, so the representation is irreducible, but its commuting endomorphisms include all $aI + bJ$, a copy of $\mathbb{C}$. Thus "irreducible implies scalar commutant" is false over $\mathbb{R}$.

For a finite-dimensional semisimple representation over an algebraically closed field, let S_i represent the distinct simple types occurring in V. The evaluation map gives the canonical isotypic description

$$\bigoplus_i S_i \otimes_k \mathrm{Hom}_G(S_i, V) \xrightarrow{\sim} V, \qquad s \otimes f \longmapsto f(s).$$

The group acts only on the first factor. To prove the map is an isomorphism, choose a decomposition $V \cong \bigoplus_i S_i^{m_i}$ and use Schur's lemma: the Hom space for S_i is k^{m_i} and contributes exactly that summand. The map was specified without this decomposition, so invertibility does not make the chosen copies canonical. The isotypic subspaces are intrinsic; a basis of each multiplicity space is not. Their commutant is a product of matrix algebras $\prod_i M_{m_i}(k)$ after choosing those bases.

7.4 Duals and tensors of representations

Define the dual action by

$$\rho^{\vee}(g)(\lambda) = \lambda \circ \rho(g^{-1}).$$

The inverse compensates for reversal under duality, ensuring $\rho^{\vee}(gh) = \rho^{\vee}(g)\rho^{\vee}(h)$. Plain transpose without the inverse would reverse the group multiplication. In dual bases its matrix is $\rho(g^{-1})^{\mathsf{T}}$, without complex conjugation.

For two representations, the diagonal tensor action is

$$g \cdot (v \otimes w) = \rho(g)(v) \otimes \sigma(g)(w).$$

Functoriality proves the group law. The canonical identification $W \otimes V^* \cong \mathrm{Hom}(V, W)$ carries the action to $f \mapsto \sigma(g) f \rho(g)^{-1}$. Its invariant vectors are the intertwiners. This is a practical use of tensor variance, not merely an alternative notation.

Characters are traces $\chi_V(g) = \operatorname{tr} \rho(g)$. Direct sums and tensor products satisfy

$$\chi_{V \oplus W} = \chi_V + \chi_W, \qquad \chi_{V \otimes W} = \chi_V \chi_W.$$

For finite groups over $\mathbb{C}$, characters determine isomorphism classes; the full orthogonality theorem proving that statement is a cited next step, not required to prove the two formulas. In modular characteristic the ordinary complex character package cannot be copied without changes.

7.5 From groups to Lie algebras: a short concrete bridge

For a differentiable representation of a matrix Lie group, differentiation at the identity yields a Lie algebra representation $d\rho : \mathfrak{g} \to \mathrm{End}(V)$ preserving brackets. Tensor actions differentiate according to the product rule:

$$X \cdot (v \otimes w) = (X \cdot v) \otimes w + v \otimes (X \cdot w).$$

Thus the Lie algebra operator is $A \otimes I + I \otimes B$, whereas the group operator is $\rho(g) \otimes \sigma(g)$. Confusing these two formulas changes the representation. Dualization differentiates to $X \cdot \lambda = -\lambda \circ d\rho(X)$.

Here is an explicit algebraic example over a characteristic-zero field. On homogeneous polynomials of degree m in x, y, set

$$E = x\frac{\partial}{\partial y}, \qquad F = y\frac{\partial}{\partial x}, \qquad H = x\frac{\partial}{\partial x} - y\frac{\partial}{\partial y}.$$

On the basis $v_j = x^{m-j}y^j$, $0 \le j \le m$,

$$Ev_j = jv_{j-1}, \qquad Fv_j = (m-j)v_{j+1}, \qquad Hv_j = (m-2j)v_j,$$

with missing endpoint vectors interpreted as zero. Direct calculation gives $[H, E] = 2E$, $[H, F] = -2F$, and $[E, F] = H$. This defines a representation of $\mathfrak{sl}_2$.

It is irreducible. The distinct eigenvalues $m-2j$ of H allow polynomial projections onto its individual weight spaces by interpolation. A nonzero invariant subspace therefore contains some v_j. Repeated application of E reaches v_0, since the intervening integers are nonzero in characteristic zero; repeated application of F then reaches every v_j. The argument explicitly explains where small positive characteristic can fail.

Over $\mathbb{C}$, these are all finite-dimensional irreducible $\mathfrak{sl}_2$ representations. Hall §4.6 proves this classification using a highest-weight vector and the commutator identity $EF^jv = j(m-j+1)F^{j-1}v$ (printed pp. 96–101, PDF pp. 109–114). The notes above prove the model and its irreducibility, while the exhaustion statement is cited.

The associated $SU(2)$ representation is $\operatorname{Sym}^m \mathbb{C}^2$. The element $-I$ acts by $(-1)^m$, so it descends through $SU(2) \to SO(3)$ exactly when m is even. Integrating a Lie algebra representation and descending through a covering are extra global questions; Lie algebra isomorphisms alone do not identify the corresponding groups or their representations. Hall's group-versus-Lie-algebra sections and Xi III §3.6 are useful precisely at this boundary.

Selective sources. Xi III §3.2 gives an actual averaging proof (Theorem 3.20, printed pp. 109–111, PDF pp. 115–117), §3.7 constructs dual and tensor representations (pp. 152–162, PDF pp. 158–168), and §4.6 introduces $\mathfrak{sl}_2$ modules (starting p. 215, PDF p. 221). Grillet IX.2 and IX.7 place these in semisimple module theory. Hall §§4.3–4.7 gives the finite-dimensional Lie-theoretic version. Compact-group averaging requires integration and continuity, whereas the finite-group proof above requires only an invertible group order.

Terminology warning. Grillet IX.7 initially uses "irreducible" to mean indecomposable, that is, admitting no nontrivial invariant direct-sum decomposition (printed p. 381, PDF p. 393). He explicitly relates that definition to indecomposable modules on p. 383 (PDF p. 395). This study and Xi/Hall use the standard invariant-subspace definition, corresponding to a simple module. They coincide in a semisimple setting, but not for a nontrivial Jordan block. Thus Grillet IX.7.1, asserting a decomposition of every finite-dimensional representation into his "irreducibles," is not the complete-reducibility theorem with the terminology of these notes. This is a verified convention difference, not a mathematical counterexample to his stated definition.

8 Bridges to arithmetic, algebraic geometry, and differential topology

These sections explain how the same constructions lead outward. They are selective derivations and prerequisite diagnoses, not compressed substitutes for the destination books. The Hadamard/Williamson bridge follows separately and uses only the directions that have a concrete role there.

8.1 Finite extensions as linear algebra: trace and norm

Let L/k be a finite field extension of degree n. Every $a \in L$ defines the k-linear multiplication map

$$m_a : L \to L, \qquad m_a(x) = ax.$$

Define $\operatorname{Tr}_{L/k}(a) = \operatorname{tr}(m_a)$ and $N_{L/k}(a) = \det(m_a)$. Since $m_{ab} = m_a \circ m_b$ and $m_{a+b} = m_a + m_b$, norm is multiplicative and trace is k-linear. For $c \in k$, $m_c = c1_L$, so trace is nc and norm is c^n. The map $a \mapsto m_a$ is faithful: evaluate at 1 to recover a.

For $L = k(\alpha)$ with monic minimal polynomial $f(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_0$, multiplication by α has the companion matrix of f. Hence

$$\mathrm{Tr}_{L/k}(\alpha) = -a_{n-1}, \qquad N_{L/k}(\alpha) = (-1)^n a_0.$$

If L/k is separable, let Ω be an algebraic closure of k. The primitive-element theorem writes $L = k(\alpha)$; it is cited here, not assumed for inseparable extensions. The polynomial f has distinct roots $\alpha_1, \dots, \alpha_n$, and

$$\Omega \otimes_k L \cong \Omega[t]/(f) \cong \prod_{i=1}^{n} \Omega$$

by scalar extension and Chinese remaindering. Multiplication by a becomes diagonal with entries $\sigma_i(a)$ for the n embeddings $\sigma_i : L \to \Omega$. Trace and determinant are unchanged by scalar extension of the matrix, so

$$\mathrm{Tr}_{L/k}(a) = \sum_i \sigma_i(a), \qquad N_{L/k}(a) = \prod_i \sigma_i(a).$$

This is the same polynomial-splitting mechanism used for operators. The new content is that the regular representation comes from a field multiplication.

The trace pairing $b(a, b) = \mathrm{Tr}_{L/k}(ab)$ is nondegenerate for finite separable extensions. In the power basis its Gram matrix is $V^{\mathsf{T}}V$, where $V_{ij} = \alpha_i^{j-1}$. The Vandermonde determinant is nonzero because the roots are distinct; therefore its square, the determinant of the trace Gram matrix, is nonzero. This proves nondegeneracy even in positive characteristic when the extension is separable; positivity has no role.

For a finite purely inseparable counterexample, take $k = \mathbb{F}_p(s)$ and $L = k(\alpha)$ with $\alpha^p = s$. In the basis $1, \alpha, \dots, \alpha^{p-1}$, multiplication by $\sum_{j=0}^{p-1} a_j \alpha^j$ has every diagonal entry a_0; the other powers produce cyclic shifts. Thus its trace is $pa_0 = 0$. The trace pairing is identically zero although $[L : k] = p$. A field extension being finite does not suffice for trace duality.

Xi III §5.8 (printed pp. 272–278, PDF pp. 278–284) is the immediate Galois-theoretic source. Lang III.1 (printed pp. 57–62, PDF pp. 71–76) uses the separable trace pairing to define complementary integral modules. The latter retains an integral lattice, which is extra information beyond the k-vector space and its nondegenerate pairing.

8.2 Why arithmetic needs more than field extensions

In a number field, the ring of integers is a $\mathbb{Z}$-module inside a $\mathbb{Q}$-vector space. Tensoring with $\mathbb{Q}$ recovers the ambient vector space but loses torsion and integral index. For example $2\mathbb{Z} \subset \mathbb{Z}$ becomes an isomorphism after tensoring with $\mathbb{Q}$, while its cokernel is $\mathbb{Z}/2$ before tensoring. Integral bases and discriminants cannot be read from the rational vector space alone.

Localizing a commutative ring R at a multiplicative set S produces $S^{-1}R$ with a map from R making all elements of S invertible. It is universal for that property. For modules, $S^{-1}M \cong S^{-1}R \otimes_R M$ and localization is exact. For instance if a localized element maps to zero, some denominator in S annihilates its original image; multiplying the numerator by that denominator reduces the kernel

assertion to the original exact sequence. This proves the middle exactness directly, with surjectivity obtained by lifting numerators.

A finitely generated torsion-free module over a Dedekind domain is projective (Lang I.9, Proposition 26, printed p. 29, PDF p. 43). It need not be free: a nonprincipal invertible ideal is a rank-one example. Its scalar extension to the fraction field is one-dimensional, so vector-space dimension misses its ideal class. Over a PID, this distinction vanishes for finitely generated torsion-free modules. That is the arithmetic boundary of the elementary structure theorem.

To read Lang efficiently, the missing bridges are integral dependence and integral closure, prime ideal factorization in Dedekind domains, discrete valuations, completions, and local ramification. Some algebraic ingredients appear in Hungerford VIII and Grillet's commutative algebra chapters; their existence in a table of contents does not establish complete preparation for Lang's local and analytic arguments. Lang I.9 explicitly says it is not used later in that book, so projectivity should be read for this conceptual bridge, not falsely advertised as a prerequisite for all his subsequent chapters.

8.3 Affine geometry: symmetric algebra, ideals, and tangent vectors

For a finite-dimensional vector space V, the affine coordinate algebra is $\mathrm{Sym}(V^*)$. Imposing polynomial equations amounts to taking a quotient by an ideal. Over an algebraically closed field, passage between radical ideals and algebraic sets requires the Nullstellensatz; it is a substantial theorem, not an immediate consequence of the quotient property. Nonradical ideals retain nilpotent information invisible to a set of points.

For a finitely generated k-algebra A and a k-rational point $x : A \to k$, put $\mathfrak{m} = \ker x$. A tangent vector at x can be defined as a k-linear derivation $D : A \to k$ satisfying

$$D(ab) = x(a)D(b) + x(b)D(a).$$

It vanishes on constants and on $\mathfrak{m}^2$. Conversely every functional $\ell : \mathfrak{m}/\mathfrak{m}^2 \to k$ defines $D(a) = \ell((a - x(a)) \bmod \mathfrak{m}^2)$. Expanding a product proves the derivation law, and these assignments are inverse. Thus

$$T_x \operatorname{Spec} A \cong \operatorname{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k).$$

This statement uses a rational point, so its residue field is k. At a general scheme point the residue field and the relative differential module must be tracked.

For $A = k[x_1, \dots, x_n]/I$ at a rational point a, expand every $f \in I$ modulo the square of the maximal ideal:

$$f(a + u) \equiv f(a) + \sum_i \frac{\partial f}{\partial x_i}(a)u_i \pmod{(u)^2}.$$

The tangent space is the common kernel of these linear equations. For the cusp $y^2 = x^3$ at the origin, there is no nonzero linear equation from its defining polynomial, so the tangent space is two-dimensional even though the curve has dimension one. For $A = k[\varepsilon]/(\varepsilon^2)$, the point has a one-dimensional tangent space despite its underlying point set having one point. Nilpotents store infinitesimal directions.

Tensor products now do geometric work: affine fiber products are represented by $A \otimes_R B$. Fibers of a module use $M \otimes_R \kappa(\mathfrak{p})$. Because this tensor operation may fail to preserve injections, flatness is needed to control families. The finite-dimensional vector-space version of exactness is insufficient once the coefficient ring varies.

8.4 What the scheme and sheaf bridge adds

Hartshorne II.1–2 introduces sheaves and schemes; II.5 develops sheaves of modules, and II.8 introduces differentials. These topics add local data and gluing, not just new names for global vector spaces. A locally free sheaf is locally a finite direct sum of the structure sheaf; a global free basis is an extra condition. On an affine scheme, finite locally free sheaves correspond to finitely generated projective modules, a theorem requiring the module–sheaf correspondence and its local criteria.

Global sections form a left-exact functor; their higher derived functors are sheaf cohomology. The obstruction here concerns sheaves and gluing even when the ground field is a field. It is not contradicted by $\operatorname{Ext}^j_k(V, W) = 0$: these are different categories and different functors. Hartshorne III.1–3 develops that transition. His introduction marks externally cited commutative-algebra and homological results with an "A" instead of proving them all in place (printed pp. xv–xvi, PDF pp. 14–15). The collection provides many algebraic ingredients, but a reader still needs localization, local rings, sheaves and stalks, and substantial commutative-algebra fluency before this becomes a comfortable route.

8.5 Differential topology: linear algebra varies smoothly

For a smooth map $F : M \to N$, differentiation gives maps $dF_x : T_xM \to T_{F(x)}N$. The chain rule makes differentiation functorial. At a regular point of a level set $F^{-1}(y)$, the implicit-function theorem identifies

$$T_xF^{-1}(y) = \ker dF_x.$$

For a smooth embedded submanifold $M \subset N$, there is an exact sequence of vector bundles over M,

$$0 \to TM \to TN|_M \to \nu \to 0,$$

where ν is the quotient normal bundle. A Riemannian metric on N identifies it with the orthogonal normal bundle. On a paracompact smooth base, a smooth positive bundle metric exists by partitions of unity, so every short exact sequence of smooth vector bundles splits. The splitting need not be natural without the chosen metric or other extra data.

This is stronger than choosing complements separately at each point: arbitrary pointwise choices need not vary continuously. Conversely splitting does not make either summand trivial. A nontrivial line bundle can be a direct summand of a trivial higher-rank bundle. Smooth variation and a global frame are different requirements.

Hirsch proves the bundle splitting theorem as IV.2.2 (printed p. 95, PDF p. 105), using partitions of unity, and defines the orthogonal structure immediately afterward. This is actual proof coverage, not an analogy inferred from the chapter title. The necessary new bridges for his development are multivariable calculus, the inverse/implicit-function theorems, general topology, manifolds, partitions of unity, and then approximation and transversality.

The Hessian of a smooth real function at a critical point is a symmetric bilinear form on the tangent space. Its inertia survives coordinate change, which explains why Sylvester's law underlies the Morse index. The Morse lemma and Morse theory additionally require analytic and topological arguments; diagonalization alone does not prove their local normal form or global conclusions. Hirsch VI is a destination for those arguments.

Exterior algebra can build differential forms once smooth bundles exist, but Hirsch's preface expressly chooses a route without tensors or differential forms. It would be misleading to force homological algebra, algebraic geometry, or tensor calculus into his prerequisite chain. The common foundation is careful linear geometry; the subsequent methods are genuinely additional.

9 A concrete destination: Hadamard and Williamson foundations

9.1 Scope and the exact question

The confirmed existing direction is the paused Hadamard/Williamson project in this repository. Its general existence question asks whether, for every positive integer k, there is a matrix

$$H \in \{-1, 1\}^{4k \times 4k}, \qquad HH^{\mathsf{T}} = 4kI_{4k}.$$

The prior work at commit `2834fea31fb4ca7203531ca220e55a2fb7640119` records three rounds of bounded research in a restricted family. Its proofs and limitations are read here as the starting point. No prior benchmark is presented as a fresh run, and no historical claim about the smallest missing order or the worldwide current status is imported without a separate literature audit. This chapter does not restart the computational search.

The structural chain actually needed is

$$\text{Gram matrices} \longrightarrow \text{tensor constructions}, \qquad \text{finite fields} \longrightarrow \text{quadratic characters},$$

$$\text{cyclic group algebras} \longrightarrow \text{Fourier decomposition} \longrightarrow \text{correlation equations and lifting constraints}.$$

It makes direct use of Xi II's metric and tensor material and Xi III's groups, representations, and finite fields. General Ext theory, schemes, and differential topology have no established role in this chain.

9.2 Gram matrices and the order restriction

The identity $HH^{\mathsf{T}} = nI$ says precisely that the rows are mutually orthogonal in the usual positive real inner product and have squared length n. It also gives $H^{-1} = n^{-1}H^{\mathsf{T}}$ and $|\det H| = n^{n/2}$, attaining the determinant bound proved in the forms chapter. Orthogonality alone does not enforce sign entries; both conditions are essential.

Multiplying rows or columns by signs and permuting them preserves the Hadamard condition. Normalize the first row to all $+1$. A second row must have equally many signs, so n is even. When $n > 2$, permute its positive entries into the first half and let a third row have a positives in the first half and b in the second. Its products with the first two rows give

$$2(a+b) = n, \qquad 2(a-b) = 0.$$

Hence $n = 4a$. This proves the necessary orders $1, 2$, or a multiple of 4. It supplies no construction at each admissible order. The proof is over the integers or reals; reducing the equation modulo 2 would collapse all signs and lose the argument.

9.3 Tensor closure and what it leaves unsolved

If H and K have orders a, b, functoriality of the tensor product and transpose gives

$$(H \otimes K)(H \otimes K)^{\mathsf{T}} = (HH^{\mathsf{T}}) \otimes (KK^{\mathsf{T}}) = abI_{ab}.$$

Its entries remain signs. Tensoring with $S = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ gives $\begin{pmatrix} H & H \\ H & -H \end{pmatrix}$ and yields all powers-of-two orders from [1]. This is a genuine infinite construction theorem. It does not produce all multiples of four without additional seeds or constructions. The universal tensor identities explain why the construction transports orthogonality, while the sign-entry requirement explains why arbitrary orthogonal matrices are not sufficient seeds.

9.4 Finite fields and a character identity proved by counting

Let $\mathbb{F}_q$ be a finite field of odd order. Its nonzero squares form an index-two subgroup: the squaring map has exactly two preimages $x, -x$ for each nonzero square. Define the quadratic character $\chi : \mathbb{F}_q \to \{0, \pm 1\}$ with $\chi(0) = 0$, $+1$ on nonzero squares, and -1 on nonsquares. The quotient-group description proves multiplicativity, and equal coset sizes prove $\sum_x \chi(x) = 0$.

The multiplicative group of a finite field is cyclic. One short proof lets e be the least common multiple of its element orders. In a finite abelian group, combine elements of maximal prime-power orders to obtain an element of order e. Every nonzero field element is a root of $X^e - 1$, so there are at most e of them, whereas Lagrange gives $e \le q - 1$. Thus $e = q - 1$. It follows that $\chi(-1) = (-1)^{(q-1)/2}$.

For $a \neq 0$, the crucial correlation identity is

$$\sum_x \chi(x)\chi(x+a) = -1.$$

Count pairs (x, y) with $y^2 = x(x+a)$. For fixed x, their number is $1 + \chi(x(x+a))$, so the total is q plus the displayed sum. Scale $x = az$, $y = aw$ and rewrite as

$$(2z + 1 - 2w)(2z + 1 + 2w) = 1.$$

Since 2 is invertible, each nonzero first factor determines a unique second factor and a unique (z, w). There are $q - 1$ pairs, proving the sum. For $a = 0$ the corresponding sum is $q - 1$. This argument uses a finite field of odd order, not the special fact that its elements are residue classes modulo a prime.

9.5 Paley constructions with explicit signs

Index Q by $x, y \in \mathbb{F}_q$ and put $Q_{xy} = \chi(y - x)$. The character identities give

$$Q\mathbf{1} = 0, \qquad QQ^{\mathsf{T}} = qI - J, \qquad Q^{\mathsf{T}} = \chi(-1)Q.$$

If $q \equiv 3 \pmod 4$, set

$$H = \begin{pmatrix} 1 & \mathbf{1}^{\mathsf{T}} \\ \mathbf{1} & Q - I \end{pmatrix}.$$

Its entries are signs, its top row is orthogonal to every lower row because $1 + (Q - I)\mathbf{1} = 0$, and its lower Gram block is

$$J + (Q - I)(Q^{\mathsf{T}} - I) = J + qI - J - Q - Q^{\mathsf{T}} + I = (q + 1)I.$$

Therefore $HH^{\mathsf{T}} = (q + 1)I$. The prior study used the equivalent sign convention with lower first column $-\mathbf{1}$ and core $Q + I$. The conventions have been derived separately here so a copied sign does not silently spoil orthogonality.

For $q \equiv 1 \pmod 4$, define the symmetric conference matrix

$$C = \begin{pmatrix} 0 & \mathbf{1}^{\mathsf{T}} \\ \mathbf{1} & Q \end{pmatrix}.$$

It has zero diagonal, off-diagonal signs, and $C^2 = qI$. The block matrix

$$\mathcal{H} = \begin{pmatrix} C + I & C - I \\ C - I & -C - I \end{pmatrix}$$

has only sign entries. The diagonal Gram block is $(C+I)^2+(C-I)^2 = 2(q+1)I$ and the off-diagonal block vanishes by commutation. It is Hadamard of order $2(q + 1)$.

These proofs add a short prime-power and conference-matrix development to the prior study's prime, $q \equiv 3 \pmod 4$ proof. They are classical constructions, not new results or claims that the earlier scripts implemented them. Finite fields of every prime-power order exist (Xi III Theorems 5.40–5.45, printed pp. 260–261, PDF pp. 266–267); these pages and their proofs were inspected visually. Small exact instances are checked separately in this deliverable, without resuming an enumeration campaign.

9.6 Circulants as an integral group algebra

Fix $m \geq 1$ and let P be the cyclic shift with $P_{ij} = 1$ when $j = i + 1 \pmod m$. The map

$$\mathbb{Z}[z]/(z^m - 1) \longrightarrow M_m(\mathbb{Z}), \qquad f \longmapsto f(P)$$

is injective: the first row of $\sum_{j=0}^{m-1} a_j P^j$ is its coefficient vector. Its image is the circulant algebra, so circulants commute. Define $f^\iota(z) = f(z^{-1})$. Transpose corresponds to this involution. For a real sequence x with polynomial $X(z) = \sum_j x_j z^j$, the coefficient of z^s in XX^ι is

$$\mathrm{PAF}_x(s) = \sum_j x_j x_{j+s}.$$

This follows by collecting terms with exponent difference s; autocorrelation is even in s. It is convolution with the reversed sequence, so the group algebra retains the integral coefficients needed for exact verification.

If $x_j = x_{-j}$, then $X = X^\iota$ and the circulant is symmetric. Four such sign sequences form a Williamson quadruple when

$$AA^{\mathsf{T}} + BB^{\mathsf{T}} + CC^{\mathsf{T}} + DD^{\mathsf{T}} = 4mI_m,$$

equivalently their four autocorrelations sum to zero at every nonzero shift. Symmetry permits replacing the four Gram matrices by A^2, B^2, C^2, D^2. Commutation and symmetry then make

$$W = \begin{pmatrix} A & B & C & D \\ -B & A & -D & C \\ -C & D & A & -B \\ -D & -C & B & A \end{pmatrix}$$

Hadamard of order $4m$. Multiplying blocks, every diagonal Gram block is the sum of the four squares, while an off-diagonal block has terms such as $-AB + BA - CD + DC = 0$. Transpose and commutation hypotheses are indispensable in that cancellation. Existence in the Williamson family is sufficient for a Hadamard matrix of that order; it is not a necessary characterization of all Hadamard matrices.

9.7 Fourier characters are Chinese remainder coordinates

Let k have characteristic not dividing m and contain a primitive mth root ζ. The roots of $z^m - 1$ are distinct, so

$$k[z]/(z^m - 1) \xrightarrow{\sim} k^m, \qquad f \longmapsto (f(1), f(\zeta), \dots, f(\zeta^{m-1}))$$

is a Chinese remainder isomorphism. Its inverse is the finite Fourier formula

$$a_j = \frac{1}{m} \sum_{r=0}^{m-1} f(\zeta^r) \zeta^{-rj}.$$

To verify it, the geometric sum $\sum_r \zeta^{r\ell}$ equals m when $m \mid \ell$ and 0 otherwise. In the regular representation, the vector $(1, \zeta^r, \dots, \zeta^{(m-1)r})$ is an eigenvector of P with eigenvalue ζ^r. Thus all circulants diagonalize in the same basis; this is the group representation decomposition into characters.

Over $\mathbb{C}$, divide these vectors by $\sqrt{m}$ to get a unitary basis. A real symmetric circulant then has real eigenvalues, since $X(\zeta^{-r}) = X(\zeta^r) = \overline{X(\zeta^r)}$. The correlation identity becomes

$$\sum_{X\in\{A,B,C,D\}} |X(\zeta^r)|^2 = 4m \qquad (0 \le r < m).$$

The frequency-zero equation is the row-sum identity $r_a^2 + r_b^2 + r_c^2 + r_d^2 = 4m$. A single row or partial tuple must satisfy the corresponding upper bounds, but separate feasible spectral values need not come from any common sign coefficients.

The field assumptions explain failures: over $\mathbb{Q}$ the algebra need not split into one-dimensional factors, although it splits after adjoining ζ; if the characteristic divides m, $z^m - 1$ has repeated roots and this semisimple Fourier argument breaks. Positivity and absolute-square inequalities require the real/complex setting; they have no automatic meaning over a finite field.

9.8 A quotient test is not a lift theorem

For $d \mid m$, the group homomorphism $C_m \to C_d$ gives an algebra quotient sending z to the generator of order d. The compressed coefficient is $y_r = \sum_{j\equiv r \ (d)} x_j$. Applying the quotient to the four correlation identity gives

$$\sum_x C_{y_x} C_{y_x}^{\mathsf{T}} = 4mI_d.$$

The right side remains $4m$, not $4d$, and compressed coefficients are integer sums rather than signs. This is a necessary consequence of a full solution. The quotient discards coefficients, so its converse needs a separate common-lift theorem.

The prior proof gives an exact illustrative failure at $m = 9$. The normalized symmetric masks $(2, 10, 10, 13)$, where bit $j-1$ negates positions $\pm j$, compress at $d = 3$ to

$$(3,1,1), \quad (3,-1,-1), \quad (3,-1,-1), \quad (-1,-1,-1).$$

Their total compressed autocorrelation is $(36, 0, 0)$ and their row-square sum is 36, so both proper cyclic quotients of C_9 pass. But their uncompressed autocorrelation at shift 1 sums to -8. This shows that these particular lifts fail; it alone would not prove that every other lift of the compressed multiset fails. The prior independent review separately records a stronger no-common-lift fixture: the compressed multiset

$$(-1,-1,-1), \quad (-1,1,1), \quad (-1,3,3), \quad (3,-1,-1).$$

It too has total compressed autocorrelation $(36, 0, 0)$. There are 16 normalized symmetric sign rows at core nine; among their four-row multisets, exactly 27 map to this specified compressed multiset. The new exact checker independently reconstructs and rejects all 27 by their full autocorrelations. Thus no common lift exists **in this normalized symmetric Williamson family**. That is stronger evidence than one failed lift, and still says nothing about nonexistence of arbitrary Hadamard matrices of order 36.

Even positive spectral residuals need not have a sign lift. At $m = 7$, let three sequences have negative positions $\{\pm1\}$, $\{\pm2\}$, and $\{\pm3\}$. Their squared circulants sum to $20I + J$, leaving $8I - J$ toward the target $28I$. This residual is positive definite: its eigenvalues are 1 on constants and 8 on their

orthogonal complement. Nevertheless the odd Williamson product constraint below forces the fourth sequence to be all positive; its row-sum square is 49, while the remaining row-square budget is 1. No fourth Williamson row exists. A positive real Gram factor does not necessarily satisfy the required discrete sign pattern.

9.9 Why four colors appear in the prior reduction

Now let $m = 2h + 1$ be odd and normalize every sequence by $x_0 = 1$. Write $T(z) = \sum_{j=0}^{m-1} z^j$ and $N_x(z) = \sum_{x_j=-1} z^j$, so $X = T - 2N_x$. The identities $T^2 = mT$ and $TN_x = |N_x|T$, substituted into the Williamson equation and divided coefficientwise by 4 in the integral group algebra, give

$$\sum_x N_x^2 + (m - R)T = m, \qquad R = \sum_x |N_x|.$$

Each $|N_x|$ is even by symmetry and normalization. Modulo 2, squaring removes cross terms and sends z^j to z^{2j}. Since doubling permutes residues modulo odd m, each nonzero position contains an odd number of negative entries among the four rows. Equivalently

$$a_j b_j c_j d_j = -1 \qquad (j \neq 0).$$

This is the odd Williamson product theorem in the prior proof. Its hypotheses are structural restrictions; it is not a constraint on arbitrary Hadamard matrices.

Let $v_c = \mathbf{1} - 2e_c \in \mathbb{Z}^4$, $c = 1, 2, 3, 4$. The eight four-entry sign columns with product -1 are uniquely εv_c with $\varepsilon = \pm 1$, and

$$v_c^{\mathsf{T}} v_d = 4[c = d], \qquad \mathbf{1}^{\mathsf{T}} v_c = 2.$$

Thus each nonzero column has a color κ_j and sign ε_j, reflected at opposite indices. Write $[j] = \{j, -j\}$ for a nonzero opposite-index class, represented in $\{1, \dots, h\}$. For a nonzero shift t, define E_t as follows, with indices modulo m. Partition $j \in \mathbb{Z}/m\mathbb{Z}$ into the orbits of the involution $j \mapsto -t - j$. Remove the two-element orbit $\{0, -t\}$ and the unique fixed point solving $2j = -t$. For each remaining two-element orbit, choose either member j and include one unordered edge $\{[j], [j + t]\}$ in E_t, retaining one edge per orbit. Replacing j by $-t - j$ swaps the endpoints, so the definition is independent of the representative. Both endpoints are nonzero and distinct; equality of the classes would force either $t = 0$ or the removed fixed point. There are $h - 1$ edges, counted with their orbit multiplicities.

In the correlation sum, the removed pair involving zero contributes $4\varepsilon_t$; the fixed term contributes 4. Each remaining pair with endpoint classes u, v contributes $8\varepsilon_u \varepsilon_v [\kappa_u = \kappa_v]$. Hence exact correlation is equivalent to

$$\varepsilon_t + 1 + 2 \sum_{\{u,v\} \in E_t} \varepsilon_u \varepsilon_v [\kappa_u = \kappa_v] = 0.$$

If c_t counts its monochromatic edges, reducing this equation modulo 4 gives $\varepsilon_t = -(-1)^{c_t}$. All signs are therefore forced by a complete coloring. The full integer equations still have to hold; the parity consequence alone is insufficient.

Permuting the four block rows merely permutes color names. Naming colors in order of first appearance, with at most four colors, gives a unique restricted-growth representative of each orbit. This is a quotient by block permutations, including repeated blocks. It is not classification up to all Hadamard matrix equivalences, nor a quotient by arbitrary coordinate changes.

9.10 Boolean polynomials and the precise relaxation boundary

Encode each color minus one by two bits a_i, b_i. In the Boolean algebra

$$B = \mathbb{F}_2[a_i, b_i \ (1 \le i \le h)]/(a_i^2 - a_i, b_i^2 - b_i),$$

every function on the bit cube has a unique squarefree polynomial. Induct on the last bit: any function is $f_0 + x(f_1 - f_0)$, which proves existence and uniqueness recursively. Equality of colors has polynomial

$$\delta_{uv} = (1 + a_u + a_v)(1 + b_u + b_v).$$

It is 1 exactly when both bits agree. Therefore

$$n_t = 1 + \sum_{\{u,v\} \in E_t} \delta_{uv}, \qquad \varepsilon_t = (-1)^{n_t}$$

has degree at most two in the bits. Substituting a partial assignment preserves this degree bound. Write a residual $n_i = \alpha_i + p_i$ with constant α_i and zero-constant polynomial p_i, and put $\eta_i = (-1)^{\alpha_i}$, $\xi_p = (-1)^p$. Then

$$\varepsilon_i = \eta_i \xi_{p_i}, \qquad \varepsilon_u \varepsilon_v = \eta_u \eta_v \xi_{p_u + p_v}.$$

These are exact identities of functions. Equal residual polynomials therefore represent the same unknown phase even before its sign is known. Summing their integer coefficients first can reveal cancellation that separate sign ranges miss. Constants and linear terms must be included when deciding equality; equal quadratic parts alone do not suffice.

The exponents are computed in B, of characteristic two. The correlation coefficients and their sums are computed in $\mathbb{Z}$. The map $p \mapsto (-1)^p$ converts addition of Boolean functions into multiplication of signs; it is not a ring map allowing integer coefficients to be reduced modulo two.

After grouping, a lag expression can have the exact form $K + \sum_p c_p \xi_p$. Treating distinct ξ_p as independent signs gives a safe superset of its attainable values, hence a sound rejection test when zero is absent. Presence of zero is only a necessary condition. Relations such as $\xi_{p+q} = \xi_p \xi_q$ remain; even preserving all such linear phase relations need not enforce the quadratic origin. For example, the values of a, b, ab cannot be $(0, 0, 1)$, though these three polynomials are linearly independent over $\mathbb{F}_2$. A prospective stronger method must specify which extra dependencies it keeps and prove its rejection rule.

The prior work's core-11 false-positive prefix is a concrete warning that separately feasible row and lag relaxations can lack a common completion. Its existing audit, not a new run here, supports that bounded observation. Neither the algebraic reduction nor the word "exact" makes every downstream relaxation complete.

9.11 What to learn next for this direction

Tool needed for the study	Source in the collection	Supplement supplied or still needed
Orthogonality, Gram matrices, positive residuals	Xi II §§3.1–3.5, 3.8	Gram-factor versus sign-factor distinction; examples above
Kronecker products and tensor identities	Xi II §6.2; Hungerford IV.5; Grillet XI.5–6	Sylvester closure proof above
Cyclic groups, quotient rings, polynomial actions	Xi III §§1.2, 4.2–4.4; Hungerford III–IV; Grillet III, VIII	Integral cyclic-algebra involution and correlation dictionary above
Finite fields and multiplicative characters	Xi III §5.6; Hungerford V.5; Grillet IV.1.6 and V.1.3	Quadratic-character sum and both Paley block proofs above
Character decomposition and invariant splitting	Xi III §§3.1–3.5; Grillet IX.2, IX.7–9	Fourier inversion as CRT, with splitting/characteristic hypotheses above
Compression and common lifting	Quotients and maps throughout the core	A real additional theorem would have to couple sign lifts; none is assumed
Quadratic Boolean constraints	Polynomial rings, quotients, functions on finite fields	Boolean normal form and phase dependence above; algorithmic feasibility remains a separate subject

The strongest useful next reading task is to reconstruct each displayed dictionary and counterexample, then read the pinned prior proofs with the coefficient rings visible. A later research proposal should target a specific lost dependency, a genuinely wider construction family, or a proved lift criterion. Such proposals are prospective; no improvement in search complexity or progress toward universal existence is claimed by this foundational chapter.

10 Invariant complements as a computable extension obstruction

Status and scope. The extension classification, Sylvester criterion, trace duality, Bézout argument, and Smith-form consequences below are classical mathematics. The organization, connecting derivations, and worked examples are original exposition, with no claim of research novelty. This chapter builds on the operator-module translation in Chapter 2 and the cyclic extension construction in Chapter 6. It supplies the missing machinery for an actual invariant-complement problem: compute the obstruction, exhibit a certificate when it is nonzero, construct a complement when it vanishes, and determine how a nonzero obstruction joins cyclic blocks. Every main claim is proved below; no new textbook proof-comparison claim is made here.

Throughout, k is a field and $R = k[t]$. Let U, W be finite-dimensional k-spaces, with operators $A : U \to U$ and $B : W \to W$. Write U_A, W_B for the corresponding R-modules. Several arguments work without finite dimension; those extensions are identified explicitly. Matrices act on column vectors, and $J_n(0)$ has ones on the **superdiagonal**.

10.1 The datum an invariant-subspace problem actually contains

Suppose an operator T on E preserves a specified subspace U. Its restriction is A; its induced operator on $W = E/U$ is B. This gives an exact sequence with specified endpoint identifications:

$$0 \longrightarrow U_A \xrightarrow{i} E_T \xrightarrow{q} W_B \longrightarrow 0. \tag{1}$$

A k-linear section $s : W \to E$ exists because k is a field. The map $(u, w) \mapsto i(u) + s(w)$ identifies the underlying vector space with $U \oplus W$. The defect of s from being an R-linear map is

$$Ts(w) - s(Bw) = i(Cw),$$

for a unique k-linear map $C : W \to U$. Thus

$$T_C = \begin{pmatrix} A & C \\ 0 & B \end{pmatrix}. \tag{2}$$

Every C gives an extension, but C depends on the chosen section. Put

$$H = \mathrm{Hom}_k(W, U), \qquad \delta_{A,B} : H \longrightarrow H, \quad \delta(h) = Ah - hB. \tag{3}$$

Replace s by $s + ih$. Direct substitution gives

$$C \longmapsto C + Ah - hB = C + \delta(h). \tag{4}$$

Equivalently, with $P_h = \begin{pmatrix} 1 & h \\ 0 & 1 \end{pmatrix}$,

$$P_h^{-1} T_C P_h = T_{C+\delta(h)}. \tag{5}$$

This conjugacy fixes the embedded U and induces the identity on the quotient W. It is much more restricted than arbitrary similarity.

For clarity about signs, an extension isomorphism written in the *direction* $F_h : E_C \to E_D$, $F_h(u, w) = (u + hw, w)$, satisfies

$$F_h T_C = T_D F_h \quad \Longleftrightarrow \quad D = C - \delta(h). \tag{6}$$

There is no discrepancy between (4) and (6): P_h maps the new coordinate model to the old one. Both say that equivalent extensions have the same class in $H/\delta H$.

10.1.1 Complements, sections, and the solution set

Any vector-space complement to $U \oplus 0$ is the graph

$$\Gamma_h = \{(hw, w) : w \in W\}$$

of a unique $h \in H$: the quotient projection restricts to an isomorphism from the complement to W. Applying T_C to a graph vector shows

$$T_C(hw, w) = (Ahw + Cw, Bw) \in \Gamma_h \quad \Longleftrightarrow \quad Ah - hB = -C. \tag{7}$$

Consequently the following are equivalent:

1. U has a T_C-invariant complement;
2. (1) admits an R-linear section;
3. the Sylvester equation $\delta(h) = -C$ has a solution;
4. $[C] = 0$ in $H/\delta H$.

If h_0 is one solution, every solution is

$$h_0 + f, \qquad f \in \ker \delta = \mathrm{Hom}_R(W_B, U_A). \tag{8}$$

The set of invariant complements is therefore either empty or an affine space modeled on the intertwiners $W_B \to U_A$. It has no distinguished origin until a complement is chosen. An invariant complement is unique exactly when a solution exists and this intertwiner space is zero.

10.2 Why the cokernel is precisely the first Ext group

The preceding calculation is already a classification by endpoint-preserving extension equivalence: any extension admits a linear section; any equivalence fixing both endpoints must have the form F_h; (6) describes exactly when it is R-linear. Two further derivations explain the group law and connect this classification to a free resolution.

10.2.1 A canonical one-step free resolution

Let $F = R \otimes_k W$. Define

$$\begin{aligned} d_B &: F \longrightarrow F, \qquad d_B(t^j \otimes w) = t^{j+1} \otimes w - t^j \otimes Bw, \\ \varepsilon &: F \longrightarrow W_B, \qquad \varepsilon(t^j \otimes w) = B^j w. \end{aligned}$$

Then

$$0 \longrightarrow F \xrightarrow{d_B} F \xrightarrow{\varepsilon} W_B \longrightarrow 0 \tag{9}$$

is exact. Here is a proof without a determinant calculation. If a nonzero vector polynomial $z = \sum_{j=0}^{m} t^j \otimes w_j$ has $w_m \neq 0$, then the coefficient of t^{m+1} in $d_B z$ is w_m. Hence d_B is injective. Surjectivity of ε follows from $\varepsilon(1 \otimes w) = w$. Finally,

$$t^j \otimes w - 1 \otimes B^j w = d_B \left(\sum_{\ell=0}^{j-1} t^{j-1-\ell} \otimes B^\ell w \right) \qquad (j \geq 1). \tag{10}$$

Thus every vector polynomial is congruent modulo $\mathrm{im}\, d_B$ to its constant evaluation; exactly the elements with evaluation zero lie in the image. Choosing a basis of W shows that F is free over R. In matrix notation $d_B = tI - B$.

An R-map $F \to U_A$ is uniquely specified by a k-map $h : W \to U$, its value on $1 \otimes W$. Precomposition with d_B sends h to

$$w \longmapsto Ah(w) - h(Bw) = \delta(h)(w).$$

Applying $\mathrm{Hom}_R(-, U_A)$ to the resolution gives

$$0 \longrightarrow \mathrm{Hom}_R(W_B, U_A) \longrightarrow H \xrightarrow{\delta} H \longrightarrow \mathrm{Ext}^1_R(W_B, U_A) \longrightarrow 0, \tag{11}$$

and $\mathrm{Ext}^j_R(W_B, U_A) = 0$ for $j \geq 2$. In particular,

$$\boxed{\mathrm{Ext}^1_{k[t]}(W_B, U_A) \cong \mathrm{coker}(h \mapsto Ah - hB).} \tag{12}$$

The sign in this identification can be checked on the extension itself. Extend its chosen section s to the R-map $\tilde{s} : F \to E_T$, $t^j \otimes w \mapsto T^j s(w)$. Then $\tilde{s}\, d_B$ takes values in $i(U)$ and its value on $1 \otimes w$ is $i(Cw)$. Therefore the resolution cocycle is C, not an unspecified sign of C.

This resolution and the cokernel computation remain valid for arbitrary k-vector spaces, assuming the usual existence of bases. Finite dimension will matter for the trace pairing and the dimension counts below.

10.2.2 Baer addition becomes addition of upper-right blocks

Take E_C and E_D. Their fiber product over W consists of triples (u_1, u_2, w), and its operator is

$$(u_1, u_2, w) \longmapsto (Au_1 + Cw, Au_2 + Dw, Bw).$$

The Baer sum pushes out the kernel $U \oplus U$ along addition $(u_1, u_2) \mapsto u_1 + u_2$. Concretely quotient the fiber product by $\{(u, -u, 0) : u \in U\}$. The resulting coordinate $(u_1 + u_2, w)$ has operator T_{C+D}. Hence

$$[E_C] + [E_D] = [E_{C+D}]. \tag{13}$$

Pushout along $\lambda 1_U$ similarly gives $[E_{\lambda C}]$. Thus (12) respects the k-vector-space structure as well as the set of extension classes. The zero class is the split sequence.

10.3 A complete certificate for failure: trace-dual intertwiners

A failed attempt to solve a matrix equation does not by itself explain why it fails. Here the dual equation supplies a finite certificate.

Put $K = \mathrm{Hom}_k(U, W)$. There is a perfect bilinear pairing

$$K \times H \longrightarrow k, \qquad \langle g, C \rangle = \mathrm{tr}_W(gC) = \mathrm{tr}_U(Cg). \tag{14}$$

The two traces agree because both are $\sum_{i,j} g_{ij}C_{ji}$ in bases. The pairing is nondegenerate: a matrix unit in one rectangular matrix space reads a single entry of the other. This proof works in every characteristic. In particular it does **not** require $\operatorname{char} k \nmid \dim U$ or $\operatorname{char} k \nmid \dim W$.

For $h \in H$,

$$\begin{aligned} \langle g, \delta(h)\rangle &= \operatorname{tr}_W(gAh - ghB) \\ &= \operatorname{tr}_W((gA - Bg)h). \end{aligned} \tag{15}$$

Thus the dual linear map is

$$\delta^\vee : K \to K, \qquad g \longmapsto gA - Bg. \tag{16}$$

Its kernel consists of the intertwiners in the **reverse direction**:

$$\ker \delta^\vee = \operatorname{Hom}_R(U_A, W_B). \tag{17}$$

Equation (15) proves that every $\delta(h)$ is annihilated by those intertwiners. Conversely, a vector C outside $\operatorname{im}\delta$ can be separated from that subspace by a linear functional: extend a basis of the image by C and prescribe the functional to be zero on the image and one on C. The perfect pairing represents that functional by some g. Equation (15) then forces $gA = Bg$. Hence

$$\boxed{C \in \operatorname{im}\delta \iff \operatorname{tr}_W(gC) = 0 \text{ for every } g : U \to W \text{ with } gA = Bg.} \tag{18}$$

In particular, a single g satisfying

$$gA = Bg, \qquad \operatorname{tr}_W(gC) \neq 0 \tag{19}$$

certifies that U has no invariant complement in E_C. A basis of the reverse-intertwiner space suffices to test all witnesses. A complete vanishing test, followed by solving $Ah - hB = -C$, gives a constructive complement.

The quotient pairing is perfect and identifies

$$\operatorname{Ext}^1_R(W_B, U_A)^* \cong \operatorname{Hom}_R(U_A, W_B), \tag{20}$$

or equivalently, in finite dimension, $\operatorname{Ext}^1_R(W_B, U_A) \cong \operatorname{Hom}_R(U_A, W_B)^*$. This identification uses the specified polynomial variable t through (9). It is independent of bases and of the section. Its endpoint orientation is not optional: maps $W \to U$ parametrize changes of a splitting, while maps $U \to W$ test the obstruction.

For completeness, the expected compatibility with endpoint maps follows from traces. Pushing an extension out along $a : U \to U'$ replaces C by aC and pairs with $g' : U' \to W$ as $\operatorname{tr}(g'aC)$. Pulling back along $b : W' \to W$ replaces C by Cb and $\operatorname{tr}_{W'}(gCb) = \operatorname{tr}_W(bgC)$. These identities explain the natural dual pairing; they do not identify a vector space with its own dual without additional data.

10.4 Coprime annihilators give an explicit inverse

Suppose $a, b \in k[t]$ satisfy

$$a(A) = 0, \qquad b(B) = 0, \qquad u(t)a(t) + v(t)b(t) = 1. \tag{21}$$

One often stops after using Bézout to prove that a complement exists. A divided-difference identity constructs the actual solution of every Sylvester equation.

For $b(t) = \sum_{j=0}^{d} b_j t^j$, define the polynomial

$$Q_b(x, y) = \sum_{j=1}^{d} b_j \sum_{\ell=0}^{j-1} x^{j-1-\ell} y^{\ell}. \tag{22}$$

It satisfies

$$(x - y)Q_b(x, y) = b(x) - b(y). \tag{23}$$

The notation $(b(x)-b(y))/(x-y)$ abbreviates this polynomial identity; no inversion of $x-y$ is being assumed. On H, let $L(h) = Ah$ and $R_B(h) = hB$. These operators commute, even though there is no meaningful product AB when $U \neq W$. Therefore (23) gives

$$(L - R_B)Q_b(L, R_B) = L_{b(A)} - R_{b(B)} = L_{b(A)}. \tag{24}$$

Evaluating (21) at A yields $v(A)b(A) = 1_U$. Every operator involved in (24) commutes with left multiplication by $v(A)$. It follows that

$$\boxed{\delta^{-1}(D) = v(A) \sum_{j=1}^{d} b_j \sum_{\ell=0}^{j-1} A^{j-1-\ell} D B^{\ell}.} \tag{25}$$

Multiplying on either side by δ gives the identity; this is a two-sided inverse, not merely a recipe for selected right-hand sides. The unique invariant complement is the graph of $h = -\delta^{-1}(C)$.

For example, if $A^2 = 0$ and $B = 1_W$, use $a = t^2$, $b = t - 1$, and $t^2 - (t + 1)(t - 1) = 1$. Equation (25) becomes

$$\delta^{-1}(D) = -(1_U + A)D, \qquad h = (1_U + A)C.$$

Directly, $A(1_U + A)C - (1_U + A)C = -C$. No Jordan basis is needed, and the formula also works in characteristic two.

A less trivial exact check is $A = N$, $B = 1 + M$, where $N^2 = M^2 = 0$. Take $a = t^2$, $b = (t - 1)^2$ and

$$(3 - 2t)t^2 + (1 + 2t)(t - 1)^2 = 1.$$

Then (25) reduces to

$$\delta^{-1}(D) = -D - ND + DM + 2NDM. \tag{26}$$

The integers in this identity are interpreted in k; no division by two or three occurs.

10.4.1 A finite geometric inverse for distinct Jordan eigenvalues

Suppose $A = \lambda 1_U + N$ and $B = \mu 1_W + M$, with $N^a = 0$, $M^b = 0$, and $d = \lambda - \mu \neq 0$. On H, put $K(D) = ND - DM$. Its two summands commute, so the binomial expansion shows $K^{a+b-1} = 0$: each term has either at least a left factors N or at least b right factors M. Since $\delta = d1_H + K$,

$$\delta^{-1} = \sum_{r=0}^{a+b-2} (-1)^r d^{-r-1} K^r,$$

where

$$K^r(D) = \sum_{\ell=0}^{r} (-1)^\ell \binom{r}{\ell} N^{r-\ell} D M^\ell.$$

Multiplication by $d1_H + K$ telescopes to the identity, with the last term zero by nilpotence. This is a finite algebraic identity, with no convergence assumption. It works in every characteristic when $d \neq 0$; cancellation of binomial coefficients in small characteristic can shorten the nilpotence bound without invalidating it. Formula (26) is its case $d = -1$, $a = b = 2$.

10.4.2 A coordinate-free description of the same complement

Let E_T be any extension in (1) under (21). Since $b(B) = 0$, $b(T)E \subseteq U$; since $a(A) = 0$, it follows that $a(T)b(T) = 0$. The candidate complement is

$$K = \ker b(T). \tag{27}$$

Indeed $b(A)$ is invertible with inverse $v(A)$, so $K \cap U = 0$. For any lift e of $w \in W$, write $b(T)e = i(u)$ and replace e by $e - i(v(A)u)$. This new lift lies in K. Thus $q|_K : K \to W$ is an isomorphism and $E = U \oplus K$.

Every invariant complement is isomorphic to W_B, hence is killed by $b(T)$ and lies in K; it must equal K. This proves uniqueness without finite-dimensional dimension counting. Both this proof and (25) work for arbitrary k-spaces with the stated coprime annihilators. Over a commutative base ring the algebraic identities also hold, but arbitrary extensions need not admit a section over that base ring; the block classification then needs that additional hypothesis.

10.5 What a shared spectral factor does and does not imply

In finite dimension, take $a = m_A$ and $b = m_B$. The following are equivalent:

- m_A and m_B are coprime;
- $\delta_{A,B}$ is invertible;
- every extension of W_B by U_A splits, with a unique invariant complement;
- the spectra of A and B are disjoint over an algebraic closure of k.

Coprimality implies invertibility by (25). Invertibility is equivalent to $\ker\delta = \operatorname{coker}\delta = 0$ by finite-dimensional linear algebra, and (8), (12) give the extension assertion. For the converse to coprimality, suppose an irreducible p divides both minimal polynomials. The primary cyclic decompositions contain blocks $R/(p^a)$ and $R/(p^b)$ for some $a, b \geq 1$. A nonzero map from the latter to the former sends 1 to $p^{\max(a-b,0)}$ modulo p^a. Extending it by zero on the other blocks gives a nonzero $W_B \to U_A$ intertwiner, so δ is not injective. Finally, two polynomials over a field share an irreducible factor exactly when they have a common root in an algebraic closure, proving the spectral equivalence.

The conclusion concerns **all possible** C. If a factor is shared, some right-hand sides are obstructed, but a particular C may still lie in $\operatorname{im}\delta$. In particular $C = 0$ always splits, and nonzero coboundaries split as well. A common eigenvalue is not a certificate for a particular nonsplitting extension; (19) is such a certificate.

Checking only eigenvalues *in the ground field* is insufficient. Over $\mathbb{R}$ let

$$A = B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad C = 1_2.$$

Neither operator has a real eigenvalue, so their real-eigenvalue sets are both empty. Nonetheless $g = 1_2$ is a reverse intertwiner and $\operatorname{tr}(gC) = 2 \neq 0$: this extension does not split. The shared factor is $t^2 + 1$.

Nor does separate semisimplicity of the endpoints suffice. For $A = B = (\lambda)$, $\delta = 0$; every $C \neq 0$ gives a nonsplit Jordan block of size two. Semisimplicity of the middle module would force every submodule to split, but semisimplicity is not preserved by arbitrary extensions.

10.6 Cyclic extensions expose exactly how blocks are joined

Let $f, g \in R$ be nonzero monic polynomials, with $U = R/(f)$ and $W = R/(g)$. The resolution $0 \to R \xrightarrow{g} R \to R/(g) \to 0$ gives

$$\operatorname{Ext}^1_R(R/(g), R/(f)) \cong (R/(f))/g(R/(f)) \cong R/(f,g) = R/(\gcd(f,g)). \tag{28}$$

The following presentation retains the endpoint maps, which (28) alone could hide. Let u be the specified generator of U, and choose a lift e of the specified generator of W. There is a polynomial c such that

$$fu = 0, \qquad ge = cu. \tag{29}$$

Conversely these relations define

$$E_c = \operatorname{coker}\begin{pmatrix} f & -c \\ 0 & g \end{pmatrix}. \tag{30}$$

Columns are relation vectors in the ordered generators (u, e). The map $R/(f) \to E_c$, $1 \mapsto u$, is injective: if $(r, 0)$ is a relation combination $x(f, 0) + y(-c, g)$, then $yg = 0$; since R is a domain and $g \neq 0$, $y = 0$, so $f \mid r$. The quotient by this copy of U is $R/(g)$ generated by the image of e.

Replacing e by $e + hu$ changes c to $c + gh$ modulo f. Therefore the extension class is precisely $c \bmod (f, g)$, with the specified cyclic generators understood. The sequence splits exactly when that residue is zero. This is a classifying statement with fixed endpoints; it is not merely an assertion about the isomorphism type of E_c.

10.6.1 Middle modules: a smaller invariant

The first Smith factor of the matrix in (30) is the monic polynomial

$$d_1 = \gcd(f, g, c),$$

because its entries generate that ideal. Its determinant is fg, so the second factor is $d_2 = fg/d_1$. Since d_1 divides both f and g, $d_1^2 \mid fg$, and hence $d_1 \mid d_2$, as required. Thus

$$E_c \cong R/(d_1) \oplus R/(d_2). \tag{31}$$

A unit factor contributes the zero module. This is an application of the Smith theorem already proved in Chapter 2. Row and column operations classify the middle module; they need not preserve its specified submodule, quotient, or their generators.

Consequently the residue $c \bmod \gcd(f, g)$ classifies the extension, while $\gcd(f, g, c)$ determines the middle-module isomorphism type in this cyclic situation. Different residues can have the same gcd and hence the same middle module.

10.6.2 Primary powers and an explicit Smith reduction

Fix a monic irreducible p and positive integers a, b. Take $f = p^a$, $g = p^b$. Define $v_p(0) = \infty$ and

$$m = \min(a, b), \qquad \nu = \min(a, b, v_p(c)). \tag{32}$$

Changing c by a multiple of p^m leaves this truncated valuation unchanged. Equations (28) and (31) become

$$\operatorname{Ext}^1_R(R/(p^b), R/(p^a)) \cong R/(p^m), \tag{33}$$

$$\boxed{E_c \cong R/(p^\nu) \oplus R/(p^{a+b-\nu}).} \tag{34}$$

The following row and column operations exhibit the mechanism in (34). When $\nu < m$, write $c = p^\nu q$, $P = p^{a-\nu}$, $Q = p^{b-\nu}$. Then $(P, q) = 1$, so choose $s, r \in R$ with $sP + rq = 1$. Both matrices below have determinant one, and

$$\begin{pmatrix} 1 & 0 \\ rQ & 1 \end{pmatrix} \begin{pmatrix} P & -q \\ 0 & Q \end{pmatrix} \begin{pmatrix} s & q \\ -r & P \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & PQ \end{pmatrix}. \tag{35}$$

Multiplying the middle matrix by p^ν produces exactly the Smith factors p^ν and $p^{a+b-\nu}$. For $\nu = m$ the class is zero: c is divisible by p^m. If $a \le b$, add (c/p^a) times column one to column two; if $b \le a$,

add (c/p^b) times row two to row one. This removes $-c$ and gives the split diagonal presentation, reordered if necessary. These operations also cover $c = 0$.

The three boundary behaviors deserve to be kept separate:

- **Zero class:** $\nu = m$, including $c = 0$ and every $c \in p^m R$. The middle module is $R/(p^a) \oplus R/(p^b)$, and the specified sequence splits.
- **Unit residue:** $p \nmid c$, so $\nu = 0$. The first Smith factor is 1 and $E_c \cong R/(p^{a+b})$. Both cyclic lengths join into one. Here "unit" means a unit modulo p^m; c need not be a constant unit of $k[t]$.
- **Intermediate valuation:** $0 < \nu < m$. A cyclic piece of length ν remains, while the other has length $a+b-\nu$. For $p = t-\lambda$, these are the two Jordan block sizes. For irreducible p of degree greater than one, these are elementary-divisor exponents; their k-dimensions are multiplied by $\deg p$.

The formulas keep total k-dimension $(a+b)\deg p$ fixed. They do not say that arbitrary simultaneous matrix conjugation preserves the chosen extension class.

10.7 Two size-two Jordan blocks: the full quotient calculation

Take $U = W = k^2$ and

$$A = B = N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix}.$$

For $h = \begin{pmatrix} x & y \\ z & w \end{pmatrix}$,

$$Nh - hN = \begin{pmatrix} z & w-x \\ 0 & -z \end{pmatrix}. \tag{36}$$

Thus $\operatorname{im}\delta$ is spanned by E_{12} and $E_{11} - E_{22}$. The two quotient coordinates are

$$\alpha = c_{21}, \qquad \beta = c_{11} + c_{22} = \operatorname{tr} C. \tag{37}$$

Every class has the unique representative

$$C_{\alpha,\beta} = \begin{pmatrix} \beta & 0 \\ \alpha & 0 \end{pmatrix}. \tag{38}$$

For example, choose $z = c_{22}$ and $w - x = -c_{12}$ in (36); then $C + \delta(h) = C_{\alpha,\beta}$. This works in every characteristic, including two.

The reverse intertwiners are exactly $g = r1 + sN$. Their pairings are

$$\operatorname{tr}(gC) = r\beta + s\alpha. \tag{39}$$

So the abstract trace criterion recovers precisely the two explicit quotient coordinates. There is a complement if and only if $\alpha = \beta = 0$. In that case the solution space has dimension two, since $\ker\delta = k1 \oplus kN$.

To connect this computation with cyclic presentations, let u_1, u_2 be the basis of U, with $Nu_2 = u_1$; the cyclic generator is $u = u_2$. Similarly choose w_1, w_2 in W and let e be the coordinate lift $(0, w_2)$. As $B^2 = 0$,

$$T_C^2 e = ACw_2 + CBw_2 = (c_{11} + c_{22})u_1 + c_{21}u_2 = (\alpha + \beta t)u. \tag{40}$$

Thus the identification with the cyclic Ext group is

$$[C] \longmapsto \alpha + \beta t \pmod{t^2}, \qquad \operatorname{Ext}^1_R(R/(t^2), R/(t^2)) \cong R/(t^2). \tag{41}$$

The three module types now follow from (34):

Extension coordinates	Representative C	Middle-module Jordan type	$(\operatorname{rank} T, \operatorname{rank} T^2, \operatorname{rank} T^3)$
$\alpha = \beta = 0$	0	$J_2(0) \oplus J_2(0)$	$(2, 0, 0)$
$\alpha = 0,\ \beta \neq 0$	βE_{11}	$J_3(0) \oplus J_1(0)$	$(2, 1, 0)$
$\alpha \neq 0$	$\beta E_{11} + \alpha E_{21}$	$J_4(0)$	$(3, 2, 1)$

These are all possibilities, over every field. Several comparisons prevent misreadings of the table:

- $C = E_{12} \neq 0$ nevertheless represents the split class, since $E_{12} = \delta(\operatorname{diag}(0, 1))$. A complement is the graph of $h = -\operatorname{diag}(0, 1)$.
- E_{11} and E_{22} are the same extension class: their difference is $\delta(E_{21})$.
- E_{21} and $E_{21} + E_{11}$ are distinct extension classes, with cyclic residues 1 and $1 + t$, but both have middle module $R/(t^4)$. This distinction persists over $\mathbb{F}_2$.
- $C = 1_2$ gives the residue $2t$. The middle module is $J_3 \oplus J_1$ if $\operatorname{char} k \neq 2$, but the extension is split in characteristic two. In that characteristic $1_2 = E_{11} - E_{22} = \delta(E_{21})$.

The last example does not invalidate the trace criterion. The pairing on *all* rectangular matrices is nondegenerate in every characteristic; a particular trace, or the pairing restricted to a smaller subspace, can vanish.

10.8 Self-extensions and the commutant: dimensions without false identities

Since $\delta : H \to H$ is an endomorphism of a finite-dimensional space, rank–nullity gives

$$\dim_k \operatorname{Ext}^1_R(W_B, U_A) = \dim_k \operatorname{Hom}_R(W_B, U_A). \tag{42}$$

This dimension equality alone supplies no preferred isomorphism between its two sides. Trace duality supplies a different, precise statement: the dual of Ext is the **reverse** Hom space in (20).

If $U_A = \bigoplus_i R/(f_i)$ and $W_B = \bigoplus_j R/(g_j)$, the block decomposition of δ, or finite additivity of the resolution, gives

$$\operatorname{Ext}^1_R(W_B, U_A) \cong \bigoplus_{i,j} R/(\gcd(f_i, g_j)). \tag{43}$$

The displayed coordinates depend on the chosen cyclic decompositions and cyclic generators. The resulting dimension is intrinsic:

$$\dim_k \operatorname{Ext}^1_R(W_B, U_A) = \sum_{i,j} \deg \gcd(f_i, g_j). \tag{44}$$

For a single module with elementary divisors $p^{a_{p,i}}$, this yields

$$\dim_k \operatorname{Ext}^1_R(V_T, V_T) = \dim_k \operatorname{End}_R(V_T) = \sum_p \deg(p) \sum_{i,j} \min(a_{p,i}, a_{p,j}). \tag{45}$$

There is also the canonical duality, relative to the fixed t,

$$\operatorname{Ext}^1_R(V_T, V_T) \cong \operatorname{End}_R(V_T)^*. \tag{46}$$

One must not replace the dual in (46) by the commutant itself merely by writing a trace pairing *inside* the commutant. For $T = N = J_2(0)$, that commutant is $k1 \oplus kN$, and

$$\operatorname{tr}\big(N(r1 + sN)\big) = 0 \quad \text{for all } r, s.$$

Its restricted trace form is degenerate in every characteristic. Yet N detects the nonzero extension represented by E_{21} because $\operatorname{tr}(NE_{21}) = 1$. The duality pairs a commutant element with a *quotient class of arbitrary matrices*, not necessarily with another commutant element. This is why choosing a direct complement to $\operatorname{im}\delta$ in the full matrix space is an extra choice.

A useful related calculation gives the same quotient another meaning. Over the dual numbers $k[\epsilon]/(\epsilon^2)$, a first-order change of basis $1 + \epsilon h$ transforms $T + \epsilon C$ into

$$(1 - \epsilon h)(T + \epsilon C)(1 + \epsilon h) = T + \epsilon(C + Th - hT). \tag{47}$$

Thus first-order perturbations modulo changes of basis that reduce to the identity have the same quotient as self-extensions. This explains the appearance of a commutator cokernel, but does not classify finite perturbations or give a multiplication on Ext matching composition in the commutant.

10.9 The determinant bridge: resultants and integral obstructions

For $m = \dim U$ and $n = \dim W$, column-by-column vectorization gives

$$[\delta_{A,B}] = 1_n \otimes A - B^{\mathsf{T}} \otimes 1_m. \tag{48}$$

With the convention $\operatorname{Res}(f, g) = \prod_{f(\alpha)=0} g(\alpha)$ for monic polynomials, roots counted with multiplicity, one has

$$\boxed{\det \delta_{A,B} = \operatorname{Res}(\chi_A, \chi_B) = \prod_{i=1}^{m} \prod_{j=1}^{n} (\alpha_i - \beta_j).} \tag{49}$$

Here α_i, β_j are the characteristic roots in an algebraic closure. To prove this without diagonalizability or separability, extend the field to an algebraic closure and triangularize A, B. Such a triangular form follows by repeatedly choosing an eigenvector and passing to the quotient; it does not require an eigenbasis. In (48), B^{T} is lower triangular, so the matrix is lower block triangular with diagonal blocks $A-\beta_j 1_m$. Its determinant is $\prod_j \det(A-\beta_j 1_m) = \prod_{i,j}(\alpha_i-\beta_j)$. Determinants are unchanged by scalar extension, proving (49) over k. Repeated roots simply repeat factors in the product.

Now let $U = \mathbb{Z}^m$ and $W = \mathbb{Z}^n$, with integral operators A, B. There are two different questions: splitting after tensoring with $\mathbb{Q}$, and splitting as $\mathbb{Z}[t]$-modules. Because W is free over $\mathbb{Z}$, every underlying exact sequence of abelian groups ending in W splits. Thus the block and section calculations still apply, with

$$\delta_{\mathbb{Z}} : \mathrm{Hom}_{\mathbb{Z}}(W, U) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(W, U), \quad h \longmapsto Ah - hB. \tag{50}$$

Moreover (9) remains a free $\mathbb{Z}[t]$-resolution: its leading-term and telescoping proofs work verbatim, and $\mathbb{Z}[t] \otimes_{\mathbb{Z}} W \cong \mathbb{Z}[t]^n$ is free. Consequently, for these endpoints,

$$\mathrm{Ext}^1_{\mathbb{Z}[t]}(W_B, U_A) \cong \mathrm{coker}\, \delta_{\mathbb{Z}}. \tag{51}$$

The first argument being free over $\mathbb{Z}$ is essential to this free resolution argument. We are not asserting that $\mathbb{Z}[t]$ is a PID or that every $\mathbb{Z}[t]$-module has a resolution of length one.

If the resultant in (49) is nonzero, $\delta_{\mathbb{Q}}$ is invertible. The integer matrix $\delta_{\mathbb{Z}}$ is then injective with finite cokernel, and integer Smith form gives

$$|\mathrm{coker}\, \delta_{\mathbb{Z}}| = |\det \delta_{\mathbb{Z}}| = |\mathrm{Res}(\chi_A, \chi_B)|. \tag{52}$$

Indeed a Smith diagonal $d_1, \ldots, d_{mn}$ identifies the cokernel with $\bigoplus_i \mathbb{Z}/d_i\mathbb{Z}$; the product of its finite orders is $|\prod_i d_i| = |\det \delta|$. Thus a nonzero resultant gives unique rational splitting, whereas resultant ± 1 gives unique integral splitting for every C. For a particular integral C with nonzero resultant, the unique rational solution $h = -\delta_{\mathbb{Q}}^{-1} C$ is integral exactly when that extension splits integrally. More generally, Smith coordinates turn the integral solvability test into divisibility by the d_i.

10.9.1 The smallest lattice defect

Take $A = (0)$, $B = (2)$, $C = (1)$, so

$$T = \begin{pmatrix} 0 & 1 \\ 0 & 2 \end{pmatrix}, \qquad \delta_{\mathbb{Z}}(h) = -2h. \tag{53}$$

The rational complement is uniquely the graph of $h = 1/2$. But its class in $\mathrm{coker}\, \delta_{\mathbb{Z}} = \mathbb{Z}/2\mathbb{Z}$ is nonzero, so the specified copy $U = \mathbb{Z}(1, 0)$ has no invariant integral complement. The integral points on the other eigenline are generated by $(1, 2)$, and

$$[\mathbb{Z}^2 : \mathbb{Z}(1, 0) + \mathbb{Z}(1, 2)] = 2. \tag{54}$$

The quotient projection sends this eigenline lattice to $2\mathbb{Z}$, not onto $W = \mathbb{Z}$. Independent rational eigenvectors have produced a sublattice of finite index, not a direct-sum decomposition of the specified lattice. The resultant is $\mathrm{Res}(t, t-2) = -2$, whose absolute value 2 is the order of the obstruction.

This also pinpoints the boundary of trace witnesses. Here the reverse equation $gA = Bg$ forces $g = 0$ over both $\mathbb{Q}$ and $\mathbb{Z}$. There is no nonzero trace witness over either ring, yet the integral extension is nonsplit. Linear functionals into $\mathbb{Z}$ cannot detect the torsion cokernel $\mathbb{Z}/2$. Reducing (53) modulo two does expose it, but complete integral tests use the Smith divisibility conditions, including higher prime powers when present.

This is the same arithmetic distinction that appears in integral CRT: polynomials can be coprime over $\mathbb{Q}[t]$ while the inverse or Bézout coefficients require denominators. The resultant measures a finite failure of invertibility on a lattice. Its nonvanishing is the field-level splitting test; its absolute value need not be one.

10.10 Exercises with worked answers

Exercise 1 — a split extension with many complements. Let $A = B = J_2(0)$ and $C = E_{12}$. Find every invariant complement, and verify invariance without using the Ext terminology.

Answer. Equation (36) must equal $-E_{12}$, so $z = 0$ and $w - x = -1$, while x, y are arbitrary. The complements are the graphs of

$$h = \begin{pmatrix} x & y \\ 0 & x-1 \end{pmatrix}.$$

For these matrices $Ah + C = hB$, hence $T_C(hw, w) = (hBw, Bw)$ belongs to the same graph. Distinct h give distinct complements because projection to W fixes the graph parameter.

Exercise 2 — produce an obstruction witness. Let $U = k^2$, $W = k$, $A = J_2(0)$, $B = 0$, and $C(1) = (c_1, c_2)^{\mathsf{T}}$. Determine exactly when a complement exists. When it does not, provide a one-line certificate.

Answer. Write $h(1) = (x, y)^{\mathsf{T}}$. Then $\delta(h)(1) = (y, 0)^{\mathsf{T}}$, so a complement exists exactly when $c_2 = 0$, with $y = -c_1$ and x arbitrary. The reverse intertwiner $g = (0, 1)$ satisfies $gA = 0 = Bg$, and $\mathrm{tr}(gC) = c_2$. It certifies nonsplitting when $c_2 \neq 0$. This example also makes the direction $g : U \to W$ visible from matrix sizes.

Exercise 3 — compute a complement without solving entries. Suppose $A^3 = 0$ and $B = 1_W$. Find the unique invariant complement for arbitrary $C : W \to U$.

Answer. $\delta(h) = (A - 1_U)h$, and $(1_U - A)(1_U + A + A^2) = 1_U$. Thus $h = (1_U + A + A^2)C$ solves $\delta(h) = -C$. The same identity proves uniqueness. This is the specialization of the divided-difference formula to $b = t - 1$.

Exercise 4 — valuation controls the middle module. Let $U = R/(p^3)$ and $W = R/(p^5)$ for a monic irreducible p. Classify the possible middle-module types, and count extension classes of each type when $k = \mathbb{F}_q$ and $\deg p = d$.

Answer. The extension group is $R/(p^3)$; put $Q = q^d$. There is one zero class, giving $R/(p^3) \oplus R/(p^5)$. For $\nu = 0, 1, 2$, the type is $R/(p^\nu) \oplus R/(p^{8-\nu})$, and the number of classes of exact valuation ν is

$$|p^{\nu}R/(p^3)| - |p^{\nu+1}R/(p^3)| = Q^{3-\nu} - Q^{2-\nu}.$$

These counts refer to endpoint-preserving extension classes. Their sum with the split class is Q^3. Each of the four rows of this classification specifies one middle-module isomorphism type, so counting middle modules would give four, not Q^3.

Exercise 5 — the characteristic-two trap. Over $\mathbb{F}_2$, let $A = B = J_2(0)$ and $C = 1_2$. Exhibit an invariant complement and reconcile it with the fact that C is invertible.

Answer. $\delta(E_{21}) = 1_2 = -1_2$ in characteristic two. The graph of E_{21} is therefore invariant. More explicitly its basis consists of (u_2, w_1) and $(0, w_2)$; T sends the first to zero and the second to the first. Invertibility of the off-diagonal map is unrelated to whether its class in $H/\delta H$ is zero.

Exercise 6 — a larger self-extension count. If $T = J_3(0) \oplus J_1(0)$, determine the dimension of its self-extension space. Explain why the answer is not yet a classification of middle modules.

Answer. Formula (45) gives $3 + 1 + 1 + 1 = 6$. These are six independent parameters for extensions with the two copies of V_T identified in advance. Changing endpoint automorphisms or forgetting the inclusion and quotient can identify different extension classes. Determining the isomorphism types of the resulting eight-dimensional middle modules is a further problem; the number six is not a number of Jordan types.

10.11 Hypotheses, failed shortcuts, and reconstructive output

Claim or construction	Hypotheses actually used	Failed shortcut or boundary
Block form and section-change classification	A field, or an explicitly split underlying sequence over the base ring	An arbitrary exact sequence over $\mathbb{Z}$ need not admit such a section; $0 \to \mathbb{Z}/2 \to \mathbb{Z}/4 \to \mathbb{Z}/2 \to 0$ is an example
One-step resolution (9)	$R = k[t]$ and a k-space W	It is a one-variable resolution; it does not automatically compute extensions with two commuting operators
Complete trace witnesses and perfect Ext–Hom duality	Finite-dimensional spaces over a field	Infinite-dimensional endomorphisms have no general trace; a trace on the commutant alone can be degenerate
Explicit inverse (25)	Stated annihilators and a Bézout identity $ua + vb = 1$	Shared factors prevent a universal inverse but do not obstruct every individual C
Spectral disjointness criterion	Finite dimension, eigenvalues taken over an algebraic closure	Empty real-eigenvalue sets can hide a shared irreducible quadratic
Cyclic Ext and Smith formulas	Nonzero $f, g \in k[t]$; fixed cyclic endpoint generators for the residue coordinate	Smith operations classify the middle module and may change endpoint data
Jordan interpretation of (34)	$p = t - \lambda$	Higher-degree irreducibles give rational elementary divisors rather than Jordan blocks over k
Equality of commutant and self-Ext dimensions	Finite dimension	Equal dimensions do not identify the spaces or their algebraic structures; the natural statement here is duality

Claim or construction	Hypotheses actually used	Failed shortcut or boundary
Integral cokernel has order equal to absolute resultant	Finite free integral endpoints and nonzero resultant	Rational splitting can leave a finite lattice-index obstruction; rational trace tests do not detect it

The reconstruction route is short enough to use repeatedly. Record the specified submodule and quotient; choose any linear section and extract C. Compute its class modulo $Ah - hB$, or test it against reverse intertwiners. If the class vanishes, solve once and add every forward intertwiner to obtain all complements. If annihilators are coprime, (25) solves immediately and the complement is uniquely $\ker b(T)$. If an obstruction remains, cyclic coordinates and Smith reduction explain what it does to the middle module. This keeps four different outputs distinct: a yes-or-no splitting test, a witness or complement, an extension class with fixed endpoints, and the coarser isomorphism type obtained after forgetting those endpoints.

Local validation. The companion script `check_sylvester.py` checks the section/equivalence signs, both sides of the Bézout inverse, the trace-adjoint identity, all size-two Jordan extensions over $\mathbb{F}_2$ and $\mathbb{F}_3$, representative polynomial Smith reductions over $\mathbb{Q}[t]$, and the integral resultant/index bridge. These finite checks supplement the field-independent proofs; they are not substitutes for the hypotheses or the general arguments.

11 Repairing linear maps: spectral projection, separability, and averaging

An arbitrary vector-space complement rarely respects an operator or a group action. There are two ways to repair it. One can solve a separate equation for the complement, as in the Sylvester calculation, or construct a single operation that repairs every linear map between modules for the algebra in question. This chapter constructs the second operation explicitly.

The distinction matters. A particular extension may split even when no repair defined by a separability tensor exists. Conversely, a repair operation for modules over a quotient algebra says nothing about extensions whose middle term is not a module over that quotient.

11.1 Two identities that repair every linear map

Let A be an associative unital algebra over a field k. For the moment, suppose that finitely many elements $a_i, b_i \in A$ satisfy

$$\sum_i a_i b_i = 1, \qquad \sum_i c a_i \otimes b_i = \sum_i a_i \otimes b_i c \quad (c \in A). \tag{A1}$$

The tensor products in this displayed identity are tensor products of k-vector spaces. The order of the two factors in $b_i c$ matters. For left A-modules M, N and a k-linear map $f : M \to N$, define

$$\mathcal{A}_{M,N}(f)(v) = \sum_i a_i f(b_i v). \tag{A2}$$

No dimension assumption on M or N is needed: the sum defining the repair is finite.

Repair lemma. The operation $\mathcal{A}_{M,N}$ is a k-linear projection

$$\mathrm{Hom}_k(M,N) \longrightarrow \mathrm{Hom}_A(M,N).$$

If $u : M' \to M$ and $v : N \to N'$ are A-linear, then

$$\mathcal{A}_{M',N'}(vfu) = v\,\mathcal{A}_{M,N}(f)\,u. \tag{A3}$$

Proof. Apply the k-linear map $a \otimes b \mapsto af(bw)$ to the second identity in (A1). It gives

$$c\,\mathcal{A}(f)(w) = \sum_i a_i f(b_i c w) = \mathcal{A}(f)(cw),$$

so the repaired map is A-linear. If f was already A-linear,

$$\mathcal{A}(f)(w) = \sum_i a_i b_i f(w) = f(w).$$

These two assertions give $\mathcal{A}^2 = \mathcal{A}$ and its stated image. For (A3), move the A-linear maps u, v through the scalar actions in (A2). □

Consider an exact sequence of A-modules

$$0 \longrightarrow U \xrightarrow{\iota} V \xrightarrow{\pi} W \longrightarrow 0.$$

Choose a k-linear section $s : W \to V$ of π. Then

$$\pi\mathcal{A}(s) = \mathcal{A}(\pi s) = \mathcal{A}(1_W) = 1_W.$$

Thus $\mathcal{A}(s)$ is an A-linear section. A k-linear retraction $r : V \to U$ of ι is repaired in the same way: $\mathcal{A}(r)\iota = 1_U$. Consequently every such exact sequence splits as a sequence of A-modules. Ordinary vector spaces admit the initial section by choosing a basis; finite-dimensionality is not the source of this splitting theorem.

The identities also explain why averaging a projection works in Maschke's proof. If $P : V \to V$ has image in an invariant submodule U and is the identity on U, each term in (A2) has image in U, and $\mathcal{A}(P)|_U = 1_U$. These facts imply $\mathcal{A}(P)^2 = \mathcal{A}(P)$.

Boundary. Averaging is not an algebra homomorphism on $\mathrm{End}_k(V)$, and it does not preserve arbitrary idempotents. Over $\mathbb{Q}$, let $T = \mathrm{diag}(1,-1)$ and average by $\mathcal{A}(f) = (f + TfT)/2$. It removes the off-diagonal entries. Hence

$$\mathcal{A}(E_{12}) = \mathcal{A}(E_{21}) = 0, \qquad \mathcal{A}(E_{12}E_{21}) = E_{11}.$$

Moreover, the projection $Q = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ averages to $I/2$, which is not a projection. Its image was not T-invariant. The valid repair argument uses the invariant target and the identity on that target; it is not an assertion about arbitrary averages of idempotent matrices.

11.2 What the tensor means

Write

$$e = \sum_i a_i \otimes b_i^{\mathrm{op}} \quad \text{in} \quad A^e = A \otimes_k A^{\mathrm{op}}.$$

The superscript records the opposite multiplication. In A^e,

$$(a \otimes b^{\mathrm{op}})(c \otimes d^{\mathrm{op}}) = ac \otimes (db)^{\mathrm{op}}.$$

It acts on A by $(a \otimes b^{\mathrm{op}})x = axb$. The multiplication map $\mu : A^e \to A$, $\mu(a \otimes b^{\mathrm{op}}) = ab$, is a homomorphism of A-bimodules, with the outer bimodule action on its source. In general it is not a homomorphism of algebras.

The identities (A1) become

$$\mu(e) = 1, \qquad (c \otimes 1)e = (1 \otimes c^{\mathrm{op}})e. \tag{A4}$$

They imply that $\sigma(c) = (c \otimes 1)e$ is an A-bimodule section of μ. Conversely, such a section gives (A4) on putting $e = \sigma(1)$. An algebra with this property is called **separable over** k. This is a condition on the algebra together with its ground field.

The name **separability idempotent** is literal:

$$\begin{aligned} e^2 &= \sum_i (a_i \otimes 1)(1 \otimes b_i^{\mathrm{op}})e \\ &= \sum_i (a_i b_i \otimes 1)e = e. \end{aligned}$$

It need not be central in A^e. The outer-action identity (A4), rather than centrality, is what the repair proof needs.

For commutative A, the opposite algebra is just A, and these notations simplify. The next construction gives the tensor without choosing an eigenbasis or passing to a splitting field.

11.3 A divided difference constructs the polynomial repair

Let $m \in k[t]$ be monic of positive degree and suppose

$$\gcd(m, m') = 1. \tag{A5}$$

Set $S = k[t]/(m)$ and identify

$$S \otimes_k S = k[x, y]/(m(x), m(y)).$$

The expression

$$q(x,y) = \frac{m(x) - m(y)}{x - y}$$

is a polynomial: division here abbreviates the polynomial identity $m(x) - m(y) = (x-y)q(x,y)$. The Euclidean algorithm supplies a polynomial h with $h(t)m'(t) \equiv 1 \pmod{m}$. Define

$$e = q(x,y)h(y) = \frac{m(x) - m(y)}{x - y}\, m'(y)^{-1} \quad \text{in } S \otimes_k S. \tag{A6}$$

The inverse in this formula exists by (A5); it is an inverse in S, not a rational function to be evaluated at a zero of its denominator.

In $S \otimes S$ we have $(x - y)e = (m(x) - m(y))h(y) = 0$. For any polynomial f, $f(x) - f(y)$ is divisible by $x - y$, so

$$(f(x) - f(y))e = 0.$$

This is precisely the second identity in (A4). The multiplication map μ substitutes $x = y = \bar{t}$, and $q(t,t) = m'(t)$; hence $\mu(e) = m'(\bar{t})h(\bar{t}) = 1$. Formula (A6) therefore proves all the repair conclusions for S-modules.

More concretely, if T_M and T_N satisfy $m(T_M) = m(T_N) = 0$ and $e = \sum_{i,j} c_{ij} x^i y^j$, then

$$\mathcal{A}(f) = \sum_{i,j} c_{ij} T_N^i f T_M^j. \tag{A7}$$

The result satisfies $T_N \mathcal{A}(f) = \mathcal{A}(f) T_M$. It is a polynomial construction in the two actions, defined over k.

Over a splitting field K, write $m(t) = \prod_{j=1}^{d} (t - \lambda_j)$ with distinct λ_j, and let $e_j(t)$ be the usual Lagrange idempotents. At the pair of roots (λ_r, λ_s), formula (A6) is 0 when $r \neq s$ and 1 when $r = s$. The evaluation description of $K[t]/(m) \otimes_K K[t]/(m)$ therefore gives

$$e = \sum_j e_j \otimes e_j, \qquad \mathcal{A}(f) = \sum_j P_j^N f P_j^M, \tag{A8}$$

where $P_j^M = e_j(T_M)$ and $P_j^N = e_j(T_N)$. The operation keeps the maps between matching eigenspaces and deletes the maps between different eigenspaces. This is the spectral meaning of the tensor, but the divided-difference proof did not require roots in k.

In the commutative case the tensor is unique. Indeed, if e, e' both satisfy (A4), moving each factor of a term of e across e' gives $ee' = \mu(e)e' = e'$. Interchanging them gives $e'e = e$, so commutativity implies $e = e'$. Interchanging the two tensor factors produces another such tensor, which is consequently the same tensor.

11.4 Three calculations that can be reconstructed

For $m = t^2 - 1$ and $\operatorname{char} k \neq 2$, $q = x + y$ and $(2y)^{-1} = y/2$ in S. Thus

$$e = \frac{1 + xy}{2}, \qquad \mathcal{A}(f) = \frac{f + T_N f T_M}{2}. \tag{A9}$$

The projections are $P_\pm = (I \pm T)/2$, and (A8) becomes $P_+^N f P_+^M + P_-^N f P_-^M$.

For a non-diagonal example over $\mathbb{Q}$, put

$$T = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}, \qquad U = \mathbb{Q}e_1, \qquad P = \begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}.$$

Then $T^2 = I$, P projects onto U, and

$$\mathcal{A}(P) = \frac{P + TPT}{2} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}.$$

Its kernel is $\mathbb{Q}(-1, 1)$, the desired invariant complement. Equivalently, the quotient has operator -1; the ordinary section $s(1) = (0, 1)$ repairs to $(-1, 1)$.

For $m = t^2 + 1$ over $\mathbb{R}$, write $J_M^2 = J_N^2 = -I$. Now $(2y)^{-1} = -y/2$, so

$$e = \frac{1 - xy}{2}, \qquad \mathcal{A}(f) = \frac{f - J_N f J_M}{2}. \tag{A10}$$

This extracts the complex-linear part of a real-linear map between complex vector spaces. Its complementary part $(f + J_N f J_M)/2$ is complex-antilinear. On $\mathbb{C}$, a real-linear map has the form $f(z) = az + b\bar{z}$; (A10) retains az. A real eigenbasis is neither available nor needed.

For a cubic example over $\mathbb{Q}$, take $m = t^3 - 2$. Since $(3y^2)^{-1} = y/6$,

$$e = \frac{2 + xy^2 + x^2 y}{6}, \qquad \mathcal{A}(f) = \frac{2f + T_N f T_M^2 + T_N^2 f T_M}{6}. \tag{A11}$$

Multiplying (A11) on the appropriate side by T_N and T_M gives

$$6(T_N \mathcal{A}(f) - \mathcal{A}(f) T_M) = 2T_N f - 2f T_M + T_N^3 f T_M - T_N f T_M^3 = 0,$$

where the final cancellation uses both cubic relations. The repair can therefore be checked before knowing the eigenvalues of a cubic operator.

Characteristic 2 itself does not preclude a polynomial repair. For $m = t^2 + t + 1$ over $\mathbb{F}_2$, the derivative is 1 and $q = x + y + 1$. Thus $e = 1 + x + y$ and $\mathcal{A}(f) = f + T_N f + f T_M$. This is a separable quadratic field example in the same characteristic where $t^2 - 1$ fails.

11.5 The derivative is the exact obstruction to this tensor

Condition (A5) is necessary as well as sufficient. A useful proof exposes the role of derivations.

For a commutative k-algebra S and an S-module L, a k-derivation $D : S \to L$ is a k-linear map satisfying

$$D(ab) = aD(b) + bD(a).$$

Suppose $e = \sum_i a_i \otimes b_i$ satisfies (A4). Apply $a \otimes b \mapsto aD(b)$ to $(c \otimes 1)e = (1 \otimes c)e$. The two sides are

$$c\sum_i a_i D(b_i) \quad \text{and} \quad c\sum_i a_i D(b_i) + \left(\sum_i a_i b_i\right) D(c).$$

Therefore $D(c) = 0$ for every c. A separability tensor kills all such derivations.

For $S = k[t]/(m)$, however, formal differentiation defines

$$D : S \longrightarrow k[t]/(m, m'), \qquad D(\bar{f}) = \overline{f'}. \tag{A12}$$

It is well-defined: replacing f by $f + mh$ changes the derivative by $m'h + mh'$, which is zero in the target. It obeys the product rule. If $\gcd(m, m') \neq 1$, its target is nonzero and $D(\bar{t}) = 1 \neq 0$. Consequently no tensor (A4) exists. We have proved

$$\boxed{k[t]/(m) \text{ is separable over } k \iff \gcd(m, m') = 1.} \tag{A13}$$

This is the elementary content of the differential formula $\Omega_{S/k} \cong S/(m')\,dt$: differentiating the relation $m(t) = 0$ imposes exactly $m'(t)\,dt = 0$. No differential-module formalism is needed for the proof (A12).

For the smallest failure, take the dual numbers $D_0 = k[\epsilon]/(\epsilon^2)$. Write a putative tensor as

$$e = a + b(\epsilon \otimes 1) + c(1 \otimes \epsilon) + d(\epsilon \otimes \epsilon).$$

The relation $(\epsilon \otimes 1)e = (1 \otimes \epsilon)e$ forces $a = 0$ and $b = c$. Its product is then $2b\epsilon$, never 1. This argument works in every characteristic. In characteristic 2, $t^2 - 1 = (t - 1)^2$, so the involution algebra itself becomes this counterexample after putting $\epsilon = t - 1$.

11.6 Splitting over one field is weaker than separability

For a monic polynomial m, factor it in $k[t]$ as $m = \prod_i p_i^{r_i}$ with distinct monic irreducibles p_i. The algebra $S = k[t]/(m)$ is semisimple precisely when every $r_i = 1$. Here is a proof that keeps the field issue visible.

If all $r_i = 1$, the polynomial CRT identifies S with the product of the fields $k[t]/(p_i)$. Every S-module decomposes by these central idempotents, and every submodule has a complement by taking a vector-space complement over each component field.

If some $r_i \geq 2$, consider the component $B = k[t]/(p_i^{r_i})$. The quotient $B \to B/(p_i)$ has no B-linear section. The image of 1 under such a section would be an element $v \equiv 1 \pmod{p_i}$ killed by p_i. But $p_i v = 0$ in B forces v to be divisible by $p_i^{r_i-1}$, contradicting $v \equiv 1 \pmod{p_i}$. Extending this module structure through the CRT projection gives a nonsplit sequence of S-modules.

This semisimplicity criterion uses irreducible factorization over k. It does not assert that the roots in an algebraic closure are distinct. The derivative criterion (A13) imposes that stronger requirement.

For example, let

$$k = \mathbb{F}_p(s), \qquad m(t) = t^p - s.$$

Eisenstein at the prime s in $\mathbb{F}_p[s]$, followed by Gauss's lemma, proves that m is irreducible over k. Thus $S = k[t]/(m)$ is a field: every sequence of S-vector spaces splits. Nevertheless $m' = 0$, so (A12) is a nonzero derivation $S \to S$ and no separability tensor exists.

For any fixed pair of modules, ordinary vector-space projections onto the subspace of intertwiners may still be chosen. What fails here is the separability-tensor construction, which repairs all pairs coherently as in (A3).

Adjoin an element α with $\alpha^p = s$, and write $K = k(\alpha)$. Then

$$K \otimes_k S \cong K[t]/((t-\alpha)^p) \cong K[u]/(u^p).$$

This algebra has a nonsplit sequence with middle term $K[u]/(u^2)$ and trivial one-dimensional end terms. The base change has exposed nilpotent structure that was invisible to factorization over k.

In contrast, a tensor satisfying (A4) survives every extension of the ground field. Its coefficients can be extended, and the same repair proof still works. For the polynomial algebras here, separability is equivalent to semisimplicity after every field extension. One direction follows from base change of the tensor. For the other, a repeated root over a splitting field gives a quotient by $(t-\lambda)^2$ and the nonsplitting example above.

Over a perfect field, every irreducible polynomial is separable, so the two criteria coincide. Over an imperfect field, "no repeated irreducible factors in $k[t]$" and "coprime to the derivative" must be kept separate.

11.7 Which extensions is the repair allowed to split?

The previous proof applies to exact sequences **of S-modules**. It requires m to annihilate the middle term, not merely the two end terms.

Take $R = k[t]$ and $S = R/(t) \cong k$. The exact sequence

$$0 \longrightarrow k \xrightarrow{1 \mapsto \bar{t}} R/(t^2) \xrightarrow{\bar{f} \mapsto f(0)} k \longrightarrow 0 \tag{A14}$$

has both end terms annihilated by t. It does not split as a sequence of R-modules: any lift of 1 in the quotient is $1 + ct$, and $t(1 + ct) = t \neq 0$. Its middle term is not an S-module, because t does not act as zero there.

Thus the correct comparison is

$$\mathrm{Ext}^1_S(k,k) = 0, \qquad \mathrm{Ext}^1_R(k,k) \cong k.$$

The latter group is computed in the Sylvester/extension chapter. There is no contradiction. Changing the scalar ring changes which middle terms are eligible extensions.

This also locates the difference between two splitting mechanisms. For operators with coprime minimal polynomials on the two end terms, the Sylvester equation gives a unique invariant section of every extension with those end actions. The repair in this chapter may allow the end terms to share eigenvalues, but restricts the entire middle operator to a separable annihilating polynomial. In that category an invariant section exists, and its choices need not be unique.

11.8 Maschke's sum is the same tensor construction

Let G be a finite group and suppose $|G|$ is invertible in k. In $A = k[G]$, put

$$e_G = \frac{1}{|G|} \sum_{g \in G} g \otimes (g^{-1})^{\mathrm{op}}. \tag{A15}$$

Its product is 1. For $h \in G$, reindexing by $r = hg$ gives

$$\sum_g hg \otimes g^{-1} = \sum_r r \otimes r^{-1}h.$$

Linear extension establishes (A1) for every element of $k[G]$. The general repair lemma now reads

$$\mathcal{A}(f) = \frac{1}{|G|} \sum_{g \in G} \rho_N(g) f \rho_M(g^{-1}). \tag{A16}$$

It repairs intertwiners, sections, and retractions simultaneously. Maschke's invariant-complement theorem is the retraction case. No irreducible-character theory, radical theory, or classification of semisimple rings is needed for this proof.

There is also a short converse for the assertion that *every* $k[G]$-module sequence splits. Give k the trivial action and consider the augmentation

$$\varepsilon : k[G] \longrightarrow k, \qquad \sum_g c_g g \longmapsto \sum_g c_g.$$

An invariant vector in the left regular representation has all its coefficients equal, since left multiplication acts transitively on the group basis. Hence it is $c \sum_g g$. If $|G| = 0$ in k, every such vector has augmentation zero. An equivariant section would have to send 1 to an invariant vector of augmentation 1, which is impossible. Thus the division by $|G|$ marks a genuine obstruction for group algebras.

11.9 A repair without division by a dimension

The group formula should not suggest that every separable algebra needs the inverse of its dimension. Let $A = M_r(k)$ and denote its matrix units by E_{ij}. For any characteristic,

$$e = \sum_{i=1}^{r} E_{i1} \otimes E_{1i}^{\mathrm{op}} \tag{A17}$$

is a separability idempotent. Its product is $\sum_i E_{ii} = I$. For $c = E_{ab}$, both sides of the centralizing identity are $E_{a1} \otimes E_{1b}$; linearity proves the identity for every matrix. The repair formula uses no denominator.

The associated module decomposition is equally concrete. Put $L = E_{11}M$. Then

$$k^r \otimes_k L \longrightarrow M, \qquad e_i \otimes v \longmapsto E_{i1}v$$

is an A-module isomorphism, with inverse

$$m \longmapsto \sum_i e_i \otimes E_{1i}m.$$

The matrix algebra acts on the k^r factor. These formulas prove that its modules are built from copies of the standard module, and show directly why the conclusion still holds when the characteristic divides r. This is the elementary matrix-unit case of the relation between matrix algebras and their coefficient fields.

11.10 The denominator also remembers integral gluing

The rational involution formula $e = (1 + xy)/2$ has an arithmetic boundary. Work over $\mathbb{Z}$ with $A_{\mathbb{Z}} = \mathbb{Z}[t]/(t^2-1)$. Rationally its two eigenspaces are separated by $(1\pm t)/2$. Integrally the evaluation map is

$$A_{\mathbb{Z}} \longrightarrow \mathbb{Z} \oplus \mathbb{Z}, \qquad a + bt \longmapsto (a + b, a - b).$$

Its image consists exactly of pairs congruent modulo 2; it has index 2. This is the resultant obstruction for $t - 1$ and $t + 1$ in the integral CRT chapter.

There is no separability tensor over $\mathbb{Z}$ for this algebra. Such a tensor would reduce modulo 2 to one for $\mathbb{F}_2[t]/((t-1)^2)$, contradicting the dual-number calculation. Thus the rational denominator, the integral index, and the repeated root modulo 2 describe three aspects of the same failure in this example. This does not mean that every rational repair of a particular map has nonintegral coefficients; particular maps may already respect the action. It means that the universal tensor repair cannot be defined integrally.

11.11 Reconstructive exercises

1. Starting only from (A1), prove the repair lemma and repair both a section and a retraction. Explain precisely where the surrounding maps must be A-linear.

2. Derive (A9), (A10), and (A11) from the divided difference, without using eigenvectors. For (A11), verify directly that the repaired map intertwines the two cubic operators.
3. Prove that a commutative algebra has at most one separability tensor. Identify the tensor for k^d and explain why it deletes maps between distinct direct factors.
4. Check that (A12) is independent of the representative of $\bar{f}$. Use it to prove the necessity in (A13). Why does this argument not prove that every nonseparable algebra has a nonsplit sequence of its own left modules over the original field?
5. Prove that (A14) is nonsplit, then identify the failed hypothesis in an attempted application of the repair for $S = k$.
6. Prove the augmentation converse to Maschke. Compare it with the denominator-free matrix-unit construction (A17).
7. Explain why the congruence $a+b \equiv a-b \pmod 2$ is the entire integral obstruction for $\mathbb{Z}[t]/(t^2-1)$. Relate it to the reduction of the derivative $2t$ modulo 2.

Answer landmarks. For 1, use the tensor identity to move the scalar from the first slot to the input slot, then apply (A3). For 2, invert the derivative modulo the polynomial before multiplying the divided difference; the inverses are $y/2$, $-y/2$, and $y/6$, respectively. For 3, $ee' = e' = e$, and for k^d the tensor is $\sum_j e_j \otimes e_j$. For 4, the ambiguity is $m'h + mh'$; the inseparable field $\mathbb{F}_p(s)[t]/(t^p - s)$ disproves the proposed stronger conclusion. For 5, t kills both ends but not the middle. For 6, the regular invariant vectors all have augmentation zero in modular characteristic, whereas the product in (A17) is I in every characteristic. For 7, a pair (u, v) reconstructs $a = (u + v)/2$, $b = (u - v)/2$.

Sources and scope. These are standard algebraic mechanisms, presented here with explicit reconstruction and boundary examples; no novelty claim is intended. Marcelo Aguiar's original paper *A note on strongly separable algebras*, §§1–2 states the separability-tensor identities and includes the group-algebra and matrix-algebra examples. The paper's additional notion of *strong* separability is not required for the repair lemma. The Stacks Project, Lemma 10.143.4 identifies finite étale algebras over a field as finite products of finite separable field extensions; its proof supplies a broader setting for the polynomial criterion. The arguments above prove the formulas used here without requiring that general theory. The source-comparison guide separately locates the directly inspected textbook proofs of polynomial decomposition and Maschke's theorem.

12 Integral spectral gluing: the lattice behind Fourier coordinates

Status and scope. This is a local continuation of Chapter 9 at baseline a4a2879. The CRT, resultant, cyclotomic, and Fourier facts below are classical. The contribution of this chapter is their organization into explicit reconstruction certificates, together with small examples separating conditions that are easily conflated. No literature novelty, search improvement, universal existence theorem, or new Hadamard obstruction is asserted. The proofs are supplied here. The arithmetic proof comparison identifies the inspected textbook arguments and the additional integral gluing proved in this chapter.

Chapter 9 identified Fourier diagonalization with CRT over a splitting field. This chapter starts at the point where that description ceases to remember enough: the original coefficients must be integers, and often signs. An isomorphism of rational algebras need not identify their integral lattices. Even an integral inverse need not have sign coefficients. The central conclusion is Theorem 6A: once the full spectral norm equations hold, sign reconstruction at every cyclic order is exactly membership in a specified affine integral lattice. The prime-order formulas and prime-power gluing explain how that lattice is tested.

Throughout, $R = \mathbb{Z}[t]$, and

$$\mathcal{A}_m = \mathbb{Z}[z]/(z^m - 1), \qquad X = \sum_{j=0}^{m-1} x_j z^j.$$

The involution $X^* = X(z^{-1})$ is defined modulo $z^m - 1$. We use the Fourier convention $\lambda_r = X(\zeta_m^r)$; hence inverse Fourier coefficients use ζ_m^{-rj}. The augmentation is $\epsilon(X) = X(1) = \sum_j x_j$, and 1 denotes the constant polynomial, not the all-ones polynomial $T = 1 + z + \cdots + z^{m-1}$.

The dependency chain is short:

Input	Reconstructive output	What remains to check
Monic factors coprime over $\mathbb{Q}$	A rational CRT inverse and an exact finite integral defect	Membership in the integral image
An integral cyclotomic tuple	Rational coefficients with an explicit divisibility certificate	The coefficient lattice congruences
A tuple in that lattice	A unique integral row	Symmetry, normalization, and signs
Integral odd coefficients and exact total energy	Every coefficient is a sign	Any additional symmetry or normalization required by the construction
Coherent four-row spectral Gram equations	The full group-algebra Gram identity	Parity or another condition forcing signs

12.1 The integral CRT defect is a finite module

Theorem 1. Let $f, g \in \mathbb{Z}[t]$ be monic of positive degrees r, s, and assume $\gcd_{\mathbb{Q}[t]}(f, g) = 1$. There is an exact sequence of R-modules

$$0 \longrightarrow R/(fg) \xrightarrow{\alpha} R/(f) \oplus R/(g) \xrightarrow{\delta} R/(f, g) \longrightarrow 0, \tag{1}$$

where

$$\alpha(h) = (h \bmod f, h \bmod g), \qquad \delta(u, v) = u - v \bmod (f, g).$$

The last map is an additive module map; it is generally not a ring map. The first map is a ring injection, and its image has index

$$[R/(f) \oplus R/(g) : \alpha(R/(fg))] = |\operatorname{Res}(f, g)|. \tag{2}$$

In particular, rational coprimality gives an integral CRT isomorphism exactly when the resultant is 1 or -1.

Proof of exactness. Monic division makes all three principal quotients free abelian groups in their usual power bases. If $h \in (f) \cap (g)$, write $h = fu$ with $u \in R$. Rational coprimality implies $g \mid u$ in $\mathbb{Q}[t]$. Divide u by the monic polynomial g in R; uniqueness of the rational remainder shows that the integral remainder is zero. Thus $h \in (fg)$, proving $(f) \cap (g) = (fg)$ and the injectivity of α.

The difference map is well defined and surjective. If $u - v = af + bg$, then

$$h = u - af = v + bg$$

maps to $(u \bmod f, v \bmod g)$. This proves exactness in the middle. The reverse inclusion follows directly from the definitions. $\square$

The nontrivial part of (2) is a determinant computation over the integers, not an application of field CRT.

Determinant proof of the index formula. Put $L_f = R/(f)$, with basis $1, t, \dots, t^{r-1}$, and let

$$M_g : L_f \longrightarrow L_f, \qquad h \longmapsto gh.$$

Then

$$\operatorname{coker} M_g \cong R/(f,g). \tag{3}$$

Over $\mathbb{Q}$, the map M_g is invertible by Bézout's identity, so its integral matrix has a nonzero determinant. An integral square matrix M with nonzero determinant has cokernel of order $|\det M|$: integral row and column operations reduce it to Smith normal form, whose nonzero diagonal entries multiply to that determinant in absolute value.

For monic f, the standard resultant convention gives

$$\det M_g = \operatorname{Res}(f,g) = \prod_{\alpha : f(\alpha)=0} g(\alpha), \tag{4}$$

with roots counted with multiplicity. To see the determinant identity, extend scalars to an algebraic closure. Multiplication by t on L_f has companion matrix with characteristic polynomial f. Triangularize that matrix; applying g gives diagonal entries $g(\alpha)$, including repeated roots.

This is also the Sylvester determinant computation. Consider the integral map

$$S : R_{<s} \oplus R_{<r} \longrightarrow R_{<r+s}, \qquad (a,b) \longmapsto af + bg.$$

Use monic division by f to identify its target with $R_{<s} \oplus R_{<r}$, by $qf + h \leftrightarrow (q,h)$. This is an isomorphism of integral lattices. In these coordinates the matrix of S has block form

$$\begin{pmatrix} I_s & Q \\ 0 & M_g \end{pmatrix}.$$

Thus its determinant is $\det M_g$, up to the sign from the chosen ordering of Sylvester columns and target basis. Consequently (3), (4), and exactness of (1) give (2).

This argument also proves the claimed equivalence: the defect vanishes exactly when its order is one, which is exactly $(f,g) = R$. □

Two cautions make the theorem more useful.

- Rational coprimality is weaker than integral comaximality. For $f = t-1,\ g = t+1$, the defect is $\mathbb{Z}/2$, and the image in $\mathbb{Z}^2$ is $\{(a,c) : a \equiv c \pmod 2\}$.
- The resultant gives the **order** of the defect, not its complete abelian-group structure. For $f = t^2-1,\ g = t^2+1$, the defect is

  $$R/(2, t^2-1) \cong \mathbb{F}_2[t]/((t-1)^2),$$

  whose additive group is $(\mathbb{Z}/2)^2$, although the resultant is 4. Smith normal form gives the additional structure.

Monicity performs two jobs: it gives finite free power-basis lattices, and it makes rational divisibility descend to integral divisibility. Primitivity is enough for the latter job by Gauss's lemma; it does not make the principal quotient a finite free abelian group. Without even primitivity, injectivity can fail: $f = 2t$ and $g = 2(t-1)$ are coprime over $\mathbb{Q}$, but $2t(t-1)$ belongs to $(f) \cap (g)$ and not to (fg).

Corollary 2: several factors. If the monic polynomials $f_1, \dots, f_k$ are pairwise coprime over $\mathbb{Q}$, the evaluation map

$$R/(f_1 \cdots f_k) \longrightarrow \bigoplus_i R/(f_i)$$

is injective and has index

$$\prod_{i<j} |\operatorname{Res}(f_i, f_j)|. \tag{5}$$

Proof. Apply Theorem 1 first to $F = f_1 \cdots f_{k-1}$ and f_k, and then refine the $R/(F)$ coordinate. Indices of nested finite-index lattices multiply. The product-over-roots formula (4) gives $\operatorname{Res}(F, f_k) = \prod_{i<k} \operatorname{Res}(f_i, f_k)$. Induction proves the assertion. □

Formula (5) does **not** say that checking the original factor pairs separately characterizes the image. Section 5 gives a small counterexample.

12.2 Prime cyclic order is a single fiber product

Let p be prime, $\zeta = \zeta_p$, and

$$K = \mathbb{Q}(\zeta), \qquad O = \mathbb{Z}[\zeta], \qquad \pi = 1 - \zeta.$$

We use the concrete order O; no theorem identifying it with the full ring of integers is needed here. The polynomial $\Phi_p(t) = 1 + t + \cdots + t^{p-1}$ is irreducible over $\mathbb{Q}$: after replacing t by $t+1$, Eisenstein's criterion at p applies. Hence $[K : \mathbb{Q}] = p - 1$.

Reduction $\zeta \mapsto 1$ identifies

$$O/(\pi) \cong \mathbb{Z}/(\Phi_p(1)) \cong \mathbb{F}_p. \tag{6}$$

Write $\rho : O \to \mathbb{F}_p$ for this reduction.

Theorem 3: exact prime gluing and reconstruction. The map

$$\mathcal{A}_p \longrightarrow \mathbb{Z} \oplus O, \qquad X \longmapsto (a, b) = (X(1), X(\zeta))$$

has image

$$\mathbb{Z} \times_{\mathbb{F}_p} O = \{(a, b) : \rho(b) = a \bmod p\}, \tag{7}$$

of index p. Its unique rational inverse has coefficients

$$\boxed{x_j = \frac{a + \operatorname{Tr}_{K/\mathbb{Q}}(b\zeta^{-j})}{p}}, \qquad 0 \le j < p. \tag{8}$$

For $a \in \mathbb{Z}$, $b \in O$, all the coefficients in (8) are integers if and only if the single congruence in (7) holds.

Proof of the image statement. Apply Theorem 1 to $t - 1$ and $\Phi_p(t)$. Their common quotient is $\mathbb{F}_p$, and their resultant is $\Phi_p(1) = p$. There is also a direct construction. Write

$$b = B(\zeta), \qquad B \in \mathbb{Z}[t], \quad \deg B \le p - 2, \qquad d = B(1).$$

If $a \equiv d \pmod p$, set $q = (a - d)/p$. Then

$$X(t) = B(t) + q\Phi_p(t) \tag{9}$$

has integral coefficients, degree less than p, and prescribed evaluations a, b. Conversely any integral polynomial takes the same residue at 1 and ζ in the common quotient (6).

For arbitrary $a \in \mathbb{Q},\ b \in K$, the same construction with a rational representative B proves existence and uniqueness of the rational inverse.

Proof of the trace formula. The $p-1$ embeddings are $\sigma_r(\zeta) = \zeta^r$, $1 \le r < p$, so

$$\mathrm{Tr}(\zeta^\ell) = \begin{cases} p-1, & p \mid \ell, \\ -1, & p \nmid \ell. \end{cases} \tag{10}$$

For the rational inverse $b = \sum_k x_k \zeta^k$, this gives

$$\mathrm{Tr}(b\zeta^{-j}) = (p-1)x_j - \sum_{k \ne j} x_k = p x_j - a.$$

This proves (8). Moreover, for every $y \in O$,

$$\mathrm{Tr}(y) \equiv -\rho(y) \pmod{p}, \tag{11}$$

as follows by expanding y in its integral power basis and using (10). Since $\rho(\zeta^{-j}) = 1$, all p numerators in (8) have the same residue $a - \rho(b)$ modulo p. They are all divisible by p exactly when (7) holds. □

The prime 2 is included without modification. Here $\zeta = -1$, $K = \mathbb{Q}$, and $O = \mathbb{Z}$, so

$$a \equiv b \pmod{2}, \qquad x_0 = (a+b)/2, \quad x_1 = (a-b)/2. \tag{12}$$

There is no extra factor of two in the field trace at this endpoint.

A tuple that does not glue. At $p = 3$, the pair $(a, b) = (1, 0)$ consists of algebraic integers and gives real symmetric Fourier coordinates $(1, 0, 0)$. Its inverse is $(1/3, 1/3, 1/3)$, because the congruence in (7) fails. Thus neither algebraic integrality of each spectral coordinate nor reality of all coordinates is enough.

12.3 Galois coherence, symmetry, normalization, and signs

One field element b encodes every nontrivial frequency:

$$\lambda_0 = a, \qquad \lambda_r = \sigma_r(b) \quad (1 \le r < p). \tag{13}$$

Those coordinates cannot be chosen independently. In particular, if one chooses real values at conjugate frequency pairs without preserving (13), the resulting coefficients may be real but irrational. At $p = 5$, the proposed tuple

$$(\lambda_0, \lambda_1, \lambda_2, \lambda_3, \lambda_4) = (0, 0, 1, 1, 0)$$

is real and invariant under $r \mapsto -r$, and each listed value is an algebraic integer. It cannot be the spectrum of a rational row: $\lambda_1 = 0$ would force every Galois conjugate of that field coordinate to be zero.

Suppose from now on that $a \in \mathbb{Z},\ b \in O$ satisfy (7), so X is already an integral row. The following are exact tests:

$$\begin{aligned} X = X^* &\iff b = \overline{b}, \\ x_0 = 1 &\iff a + \mathrm{Tr}(b) = p, \\ x_j \in \{-1, 1\} \text{ for all } j &\iff a + \mathrm{Tr}(b\zeta^{-j}) \in \{-p, p\} \text{ for all } j. \end{aligned} \tag{14}$$

For the first equivalence, the evaluations of X^* are $(a, \bar{b})$, and rational CRT is injective. The other two statements follow from (8). Since cyclotomic Galois automorphisms commute with complex conjugation, $b = \bar{b}$ also says that every $\sigma_r(b)$ is real.

The trace in (8) is the full trace from K. If p is odd and $b = \bar{b}$, one may instead use $K^+ = \mathbb{Q}(\zeta + \zeta^{-1})$:

$$\mathrm{Tr}_{K/\mathbb{Q}}(b\zeta^{-j}) = \mathrm{Tr}_{K^+/\mathbb{Q}}(b(\zeta^j + \zeta^{-j})). \tag{15}$$

Indeed the relative trace from K to K^+ sums a term and its complex conjugate. The expression $b\zeta^{-j}$ itself need not be real; replacing its trace indiscriminately by twice a real-subfield trace is invalid. Formula (15) does not apply at $p = 2$.

There is a useful substitute for checking all the sign conditions in (14), provided parity and energy are checked exactly.

Lemma 4: spectral parity for odd prime order. For odd p, an integrally glued pair (a, b) has every coefficient x_j odd if and only if

$$a \equiv 1 \pmod 2, \qquad b \in 2O. \tag{16}$$

Proof. If all coefficients are odd, write $X = T - 2N$ with $N \in \mathcal{A}_p$. Since $T(1) = p$ and $T(\zeta) = 0$, (16) follows. Conversely assume (16), and put

$$c = (p - a)/2 \in \mathbb{Z}, \qquad d = -b/2 \in O.$$

Because 2 is invertible modulo p and $\rho(b) = a$, $\rho(d) = -a/2 = (p - a)/2 = c$ in $\mathbb{F}_p$. Theorem 3 therefore glues (c, d) to an integral N. The evaluations of $T - 2N$ equal those of X; injectivity gives $X = T - 2N$, so all its coefficients are odd. □

At $p = 2$, the exact replacement is

$$a + b \equiv a - b \equiv 2 \pmod 4. \tag{17}$$

Merely requiring a, b even fails: $(a, b) = (2, 2)$ glues to $(2, 0)$.

Lemma 5: parity and the minimum energy force signs. If $X_1, \ldots, X_q \in \mathcal{A}_m$ have odd coefficients and

$$\sum_{\ell,j} x_{\ell j}^2 = qm, \tag{18}$$

then every coefficient of every row is 1 or -1.

Proof. There are qm odd integers in the sum; each has square at least one. Equality in their summed lower bound forces every square to be one. □

For one prime-order row, exact Parseval reads

$$a^2 + \mathrm{Tr}(b\bar{b}) = p \sum_j x_j^2. \tag{19}$$

To prove it, expand $\sum_{r=0}^{p-1} |X(\zeta^r)|^2$; the geometric sum kills each term $x_j x_k$ with $j \neq k$, and leaves $p \sum_j x_j^2$. The nontrivial frequencies give the trace term. This is the positive quadratic trace form, not the multiplicative field norm $\mathrm{N}_{K/\mathbb{Q}}(b)$.

Consequently, at an odd prime, an integrally glued pair represents a sign row exactly when

$$a \text{ is odd}, \qquad b \in 2O, \qquad a^2 + \mathrm{Tr}(b\bar{b}) = p^2. \tag{20}$$

Necessity follows from signs, and sufficiency follows from Lemmas 4–5. To represent a **normalized symmetric** sign row, add $b = \bar{b}$ and $a + \mathrm{Tr}(b) = p$. These are finite exact certificates, not an assertion that a tuple with prescribed extra Gram constraints exists.

Energy and every individual spectral bound still do not suffice. At $p = 11$, take

$$x_0 = 1, \qquad x_{\pm 1} = 2, \qquad x_{\pm 2} = -1, \qquad x_j = 0 \text{ otherwise.} \tag{21}$$

This row is integral, normalized, and symmetric. Its augmentation is $a = 3$, so even the necessary normalized sign-row congruence $a \equiv p \pmod 4$ holds. Its energy is exactly $1 + 2 \cdot 4 + 2 \cdot 1 = 11$. Every Fourier value has the form

$$1 + 4\cos\theta - 2\cos 2\theta = 3 + 4c - 4c^2, \qquad -1 \le c \le 1.$$

The concave quadratic has maximum 4 at $c = 1/2$ and minimum -5 at $c = -1$. Thus every Galois conjugate satisfies

$$|\lambda_r|^2 \le 25 < 44,$$

which is stronger than the individual bound required of a row in a four-row order-eleven Gram identity. The pair

$$b = 1 + 2(\zeta + \zeta^{-1}) - (\zeta^2 + \zeta^{-2}), \quad a = 3$$

glues integrally, and (19) has the exact sign-row energy 121. Nevertheless the row is not a sign row. Its failure is visible in parity, not in the energy or these individual spectral inequalities. No completion of this row is being asserted.

12.4 Full inverse Fourier divisibility for composite order

For general m, take one coordinate

$$b_d \in O_d = \mathbb{Z}[\zeta_d] \qquad (d \mid m), \quad O_1 = \mathbb{Z}.$$

Here the standard cyclotomic factorization and irreducibility give

$$\mathbb{Q}[z]/(z^m - 1) \cong \prod_{d|m} \mathbb{Q}(\zeta_d).$$

The new issue is the integral image of this familiar rational decomposition.

Proposition 6: an integral inverse Fourier certificate. Define

$$C_j = \sum_{d|m} \mathrm{Tr}_{\mathbb{Q}(\zeta_d)/\mathbb{Q}}(b_d \zeta_d^{-j}), \qquad 0 \le j < m. \tag{22}$$

Every C_j is an integer. The unique rational inverse tuple has coefficients $x_j = C_j/m$, and the tuple comes from $\mathcal{A}_m$ if and only if

$$C_j \equiv 0 \pmod m \quad (0 \le j < m). \tag{23}$$

Proof. Choose $\zeta_d = \zeta_m^{m/d}$. For a frequency r whose character has order d, write $r = (m/d)u$ with u a unit modulo d, and prescribe the coherent value $\lambda_r = \sigma_u(b_d)$. The zero frequency is b_1. Partitioning

the ordinary inverse Fourier sum by character order gives exactly (22). Orthogonality of the m characters proves that the resulting x_j have the prescribed evaluations. They are rational because (22) is a sum of traces.

Each trace is an integer: multiplication by $b_d\zeta_d^{-j}$ preserves the free integral power-basis lattice O_d, so its matrix has integer trace, equal to the field trace. The coefficient criterion (23) now follows. □

The certificate can contain redundant congruences; its advantage is that it is exact and applies uniformly. It does not replace the task of finding a tuple satisfying sign, symmetry, or Gram constraints. If signs are required, one can check $C_j \in \{-m, m\}$ directly. Alternatively, after (23), check that every C_j/m is odd and use the exact energy certificate of Lemma 5.

For odd m, parity itself can be tested componentwise: an already integrally glued tuple represents $X \equiv T \pmod 2$ if and only if its components agree with those of T modulo $2O_d$. To justify the converse, reduce

$$z^m - 1 = \prod_{d|m} \Phi_d(z)$$

modulo 2. Its derivative is z^{m-1}, relatively prime to the polynomial, so the product is squarefree. Its displayed factors are therefore pairwise coprime in $\mathbb{F}_2[z]$, where ordinary CRT is injective. This argument fails when m is even, precisely as the $p = 2$ example already shows.

The following lattice formulation includes even order and avoids making parity look like a collection of unrelated special cases. Keep the cyclotomic factors **labelled by** d even when two underlying number fields happen to be isomorphic. Set

$$M_m = \bigoplus_{d|m} O_d, \qquad L_m = \operatorname{image}(\mathcal{A}_m \longrightarrow M_m), \qquad J_m = (m, 0, \dots, 0).$$

Thus J_m is the image of T.

Theorem 6A: the affine lattice and energy certificate, all orders. Let $m \geq 1$, and let four tuples $b_\ell = (b_{\ell,d})$ belong to M_m. Assume the exact identities

$$\sum_{\ell=1}^{4} b_{\ell,d}\overline{b_{\ell,d}} = 4m \quad \text{in } \mathbb{Q}(\zeta_d), \qquad d \mid m. \tag{23a}$$

Then the tuples reconstruct four sign rows with summed cyclic Gram $4m\,1$ if and only if

$$b_\ell \in J_m + 2L_m \quad (1 \leq \ell \leq 4). \tag{23b}$$

Symmetry holds exactly when every component is fixed by complex conjugation. Normalization $x_{\ell 0} = 1$ holds exactly when

$$\sum_{d|m} \operatorname{Tr}(b_{\ell,d}) = m. \tag{23c}$$

These last two conditions are additional requirements.

Proof. If $b_\ell = J_m + 2c_\ell$ with $c_\ell \in L_m$, the corresponding row is $T + 2Y_\ell$ with integral Y_ℓ. Its coefficients are therefore odd integers. Apply every cyclotomic embedding to (23a); all Fourier norm equations follow. Fourier injectivity gives the full Gram identity. Its constant coefficient is total energy $4m$, so Lemma 5 forces signs. Conversely every sign row has the form $T + 2Y_\ell$ with integral Y_ℓ, establishing (23b). The symmetry test follows from the involution and injectivity; (23c) is Proposition 6 at $j = 0$. This also covers $m = 1$, where there is only the rational coordinate. □

This certificate requires no separate assumption of integral gluing: membership in the affine lattice already includes it. For practical separation of the two tests, observe that

$$mM_m \subseteq L_m. \tag{23d}$$

Indeed Proposition 6 reconstructs mb with coefficients C_j, which are integers. Thus M_m/L_m is annihilated by m. For odd m, it has no nonzero two-torsion, and consequently

$$b \in J_m + 2L_m \quad \Longleftrightarrow \quad b \in L_m \ \text{ and } \ b - J_m \in 2M_m. \tag{23e}$$

For the reverse implication, put $c = (b - J_m)/2 \in M_m$. Both $2c$ and mc lie in L_m; an integer combination of 2 and m equal to one gives $c \in L_m$. At odd order the component conditions in (23e) are exactly: the augmentation coordinate is odd, and every $d > 1$ coordinate belongs to $2O_d$.

At even order one must retain the actual half-shift test $(b - J_m)/2 \in L_m$. For $m = 4$, take four copies of the integral row $2 \cdot 1 = (2, 0, 0, 0)$. Every row has energy 4, every spectral coordinate is 2, and (23a) holds. Each tuple lies in L_4 and satisfies $b - J_4 \in 2M_4$. Nevertheless its half-shift reconstructs $(1, -1, -1, -1)/2$, so (23b) fails and the rows are not signs. This example does not impose normalization: the constant coefficient is 2, showing again why (23c) is a separate condition.

Integral gluing cannot be dropped at odd order either. At $m = 3$ take augmentation coordinates $(3, 1, 1, 1)$ and nontrivial coordinates $(0, 2, 2, 2)$. These are odd/even as required, satisfy both exact spectral Gram identities, and give individual rational row energies 3. The first row reconstructs $(1, 1, 1)$; each of the other three reconstructs $(5, -1, -1)/3$. Their failure is precisely the prime gluing congruence $1 \not\equiv 2 \pmod 3$.

If a procedure generates coefficient rows as signs from the outset, their spectra automatically satisfy the integral and parity conditions. In that setting the certificate explains existing structure; it is not a new filter or a claimed speedup. No implementation claim about the earlier search scripts is needed for this conditional observation.

12.5 Why separate pairwise compatibility loses a condition at order four

Factor

$$z^4 - 1 = (z - 1)(z + 1)(z^2 + 1).$$

Write the three rational coordinates as

$$a = X(1), \qquad c = X(-1), \qquad b = X(i) = u + iv.$$

The inverse is

$$\begin{aligned} x_0 &= (a + c + 2u)/4, \quad x_1 = (a - c + 2v)/4, \\ x_2 &= (a + c - 2u)/4, \quad x_3 = (a - c - 2v)/4. \end{aligned} \tag{24}$$

It follows that the exact integral image is

$$\boxed{a + c \equiv 2u \pmod 4, \qquad a - c \equiv 2v \pmod 4.} \tag{25}$$

Indeed these congruences make the two displayed plus numerators divisible by four; subtracting $4u$ or $4v$ proves the same for the minus numerators. Necessity is immediate.

The evaluation matrix in coordinates (a, c, u, v) is

$$E_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}, \qquad \det E_4 = -8.$$

Thus the integral image has index 8. This agrees with (5), since the three pairwise resultants each have absolute value 2.

Nevertheless, the common quotient for **each original factor pair** is only $\mathbb{F}_2$. The pairwise compatibility conditions are

$$a \equiv c \equiv u + v \pmod 2. \tag{26}$$

There are two independent mod-two equations, so this pairwise-compatible lattice has index 4, not 8. For example,

$$(a, c, b) = (0, 0, 1 + i)$$

passes (26), but its inverse is $(1, 1, -1, -1)/2$. Even the real tuple $(2, 0, 0)$ passes (26) and reconstructs $(1, 1, 1, 1)/2$.

The missing information has an explicit explanation. Glue a, c first, requiring $a \equiv c \pmod 2$. Their integral polynomial modulo $z^2 - 1$ is $A + Bz$, where

$$A = (a + c)/2, \qquad B = (a - c)/2.$$

The next gluing step is between $z^2 - 1$ and $z^2 + 1$. Its common quotient is

$$\mathbb{Z}[z]/(z^2 - 1, z^2 + 1) \cong \mathbb{F}_2[z]/((z - 1)^2), \tag{27}$$

and the necessary and sufficient condition is

$$b \equiv A + Bi \pmod{2\mathbb{Z}[i]}.$$

Its two coefficient conditions are exactly (25). Replacing the common quotient (27) by its residue field $\mathbb{F}_2$ discards a nilpotent direction. Thus the failure of the naive pairwise test is already visible in a ring of four elements.

Order nine gives another exact scale for the lattice issue:

$$z^9 - 1 = \Phi_1\Phi_3\Phi_9, \qquad |\operatorname{Res}(\Phi_1, \Phi_3)| = 3, \quad |\operatorname{Res}(\Phi_1, \Phi_9)| = 3, \quad |\operatorname{Res}(\Phi_3, \Phi_9)| = 9.$$

For the last equality, $\Phi_9(\zeta_3) = 3$, so its multiplication map on the rank-two lattice O_3 is $3I$. The integral cyclotomic image therefore has index 81. This calculation is distinct from the order-nine compression fixture in Chapter 9: an index of 81 neither proves nor disproves any sign-lifting claim about that fixture.

12.5.1 Prime-power gluing records a truncated Taylor expansion

There is an elementary general version of (27). Let p be prime, $a \geq 1$, $q = p^{a-1}$, and set

$$f = z^q - 1, \qquad g = \Phi_{p^a}(z) = 1 + z^q + \cdots + z^{(p-1)q}.$$

Since $g \equiv p \pmod f$,

$$(f, g) = (f, p), \qquad R/(f, g) \cong \mathbb{F}_p[z]/((z - 1)^q) \cong \mathbb{F}_p[u]/(u^q). \tag{27a}$$

Thus a lower-order integral row $A \in \mathcal{A}_q$ and an upper coordinate $b = B(\zeta_{p^a}) \in O_{p^a}$ glue exactly when their images in this ring agree. This is a direct application of Theorem 1, so it is both necessary and sufficient. The defect of this stage has order p^q.

To give all its conditions without factorial denominators, define the Hasse derivative

$$D^{[k]}\left(\sum_j c_j z^j\right)(1) = \sum_{j\ge k}\binom{j}{k}c_j.$$

The binomial identity gives the exact expansion

$$C(1+u) = \sum_{k\ge 0} D^{[k]}C(1)\,u^k.$$

Therefore the gluing conditions are precisely

$$D^{[k]}A(1) \equiv D^{[k]}B(1) \pmod p, \qquad 0 \le k < q. \tag{27b}$$

They are independent of the chosen representatives: modulo p, $f = (z-1)^q$ and $g = (z-1)^{(p-1)q}$, so adding a multiple of either defining relation does not change the relevant truncated expansion.

The word "jet" here means only this finite list of coefficients of $C(1+u)$ modulo u^q. It adds no constraints beyond equality in (27a). On polynomials of degree less than q, substitution $z = 1+u$ is an invertible triangular integer basis change, with inverse $u = z-1$. Thus the jet congruences and the coefficient congruences in the common quotient are equivalent descriptions.

At order 9, the final stage has $q = 3$, so it remembers the constant, linear, and quadratic coefficients in $\mathbb{F}_3[u]/u^3$. Taking $A = 0$ and $b = (\zeta_9 - 1)^2$ passes the constant and linear tests but fails the quadratic test. At order 8, taking $A = 0$ and $b = (\zeta_8 - 1)^2$ shows why ordinary derivatives are unsafe in characteristic two: every positive ordinary derivative of $(z-1)^2$ vanishes modulo two, as does its value at 1, but its second Hasse derivative is 1.

Iterating the stage indices from 1 to p^a gives

$$[M_{p^a} : L_{p^a}] = p^{1+p+\cdots+p^{a-1}}. \tag{27c}$$

For order 9, the full evaluation lattice has Smith invariants

$$1, 1, 1, 1, 1, 1, 3, 3, 9.$$

For an explicit reduction, write a degree-less-than-nine polynomial uniquely as $B + C\Phi_9$, with $\deg B < 6$, $\deg C < 3$. The Φ_9 coordinate is B; since Φ_9 equals 3 in both lower factors, subtracting the lower evaluations of B leaves the block $3E_3$, where E_3 is the order-three integral evaluation matrix. Its Smith invariants are $1, 1, 3$ (an entry and a two-by-two minor are units, and its determinant has magnitude 3). The full order-nine matrix is therefore equivalent over integral unimodular basis changes to $I_6 \oplus 3E_3$, proving the displayed invariants.

12.6 What the joint Gram equations actually enforce

For integral or real coefficient rows, Fourier evaluation gives

$$\sum_{\ell=1}^{q} X_\ell X_\ell^* = qm\,1 \quad\Longleftrightarrow\quad \sum_{\ell=1}^{q} |\lambda_{\ell,r}|^2 = qm \quad \text{at every frequency } r. \tag{28}$$

The reverse implication uses injectivity of the full complex Fourier map. It requires a coherent spectrum for each row. Independent feasibility intervals at different frequencies do not supply such rows.

For $q = 4$ at prime order, exact algebraic coordinates make (28) particularly economical:

$$\sum_{\ell=1}^{4} a_\ell^2 = 4p, \qquad \sum_{\ell=1}^{4} b_\ell \bar{b}_\ell = 4p \quad \text{in } K. \tag{29}$$

Applying every Galois embedding to the second identity gives every nonzero-frequency equation. For symmetric rows one may replace $b_\ell \bar{b}_\ell$ by b_ℓ^2. One equality proved **as an algebraic equality** in a faithful embedding is enough; testing a numerical approximation or only an inequality at one embedding is not the same thing.

Corollary 7: an exact prime-order four-row sign certificate. Let p be odd, and choose four pairs $(a_\ell, b_\ell) \in \mathbb{Z} \oplus O$. They reconstruct a normalized symmetric Williamson quadruple if and only if all of the following hold:

1. $\rho(b_\ell) = a_\ell \bmod p$ for each row.
2. $b_\ell = \bar{b}_\ell$ and $a_\ell + \mathrm{Tr}(b_\ell) = p$ for each row.
3. Every a_ℓ is odd and every b_ℓ belongs to $2O$.
4. The two exact Gram identities in (29) hold.

Proof. Necessity follows by evaluation of a normalized symmetric sign quadruple. Conversely, condition 1 reconstructs integral rows, condition 2 gives symmetry and normalization, and condition 3 gives odd coefficients by Lemma 4. Condition 4 gives (28), whose constant coefficient is total energy $4p$. Lemma 5 makes every entry a sign. The resulting group-algebra identity is exactly the Williamson condition from Chapter 9. □

This is an equivalent exact formulation, not a relaxation and not an existence proof. Its value is that each obstruction has an identified location: a lattice congruence, a linear condition, parity, or a coupled quadratic identity.

Even the full Gram identity does not force signs without a lower bound on coefficient size. In $\mathcal{A}_3$, put

$$X_1 = 1 - 2z - 2z^2, \qquad X_2 = X_3 = X_4 = 1. \tag{30}$$

All four rows are integral, symmetric, and normalized by $x_0 = 1$. Writing $S = z + z^2$, the relation $S^2 = 2 + S$ gives

$$X_1^2 = 1 - 4S + 4(2 + S) = 9, \qquad \sum_{\ell=1}^{4} X_\ell X_\ell^* = 12.$$

Their prime spectral coordinates are

$$(a_1, a_2, a_3, a_4) = (-3, 1, 1, 1), \qquad (b_1, b_2, b_3, b_4) = (3, 1, 1, 1).$$

They satisfy integral gluing, normalization, symmetry, and both equations in (29), exactly. Yet the coefficient rows are $(1, -2, -2)$ and three copies of $(1, 0, 0)$. Their separate energies are $9, 1, 1, 1$; the example does not claim to enforce energy 3 on each row. Condition 3 of Corollary 7 fails. This example and (21) expose different missing constraints.

More generally, integrality, odd coefficients, and the full identity (28) force signs at **any** order by Lemma 5. Oddness can be weakened to the requirement that all integral coefficients be nonzero, but nonzeroness has a less convenient spectral description.

The nonsymmetric boundary. Without symmetry, the exact norm equations produce complementary circulant sign rows. They do not by themselves justify the standard Williamson block array

displayed in Chapter 9. For that array its block $(1,2)$ in the Gram matrix is

$$-AB^{\mathsf{T}} + BA^{\mathsf{T}} - CD^{\mathsf{T}} + DC^{\mathsf{T}}.$$

Commutation of circulants does not make each term cancel its partner: one sufficient extra condition is pairwise amicability $UV^{\mathsf{T}} = VU^{\mathsf{T}}$. Symmetric circulants satisfy this automatically.

For a concrete check, at length three take first rows

$$A = (1,1,1), \quad B = C = (1,1,-1), \quad D = (1,-1,1).$$

The sum of their cyclic Gram matrices is $12I_3$. With the Chapter 9 convention $P_{ij} = 1$ when $j = i+1$, the standard block array nevertheless has Gram block $(1,2)$

$$\begin{pmatrix} 0 & -4 & 4 \\ 4 & 0 & -4 \\ -4 & 4 & 0 \end{pmatrix} \neq 0.$$

Indeed $D = B^{\mathsf{T}}$, while A commutes amicably with every circulant; the offending block is $(B^{\mathsf{T}})^2 - B^2$. Thus Theorem 6A gives complementarity without symmetry, whereas Corollary 7 deliberately retains symmetry to give a Williamson quadruple and the stated block construction.

12.7 Negative supports retain the discrete information

For actual sign rows define

$$N_\ell = \sum_{x_{\ell j} = -1} z^j, \qquad r_\ell = \epsilon(N_\ell), \qquad R = \sum_{\ell=1}^{4} r_\ell.$$

Then $X_\ell = T - 2N_\ell$. The essential assertion is that the coefficients of N_ℓ lie in $\{0,1\}$; it is **not** the group-algebra equation $N_\ell^2 = N_\ell$. Coefficientwise multiplication and cyclic convolution are different.

Using $T^2 = mT$ and $TN_\ell = N_\ell T = r_\ell T$, expansion gives

$$\sum_{\ell=1}^{4} X_\ell X_\ell^* = 4m\,1 \quad \Longleftrightarrow \quad \sum_{\ell=1}^{4} N_\ell N_\ell^* + (m-R)T = m\,1. \tag{31}$$

No symmetry is required for this equivalence. At a nonzero shift, the equation says

$$\sum_{\ell=1}^{4} |\operatorname{supp}(N_\ell) \cap (\operatorname{supp}(N_\ell) + s)| = R - m. \tag{32}$$

At shift zero the equation holds automatically because the N_ℓ are indicators. Thus (31) is a joint system of cyclic difference counts, not just four independent row conditions.

For odd m, impose symmetry and normalization $x_{\ell 0} = 1$. Each negative support is a union of nonzero pairs $\{j, -j\}$; therefore every r_ℓ is even and R is even. The row sums satisfy

$$\epsilon(X_\ell) = m - 2r_\ell \equiv m \pmod 4, \qquad 2 - m \le \epsilon(X_\ell) \le m. \tag{33}$$

These are necessary conditions. They do not characterize either individual sign rows in spectral space or a joint solution.

Symmetry lets us replace $N_\ell N_\ell^*$ by N_ℓ^2. Reduce (31) modulo two. Since m and $m-R$ are odd,

$$\sum_\ell N_\ell^2 = T+1 \quad \text{in } \mathbb{F}_2[C_m]. \tag{34}$$

In characteristic two, squaring sends $\sum n_j z^j$ to $\sum n_j z^{2j}$. Because m is odd, doubling permutes the residue classes, so this map is an automorphism fixing T and 1. Applying its inverse to (34) yields

$$\sum_\ell N_\ell = T+1. \tag{35}$$

At every nonzero position an odd number of the four entries are negative. Equivalently,

$$x_{1j}x_{2j}x_{3j}x_{4j} = -1 \qquad (j \neq 0). \tag{36}$$

This recovers the odd normalized symmetric Williamson product condition with every ring and divisibility step visible.

The converse is absent: reducing (31) modulo two throws away its integer difference counts. The order-nine fixture in Chapter 9, and its separate continuation analysis, illustrates this loss. The present argument does not need or re-run that enumeration.

At prime order the support coordinates are

$$\epsilon(N) = (p-a)/2, \qquad N(\zeta) = -b/2. \tag{37}$$

Lemma 4 explains exactly when these data glue to an **integral** N. To obtain an actual negative support, its coefficients must also lie in $\{0,1\}$. A useful energy reformulation makes the last issue explicit: if $X = T - 2N$ has integral N, then

$$\sum_j x_j^2 = m \quad \Longleftrightarrow \quad \sum_j n_j(n_j-1) = 0.$$

Every summand on the right is nonnegative for integral n_j; equality is therefore equivalent to $n_j \in \{0,1\}$ for all j. The same proof works after summing over several rows.

12.8 Hypotheses, failed replacements, and bounded conclusions

Hypothesis or exact requirement	Tempting replacement	Counterexample or precise failure
Integral comaximality	Coprimality over $\mathbb{Q}$	$t-1, t+1$ have defect $\mathbb{Z}/2$
Monic/primitive control of divisibility	Arbitrary rationally coprime integral polynomials	$2t, 2(t-1)$: the product ideal is smaller than the intersection
Full Galois coherence	Independently chosen real algebraic-integer eigenvalues	The order-five tuple $(0,0,1,1,0)$ in Section 3
Integral inverse Fourier congruences	Each cyclotomic coordinate is an algebraic integer	$(a,b) = (1,0)$ at $p=3$ gives coefficients $1/3$
Simultaneous multi-factor gluing	All original pairwise tests	$(a,c,b) = (0,0,1+i)$ at order four
Parity plus exact energy	Exact energy, symmetry, normalization, and separate spectral upper bounds	The order-eleven row (21)
Parity plus full Gram	Full Gram, integrality, symmetry, and normalization	The order-three quadruple (30)

Hypothesis or exact requirement	Tempting replacement	Counterexample or precise failure
The affine lattice $J_m + 2L_m$ at even order	Integral gluing and componentwise congruence modulo 2	Four copies of $2 \cdot 1$ at order four satisfy every norm equation but fail the half-shift gluing test
Odd p in the componentwise parity lemma	The same mod-two conditions at $p = 2$	$(a, b) = (2, 2)$ gives $(2, 0)$
Symmetry or sufficient amicability for the stated Williamson array	Summed Gram equations for nonsymmetric circulant sign rows	The explicit order-three array in Section 6 has a nonzero off-diagonal Gram block
Odd m in the normalized product theorem	Arbitrary m	At $m = 2$, rows $(1, 1), (1, 1), (1, -1), (1, -1)$ satisfy the full Gram identity but have product $+1$ at position 1
Normalization in the stated product theorem	Symmetry alone	At $m = 3$, start with $(1, 1, 1)$ and three $(1, -1, -1)$; negate the first entire row. Gram and symmetry survive, while the nonzero-position products become $+1$

None of these examples rules out a Hadamard matrix at any order. They identify failures of specified implications and specified representations. Conversely, Corollary 7 is sufficient precisely because it restores the lost integral and discrete conditions.

12.9 Exercises with answers

Exercise 1. For $f = t - 1,\ g = t^2 + t + 1$, describe the image of $R/(fg)$ in $\mathbb{Z} \oplus \mathbb{Z}[\zeta_3]$. Reconstruct $(a, b) = (1, 1 + \zeta_3)$, and decide whether it is integral.

Answer. Write $b = u + v\zeta_3$. The condition is $a \equiv u + v \pmod 3$. Here $u + v = 2$, so the pair does not glue. Formula (9) gives

$$X = (1 + t) - \tfrac{1}{3}(1 + t + t^2) = \tfrac{2}{3} + \tfrac{2}{3}t - \tfrac{1}{3}t^2.$$

This example also shows why taking a small integral power-basis representative for b is not enough.

Exercise 2. At $p = 5$, reconstruct the pair

$$a = 1, \qquad b = -2(\zeta + \zeta^{-1}).$$

Check symmetry, normalization, and signs by both direct coefficients and the spectral tests.

Answer. The row is $(1, -1, 1, 1, -1)$, since $X = T - 2(z + z^4)$. Its sum is 1, its trace $\mathrm{Tr}(b) = 4$, and $\rho(b) = -4 = 1$ in $\mathbb{F}_5$. Also $b = \bar{b}$, $b \in 2O$, and

$$\mathrm{Tr}(b^2) = 4\,\mathrm{Tr}(\zeta^2 + 2 + \zeta^{-2}) = 4(-1 + 8 - 1) = 24.$$

Thus $a^2 + \mathrm{Tr}(b^2) = 25$, as required by (20).

Exercise 3. Explain why a four-row prime spectral certificate cannot check only $\sum_\ell |b_\ell|^2 \le 4p$ in one complex embedding, even when the b_ℓ are individually integral.

Answer. The proposed inequality imposes neither equality nor the augmentation equation, integral gluing, parity, normalization, or coherent constraints at the other embeddings. The exact identity

$\sum b_\ell \bar{b}_\ell = 4p$ in K transports to all embeddings; a numerical upper bound in one embedding does not. The examples (21) and (30) show that even much stronger subsets of the exact conditions can fail to force signs.

Exercise 4. Prove that the order-four defect has additive group $\mathbb{Z}/4 \oplus \mathbb{Z}/2$, not $(\mathbb{Z}/2)^3$.

Answer. The cokernel of E_4 has order 8. Its first Smith invariant is 1, since it has an entry 1; the second is 1, since a 2×2 minor from rows $1, 3$ and columns $2, 3$ has determinant -1. Every 3×3 minor is even because E_4 has rank 2 modulo 2, and a 3×3 minor (rows $1, 2, 3$, columns $1, 2, 3$) equals 4, while the minor using rows $1, 3, 4$, columns $1, 2, 3$, equals 2. Thus the third determinantal divisor is 2. The Smith invariants are $1, 1, 2, 4$.

Exercise 5. Why does the positivity argument in Lemma 5 require integrality or an equivalent lower bound? Give a real symmetric normalized row of odd length with energy equal to its length that is not a sign row.

Answer. At length 5, take $(1, \sqrt{2}, 0, 0, \sqrt{2})$. It is symmetric and normalized, and its energy is $1 + 2 + 2 = 5$, but its entries are not signs. Without the integer odd/nonzero condition, an entry can have square strictly between zero and one, and the minimum-energy proof does not apply. The example itself also has zero entries and entries of magnitude greater than one.

Exercise 6. Show directly that (35) is weaker than (31), without using a search or the order-nine fixture.

Answer. At odd length $m = 5$, take $N_1 = T - 1$ and $N_2 = N_3 = N_4 = 0$. These are normalized symmetric negative supports, and their sum modulo two is $T + 1$, so (35) and (36) hold. The corresponding row sums are $(-3, 5, 5, 5)$, whose squared sum is 84, not $4m = 20$. Thus the frequency-zero consequence of (31) already fails.

12.10 Verification and source scope

The accompanying checker uses exact integer and rational arithmetic. All twelve check groups passed with SymPy 1.14.0: 129 prime trace reconstructions (including $p = 2$); 150 bounded coefficient parity cases; all component-basis trace inverses at orders $1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12$; 81 order-four congruence cases; 66 prime-power gluing probes; Smith invariants at orders four and nine; and the fixed energy, Gram, affine-lattice, and nonsymmetric-array counterexamples and exercise calculations involving explicit rows. The checks are consistency evidence for this exposition; the general proofs above do not depend on testing finitely many cases.

The strongest formulation here is the all-order affine-lattice certificate in Theorem 6A, with the normalized symmetric prime specialization in Corollary 7. These reformulate classical identities and elementary parity/energy reasoning; no novelty claim is made. Independent mathematical reviews checked the all-order affine theorem, the odd-order torsion argument, prime trace formula, even-order boundary, order-four gluing, prime-power jets, order-nine Smith reduction, and the nonsymmetric array counterexample. A separate computation using Ramanujan sums verified both products of the trace-inverse and evaluation matrices at eleven fixed orders. See the continuation review for the evidence boundary.

For the broader classical integral CRT theory, see Kamalakshya Mahatab and Kannappan Sampath, "Chinese Remainder Theorem for Cyclotomic Polynomials in Z[X]", *Journal of Algebra* 435 (2015), 223–262. Their paper studies the full cyclotomic cokernel and its Smith structure. The elementary special cases and reconstruction certificates used here are proved explicitly above; no claim is made

that the inspected textbook passages already contain them.

13 Compression, orthogonal ideals, and a common-lift obstruction

The earlier Hadamard chapter proves that compression preserves correlation equations. Here we determine exactly what it retains, count the individual sign lifts, and explain the fixed core-nine failure by one polynomial identity. These are three different tasks: lifting each row, satisfying a quotient equation, and lifting all rows so that their full equations hold. Their separation is useful even when every underlying algebra is semisimple.

13.1 The quotient has an explicit orthogonal splitting

Write $m = dr$ with d, r positive integers, and work over the reals:

$$A = \mathbb{R}[z]/(z^m - 1), \qquad B = \mathbb{R}[w]/(w^d - 1), \qquad q(z) = w.$$

The involution is $z^* = z^{-1}$. Put

$$e = \frac{1}{r} \sum_{\ell=0}^{r-1} z^{\ell d}.$$

Each element of the subgroup generated by z^d occurs r times in the square of the numerator, so $e^2 = e$. Negation permutes that subgroup, so $e^* = e$, and $q(e) = 1$. Also $ez^d = e$. It follows that

$$\iota : B \longrightarrow eA, \qquad \iota(w^s) = ez^s$$

is well defined and is inverse to $q|_{eA}$. The identity of eA is e, not 1_A. Since $e(1-e) = 0$, multiplication gives the algebra decomposition

$$A = eA \oplus (1 - e)A, \qquad \ker q = (1 - e)A.$$

It is also orthogonal for the coefficient inner product $\langle X, Y \rangle = [1](XY^*)$. Indeed, $\langle eX, (1 - e)Y \rangle = [1](e(1 - e)XY^*) = 0$.

If $X = \sum_j x_j z^j$ and $y_s = \sum_{j \equiv s \pmod d} x_j$, direct coefficient collection gives

$$(eX)_j = \frac{y_{j \bmod d}}{r}, \qquad \|eX\|^2 = \frac{1}{r} \sum_{s=0}^{d-1} y_s^2.$$

The retained component eX is the mean on each fiber; compression records r times those means. The remaining coefficients record deviations from them; they are not arbitrary once signs and symmetry are imposed.

For four rows put $L_i = eX_i$ and $R_i = (1 - e)X_i$. Cross products vanish, so the full Gram identity $\sum_i X_i X_i^* = 4m1_A$ is equivalent to two independent algebra equations:

$$\sum_i L_i L_i^* = 4me, \qquad \sum_i R_i R_i^* = 4m(1-e).$$

The first is precisely $\sum_i q(X_i)q(X_i)^* = 4m$ in B. There is no extra factor of r in the right side. Taking the coefficient of the identity gives total energies $4d$ and $4(m-d)$ respectively. For example, the first follows from $\sum_{i,s} y_{i,s}^2 = 4m$ and the factor $1/r$ above.

The algebraic decomposition holds for all positive d, r. Oddness, normalization, and signs enter the following discrete problem.

13.2 The exact count of normalized symmetric row lifts

Assume now that m, d, r are odd. A normalized symmetric sign row satisfies

$$x_j \in \{-1, 1\}, \qquad x_{-j} = x_j, \qquad x_0 = 1.$$

An integer row $y \in \mathbb{Z}^d$ has such a lift if and only if:

1. $y_s = y_{-s}$ for all s;
2. for $s \neq 0$, $|y_s| \leq r$ and $y_s \equiv r \pmod 2$;
3. $y_0 = r - 4a$ for an integer $0 \leq a \leq (r-1)/2$.

When these conditions hold, the number of lifts is

$$\binom{(r-1)/2}{(r-y_0)/4} \prod_{s=1}^{(d-1)/2} \binom{r}{(r-y_s)/2}.$$

Proof. Negation exchanges the two nonzero fibers F_s and F_{-s}. Choose $(r-y_s)/2$ negative positions among the r positions of F_s; reflection then fixes the entire opposite fiber. Different pairs of fibers are independent. The zero fiber contains the fixed coordinate 0, with value 1, and $(r-1)/2$ opposite pairs. Making a pairs negative changes its all-positive sum r by $4a$. This proves both necessity and sufficiency, as well as the product count.

Notice that the smallest possible zero-fiber sum is $2-r$, not $-r$. For $d = 1$ the product is empty; for $r = 1$ the conditions say exactly that y itself is the unique normalized symmetric sign row.

This proof gives explicit reconstruction. Choose the negative-position subsets on representative fibers, reflect them, and independently choose the negative pairs on the zero fiber. It does not solve the equations that couple the four rows.

13.3 What remains after subtracting the fiber means

On a nonzero representative fiber, write $x = \mathbf{1} - 2u$, where $u \in \{0,1\}^r$ and $\sum u_j = b = (r - y_s)/2$. Then

$$x - \frac{y_s}{r}\mathbf{1} = -2\left(u - \frac{b}{r}\mathbf{1}\right).$$

These are the translated and rescaled vertices of the hypersimplex

$$\Delta(r,b) = \{u \in [0,1]^r : \sum u_j = b\}.$$

The opposite fiber is determined by reflection, not chosen independently. The zero fiber has one fixed coordinate and duplicated opposite pairs. Its choices are an affine image of the vertices of $\Delta((r-1)/2, a)$; treating it as an unrestricted $\Delta(r,(r-y_0)/2)$ would count forbidden rows. Explicitly, if u records the a negative pairs, the residual coordinate at zero is $4a/r$, and both coordinates of pair j are $4a/r - 2u_j$. Use $\Delta(0,0) = \{()\}$ when $r = 1$.

Consequently the unresolved equation $\sum_i R_i R_i^* = 4m(1-e)$ is imposed on a finite product of specified vertex sets. Real positivity alone does not enforce those sets. Nor does the existence of an individual lift for every compressed row guarantee a simultaneous solution.

There is a particularly transparent real relaxation. Suppose real compressed rows $Y_i \in B$ already satisfy $\sum_i Y_i Y_i^* = 4m$. Set $L_i = \iota(Y_i)$, choose

$$R_1 = 2\sqrt{m}(1-e), \qquad R_2 = R_3 = R_4 = 0,$$

and put $X_i = L_i + R_i$. Then $q(X_i) = Y_i$ and $\sum_i X_i X_i^* = 4m$, because $R_1 R_1^* = 4m(1-e)$. If the Y_i are symmetric, the X_i are symmetric too. This proves existence of an unrestricted real lift, not of normalized sign rows. It explains why an obstruction can disappear completely after replacing a discrete coefficient problem by a real algebra problem.

13.4 A proof of the fixed core-nine obstruction

Consider the four compressed rows, with labels fixed in this order,

$$Y_A = (-1,-1,-1), \quad Y_B = (-1,1,1), \quad Y_C = (-1,3,3), \quad Y_D = (3,-1,-1).$$

They satisfy the compressed Gram equation with target 36. The lift-count formula gives $3, 3, 1, 3$ individual lifts, hence 27 labeled four-row lifts. Since the compressed rows are distinct, there are also 27 unordered four-row multisets; if only their compressed multiset is prescribed and row order matters, there are $24 \cdot 27 = 648$ ordered tuples. These are different counting conventions.

A normalized symmetric row of length nine is determined by positions $1, 2, 3, 4$, and its compression is

$$y_0 = 1 + 2x_3, \qquad y_1 = y_2 = x_1 + x_2 + x_4.$$

Let $P = \{1, 2, 4\}$. On P, rows A, D each have one positive entry, row B has one negative entry, and row C is entirely positive. At position 3 the column is $(-1,-1,-1,1)^{\mathsf{T}}$. Write p, q for the positive exceptions of A, D, and k for the negative exception of B.

The odd Williamson product theorem, proved in the earlier chapter, requires the product of every nonzero column to be -1. If $p = q$, the product of A, D is positive everywhere on P, which would require B to be negative everywhere there. If $p \neq q$, that product is negative at p, q and positive at the remaining position. Thus the product condition holds exactly when p, q, k are the three different elements of P. It rejects 21 lifts and leaves six.

All six fail for the same algebraic reason. In $\mathbb{Z}[z]/(z^9-1)$ put $S_j = z^j + z^{-j}$. Let $\mathbf{1} \in \mathbb{Z}^4$ and $v_i = \mathbf{1} - 2e_i$, with labels $i = A, B, C, D$. Then

$$v_i \cdot v_j = 4[i = j], \qquad \mathbf{1} \cdot v_i = 2.$$

For the surviving pattern, the columns at $p, q, k, 3$ are respectively $v_D, v_A, -v_C, -v_D$. Hence the vector of four row polynomials is

$$\mathbf{X} = \mathbf{1} + v_D S_p + v_A S_q - v_C S_k - v_D S_3.$$

Since each $X_i^* = X_i$ by symmetry, taking its dot product with itself gives the relevant Gram polynomial:

$$\sum_i X_i^2 = 4 + 4(S_p + S_q - S_k - S_3) + 4\sum_{j=1}^{4} S_j^2 - 8S_pS_3.$$

Doubling permutes the four nonzero opposite-index classes modulo nine, so

$$\sum_{j=1}^{4} S_j^2 = 8 + S_p + S_q + S_k + S_3.$$

For $p \in \{1, 2, 4\}$, the two classes represented by $3-p, 3+p$ are exactly the other two members q, k of P. Therefore $S_pS_3 = S_q + S_k$, and substitution gives the precise defect

$$\boxed{\sum_i X_i^2 - 36 = 8(S_p - S_k) \neq 0.}$$

The supports of S_p and S_k are disjoint. The summed autocorrelation is 8 at shifts $\pm p$, -8 at shifts $\pm k$, and zero at all other nonzero shifts. Compression kills this defect: $q(S_p) = q(S_k) = w + w^{-1}$. Thus the remaining obstruction lies exactly in the complementary ideal discarded by the quotient.

This proves that none of the 27 lifts in the specified normalized symmetric family works. The exact checker independently reconstructs the rows and their correlations, including all six polynomial defects. It is a bounded verification of this proof and fixture, not a proof about all Hadamard matrices of order 36 or a new enumeration campaign. The odd-product theorem is classical; see Curtis Bright, "A New Form of Williamson's Product Theorem," Theorem 1. No literature-priority claim is made for the explanatory calculation.

13.5 What the obstruction suggests, and what it does not establish

Compression gives exact necessary equations and an explicit real splitting. Integral Fourier reconstruction gives exact coefficient lattice conditions. The affine-lattice sign theorem in the preceding chapter combines those conditions with parity and energy. Each is a complete statement about its specified data. None proves that feasible partial data have a common completion.

For an algorithm, the remaining question is operational: can the constraints be represented and propagated more cheaply than retaining all original sign variables? Rewriting a condition as an exact

lattice membership test does not answer that question. A useful next experiment would compare two implementations on the same fixed fixtures while recording time, memory, and rejected states. That experiment is outside this foundation study; no improvement in search complexity is claimed.

13.6 Reconstruction exercises with solutions

1. Recover the scale. For $m = 15, d = 5$, a compressed four-row solution has total coefficient energy 60. Find the total energies of its retained and complementary components.

Solution. Here $r = 3$. Fiber averaging divides the retained energy by three, giving $20 = 4d$; the complementary target is $60 - 20 = 40 = 4(m - d)$. The quotient Gram target remains 60.

2. Detect a forbidden zero fiber. Can a normalized symmetric sign row with fiber size five have compressed zero coordinate -5, -3, or 1?

Solution. The possible zero sums are $5 - 4a$ for $a = 0, 1, 2$, namely $5, 1, -3$. Thus -5 is forbidden even though it has the correct ordinary sign-sum parity and bound.

3. Explain a false converse. Every compressed row in the core-nine fixture has a sign lift and the compressed Gram equation holds. Which step fails in inferring a Williamson quadruple?

Solution. Individual choices must also solve a common equation in $(1 - e)A$. The product theorem leaves six choices, each with nonzero defect $8(S_p - S_k)$ in that ideal. Neither individual feasibility nor the quotient equation couples those choices adequately.

4. Separate algebra and arithmetic. Why does the real lift above not contradict the no-sign-lift proof?

Solution. It imposes neither the finite fiber vertex sets nor the fixed coefficient $x_0 = 1$. The real group algebra splits by an idempotent containing $1/r$; the normalized integral sign domain is not the product of unrestricted real subsets in its two ideals.

14 Selective comparison: what can be reorganized, what must be retained

The continuation adds two proof-level supplements: operators and multilinear constructions and arithmetic and representations. They broaden the actual proof reading behind this initial overview and give precise substitution boundaries, dependency risks, and stopping tests. In particular, Xi supplies two Jordan constructions and both Maschke proofs; the supplements distinguish those methods from the graduate texts' routes.

14.1 How to read the comparison

The criterion is the user's intended use: rebuilding algebra around reusable principles, then returning to linear algebra and the Hadamard/Williamson study. "Covered" has several meanings. A theorem may be stated, proved through a different construction, left as an exercise, or illustrated geometrically. These are distinct contributions. A chapter list can establish where to look; it cannot establish that two books are interchangeable.

Evidence levels and edition/page conversions are in the source ledger. The comparisons below are selective, and pedagogical judgments are identified as judgments based on the inspected material. No global difficulty ranking or exercise-by-exercise replacement audit is claimed.

14.2 Xi I versus the graduate algebra books

Xi I starts with equations and low-order determinants, develops sets and maps, then connects vector rows/columns, rank, and linear maps before abstract group/ring/field language. Its Chapter 4 does more than provide a determinant formula: oriented-volume motivation and multilinearity appear on printed pp. 75–76 (PDF pp. 81–82), and §4.4 gives multiple characterizations on p. 93 (PDF p. 99). The surrounding exercises include minor identities. That material has structural value even when its elementary calculations are already familiar.

Hungerford III and Grillet III give a broader ring-theoretic organization of polynomial division and factorization; their later module chapters explain why a PID is the right setting for one-operator structure. The present study therefore reorganizes the **use** of Xi I's polynomial arithmetic into Bézout, Chinese remainder, and annihilator arguments. It does not infer that Xi I's derivations, examples, or root-location exercises are redundant.

Xi I §7.4 starts with symmetry in roots and elementary symmetric polynomials (p. 177, PDF p. 183); §7.6 makes common-factor detection into a resultant matrix (p. 190, PDF p. 196). These are useful links from coefficient calculations to invariant expressions and elimination. Grillet IV.7's resultant/discriminant location is verified from contents (p. 176, PDF p. 188), but its entire proof and exercise treatment was not compared against Xi's. No preference about replacement follows from that heading alone.

For a returning reader who can prove quotient factorization, rank–nullity, determinant multilinearity, Euclidean division, and Bézout, much of Xi I can be used as a reference rather than reread sequentially. That is a competence criterion for selective study, not a judgment that introductory material lacks content.

14.3 Xi II and two routes to Jordan form

There is substantial, verified theorem overlap. Xi II Theorem 2.51 (printed pp. 72–73, PDF pp. 78–79) proves the coprime-annihilator split using Bézout. Theorems 2.52–2.53 build the generalized eigenspaces. Theorems 2.56–2.57 then establish Jordan form under the actual hypothesis that the characteristic polynomial splits in the base field. Existence is proved by quotient induction for nilpotent operators; uniqueness comes from dimensions of images of powers (pp. 75–77, PDF pp. 81–83). These proof pages were read, not inferred from the section title.

Grillet VIII.6 proves the PID structure theorem (pp. 336–342, PDF pp. 348–354) and VIII.7 makes V a $k[t]$-module, proves its annihilator has a monic generator, and derives primary cyclic blocks and Jordan bases (pp. 342–346, PDF pp. 354–358). Its sampled proof gives a real alternative construction, with rational canonical forms and general coefficient rings naturally close at hand. Hungerford's suggested linear-algebra course explicitly takes torsion PID modules before VII.1–5 (printed p. xviii, PDF p. 15); his IV.6 and VII.4–5 locations support the same broad organization. A detailed proof-by-proof audit of Hungerford VII was not made.

For the user's purpose, the module route is an efficient **structural spine**: one remembers a module and its annihilator, then recovers the canonical forms. Xi's route retains pedagogical value: it shows what the projections do inside the actual vector space and constructs chains through quotients. A reader who knows only the abstract theorem should still reconstruct one chain and the kernel-growth count. The two proofs illuminate different failure cases.

14.4 Xi II tensor constructions versus Grillet and Mac Lane

Xi II Definition 6.11 constructs tensor products from a basis criterion, proves uniqueness, and **proves the universal property** in Theorem 6.15 (pp. 273–275, PDF pp. 279–281). It is inaccurate to say that this book has only coordinates or no universal-property treatment. Its Hom identification, scalar extension, functorial maps, Kronecker products, contraction as trace, and contraction as operator composition are explicit examples (pp. 274–283, PDF pp. 280–289).

Grillet XI.5 constructs tensor products by generators and relations for a right module and a left module, proving the balanced universal property (pp. 434–437, PDF pp. 446–449). XI.6–8 adds right exactness, dual modules, and flatness. This is a genuine extension of scope: vector-space bases do not exist for arbitrary modules, and tensoring can destroy injections. The simplest $\mathbb{Z}/2$ counterexample explains why that generality matters.

Mac Lane III.1–2 (pp. 55–62, PDF pp. 68–75) identifies universal arrows, universal elements, and representability, and proves Yoneda. IV.1 (pp. 79–80, PDF pp. 92–93) formulates adjunction, including the Hom–tensor example. Its contribution is a common recognition language, with careful naturality. It does not replace tensor calculations or tell the reader which finite-dimensionality hypotheses can be forgotten.

There is one confirmed scope error: the remark after Xi II Theorem 6.19 (p. 278, PDF p. 284) extends its identification of **all** mixed multilinear forms to infinite-dimensional spaces. The identity-operator/finite-rank counterexample in the Hom chapter disproves that extension for the preceding definition (p. 263, PDF p. 269). This local correction does not invalidate the finite-dimensional construction, its universality theorem, or the chapter's examples. The definition and remark were checked as images.

14.5 Exterior algebra: a real change in construction and scope

Xi II §6.4 begins by explicitly assuming characteristic zero and models exterior powers inside antisymmetric tensors (p. 292, PDF p. 298). Its exterior product uses the alternation operator. This model naturally continues the chapter's tensor calculations and geometric language.

Grillet XIII.2–4 builds tensor, symmetric, and exterior algebras by universal properties and quotient ideals over a commutative ring; the exterior relation is $v \otimes v = 0$ (pp. 518–527, PDF pp. 530–539). The basis and alternating-map arguments were sampled in the actual proof pages. This quotient construction is better suited to retaining characteristic 2 and avoiding factorial denominators. In characteristic zero, the models are connected by normalized alternation; normalizations must be translated.

For determinants and minors, Xi I's characterizations and Xi II's geometric motivation remain helpful. The notes choose the quotient exterior construction so the rank and Cauchy–Binet arguments work in arbitrary field characteristic. That is a decision about the desired scope, not an assertion that Xi's stated characteristic-zero treatment is incomplete on its own terms.

14.6 Metric and geometric material is not superseded by module theory

Xi II Chapter 3 develops real and Hermitian inner products, self-adjoint, isometric and normal operators, complexification, orthogonal expansion, projections, least squares, and orthogonal-polynomial examples. The normal spectral theorem, Theorem 3.62 (pp. 126–128, PDF pp. 132–134), was read with its proof. This supplies an orthonormal conclusion under a positive metric; Jordan form supplies a different conclusion under arbitrary basis changes.

The inspected metric exercises include dual-basis transformations and norm comparisons, providing practice that is not supplied by merely knowing the module structure theorem. Xi's Chapters 4–5 add affine and Euclidean geometry, transformation groups, indefinite spaces, quadrics, and projective geometry. Only their contents and selected contextual passages were audited here, so no claim that a graduate algebra book covers all their proofs or exercises is warranted. Their relevance is already concrete in the Hadamard direction: positive real Gram matrices and exact sign constraints are central, while similarity alone discards the metric problem.

14.7 Xi III, Grillet, and Hall: symmetry with a terminology trap

Xi III has an actual averaging proof for finite complex representations (Theorem 3.20, pp. 109–111, PDF pp. 115–117), followed by examples and remarks about compact groups. Its §3.7 gives dual and tensor representations (pp. 152–162, PDF pp. 158–168). Its module definitions in §4.4 (pp. 186–188, PDF pp. 192–194) connect groups, algebras and actions. The rendered pages show substantive construction coverage.

Grillet IX.7 proves the group-algebra correspondence and Maschke under the field-characteristic condition (pp. 380–384, PDF pp. 392–396). It initially calls a representation "irreducible" when it is **indecomposable** (p. 381, PDF p. 393), explicitly making that correspondence on p. 383. These notes, Xi and Hall use irreducible to mean a nonzero representation with no proper nonzero invariant subspace. Outside a semisimple setting, Grillet IX.7.1 must therefore not be read as an unconditional complete reducibility theorem in the latter terminology. This is a convention difference verified against a rendered page, not an OCR mistake or a false theorem under his definition.

Hall §§4.3–4.7 proves unitary splitting, Schur's lemma, and the $\mathfrak{sl}_2$ classification (pp. 90–101, PDF pp. 103–114). Its unipotent representation of $\mathbb{R}$ demonstrates failure of complete reducibility even over $\mathbb{C}$. Xi III's §4.6 starts from Lie module conditions and explicitly points outside that section for enveloping algebra and PBW theory (pp. 215–216, PDF pp. 221–222). Hall's later PBW, root-system and highest-weight chapters are verified destinations from contents; their full proofs were not compared against Xi's later examples or appendix. Thus "Hall develops a systematic further theory" is supported, while "all Xi representation examples are replaced" is not established.

For Hadamard/Williamson, the immediately useful representation theory is finite cyclic characters and simultaneous splitting of the shift operator. Full compact Lie theory and $\mathfrak{sl}_2$ are enriching branches that can be deferred for that project.

14.8 Homological and geometric graduate books: bridges, not substitutes

Hilton–Stammbach III.1–4 carefully distinguishes extension equivalence from middle-module isomorphism and computes cyclic Ext groups (pp. 84–99, PDF pp. 95–110). Its introduction explains why extensions precede general derived functors. This is the right added construction when invariant splitting fails. Over the field alone, positive Ext/Tor vanish; complex homology remains meaningful. No need for advanced homological machinery has been demonstrated in the sign-lifting constraints of the Hadamard study.

Hartshorne II.5 and II.8 develop local module data and differential universal properties (pp. 108–111, 172–174; PDF pp. 123–126, 187–189). They require sheaves, localization and local rings beyond one-vector-space arguments. His "A" references make clear that some algebra is cited rather than rederived. Lang I.9 and III.1 add projectivity over Dedekind rings and integral trace duality (pp. 29, 57; PDF pp. 43, 71). They require integral and local arithmetic information that scalar extension forgets.

Hirsch IV.2.2 proves splitting for vector bundles on a paracompact base using partitions of unity (p. 95, PDF p. 105). This is an important extension of the linear splitting discussion, while his preface deliberately chooses not to use tensors or differential forms (printed p. vii, PDF p. 8). An algebraic prerequisite ladder that makes him depend on those subjects would misrepresent the actual book.

The practical decision is selective: use Xi for the concrete, metric, and geometric language; choose Grillet or Hungerford for the module spine; use Mac Lane to sharpen universal reasoning and Hilton–Stammbach when a real exactness obstruction appears. Enter Hall, Lang, Hartshorne, and Hirsch along their particular missing bridges. This is a branching reading plan, not eleven sequential complete readings.

15 Proof-level comparison: operators and multilinear constructions

This comparison supports a smaller route through the specified topics, not replacement of any whole book. It is based on newly read contiguous proof passages in the supplied editions: Xi Nanhua, *Basic Algebra*, volume II (Science Press, 2018); Thomas W. Hungerford, *Algebra* (the supplied Springer reprint of the 1974 text); and Pierre Antoine Grillet, *Abstract Algebra*, second edition (2007). Page references below give printed pages first and one-based physical PDF positions second. The offsets at these locations are respectively +6, +19, and +12.

P means the passage itself was read, including its argument; **V** means a rendered page was inspected; **T** means only a location or declared scope was checked. An image request returning a pointer or a warning does not earn V. The new proof readings below are P. The exact glyphs in Xi's quotient-chain correction and normalized-antisymmetrization displays remain **unverified visually** because the available reader returned extracted text and image pointers or explicit image-unavailable warnings; the prescribed download helper was unavailable. The reconstructed arguments are written independently in consistent notation and checked mathematically; they are not transcriptions.

15.1 Three existence proofs, one closely related uniqueness mechanism

Source and inspected passage	Hypotheses and actual construction	What it supplies and what it costs
Xi II §2.4, Theorem 2.44, pp. 68–69; PDF 74–75, P	Finite-dimensional vector space over a field. Proves Cayley–Hamilton by dimension induction. A proper invariant subspace and its quotient give two annihilators whose product kills the whole space. If no proper nonzero invariant subspace exists, every nonzero vector is cyclic; its successive images form a basis, and the companion relation gives the characteristic polynomial.	A linear-algebra proof that precedes Jordan form. Requires quotient operators, the characteristic-polynomial product for a block triangular matrix, and polynomial division, without a PID module classification theorem.

Source and inspected passage	Hypotheses and actual construction	What it supplies and what it costs
Xi II §2.5, Theorems 2.51–2.53, pp. 72–74; PDF 78–80, P	A polynomial fg annihilates T, with $\gcd(f, g) = 1$. Evaluating a Bézout identity splits the space as $\ker f(T) \oplus \ker g(T)$. The proof also proves that the other factor acts invertibly on each kernel. Applying this to a split characteristic polynomial gives generalized eigenspaces and their dimensions.	The essential coprime-primary mechanism already appears in operator language. The two-factor lemma itself does not need eigenvalues or algebraic closure. The subsequent Jordan theorem needs splitting over the ground field.
Xi II §2.5, Theorems 2.56–2.57, pp. 75–77; PDF 81–83, P	Reduces to nilpotence on each generalized eigenspace. Quotients by a line in the kernel, uses induction to obtain chains downstairs, then changes their lifted generators so that at most one chain absorbs the kernel line.	A direct existence proof over any field for a nilpotent operator; split characteristic polynomial suffices for general Jordan form. It teaches the actual obstruction in lifting a quotient basis.
Hungerford IV.6, Lemma 6.8 and Theorem 6.9, pp. 222–223; PDF 241–242, P	For a module killed by p^n, chooses an element of order p^n. An elementary correction produces a nonzero cyclic submodule disjoint from it whenever the module is not cyclic. A maximal disjoint submodule, obtained using Zorn's lemma, is shown to be a complement by applying the same argument in the quotient. Finitely generated primary modules then decompose by induction on their number of generators.	A maximal-order cyclic splitting argument over every PID, not just $K[t]$. The splitting lemma itself permits modules that are not finitely generated, provided their p-power exponent is bounded. That extra scope should not be erased by calling it merely another Jordan proof.
Grillet VIII.6, Theorems 6.1–6.3, pp. 336–339; PDF 348–351, P	First proves submodules of free PID modules free, with a Zorn argument in the arbitrary-rank statement. For finite free F and $M \subseteq F$, maximizes an ideal $\mu(M)$ among images under maps $F \to R$. Bézout combinations force divisibility of all coordinates of a suitable $m \in M$; writing $m = ae$ gives $\mu(e) = 1$. This splits both F and M, and rank induction yields a relative basis $a_i e_i$, with $a_i \mid a_{i+1}$. Taking a free presentation gives invariant factors.	The relative-basis theorem supplies integral presentations and arbitrary-field rational forms in one route. It needs free modules, quotients, principal ideals, the ascending chain condition for ideals of a PID, and rank. For the finite route, the arbitrary-rank Zorn theorem can be replaced by its finite-rank induction.

The two PID treatments also use Bézout to obtain primary decomposition: Hungerford IV.6, Theorem 6.7, p. 222/PDF 241; Grillet VIII.6, Proposition 6.5, p. 340/PDF 352. Their statements concern arbitrary torsion modules over a PID. They factor an annihilator of each individual element; they do not assume one polynomial annihilates the entire module. Xi's finite-dimensional application has a common annihilator available from Cayley–Hamilton. These are the same coprime splitting mechanism at different scopes.

15.1.1 The quotient-lifting step worth retaining

Here is a reconstruction of Xi's existence mechanism, with the delicate exponent made explicit. Let N be nilpotent and choose $0 \neq w \in \ker N$. In V/Kw, take chains with lifted top vectors e_j and lengths m_j. Their defects satisfy

$$N^{m_j} e_j = \alpha_j w.$$

If every $\alpha_j = 0$, the lifted chains together with w already give a chain basis. Otherwise choose r with $\alpha_r \neq 0$ and m_r maximal among indices having nonzero defect. Keep e_r, extend its chain by one place, and, for $j \neq r$ with $\alpha_j \neq 0$, replace

$$e_j \quad \text{by} \quad e'_j = e_j - \frac{\alpha_j}{\alpha_r} N^{m_r - m_j} e_r.$$

The exponent is nonnegative because of the choice of r, and $N^{m_j} e'_j = 0$. These are invertible elementary changes among the old chain vectors; $w = \alpha_r^{-1} N^{m_r} e_r$ is now in the extended chain. This explains both why a quotient decomposition need not lift unchanged and how the correction repairs it. It is a useful finite-dimensional precursor to extension problems.

This construction and Hungerford's maximal-order splitting should not be conflated. Xi starts from an arbitrary one-dimensional submodule and repairs the lifted chains. Hungerford first chooses a cyclic submodule of maximal possible order and proves that this chosen submodule splits. The false assertion that **every** cyclic invariant subspace has an invariant complement is used by neither proof.

15.1.2 A second Xi construction missed by a theorem-only comparison

Xi continues beyond the existence theorem. On pp. 79–81/PDF 85–87, P, he constructs chains from

$$W_k = \operatorname{im} N^{k-1} \cap \ker N.$$

One starts with a basis of the deepest nonzero W_k, extends it through the descending filtration, and chooses preimages under the appropriate powers of N. Applying high powers of N to a putative dependence eliminates successive coefficients. The number of vectors is justified by

$$\dim W_k = \dim \ker N^k - \dim \ker N^{k-1},$$

using the surjection $N^{k-1} : \ker N^k \to W_k$, and then telescoping. This supplies a practical basis-finding procedure and an independence proof. The PID theorem's existence statement alone does not replace that skill.

15.1.3 Exactly how uniqueness agrees

Xi pp. 77–78/PDF 83–84, P, counts blocks by differences of image dimensions. Grillet VIII.6, Lemma 6.7 and the completion of Theorem 6.3, pp. 340–342/PDF 352–354, P, first counts primary cyclic summands using M/pM over $R/(p)$, then applies induction to pM. Hungerford IV.6, Lemmas 6.10–6.11 and Theorem 6.12, pp. 224–225/PDF 243–244, P, explains how to adapt the earlier abelian-group uniqueness argument. That referenced argument was also read: II.2, Theorem 2.6, pp. 78–81/PDF 97–100, P. It detects the number of summands through p-torsion, then multiplies by a power of p to contradict the first discrepancy between exponent lists. Over a general PID, residue-field dimension replaces finite cardinality.

An economical common formulation, reconstructed from these arguments, is this. For a finite primary module $M \simeq \bigoplus_j R/(p^{a_j})$, set

$$d_k = \dim_{R/(p)} p^{k-1} M / p^k M.$$

Each summand contributes one dimension precisely when $a_j \geq k$. Thus $d_k - d_{k+1}$ is the number of summands of exact exponent k. For $R = K[t]$ and $p = t - \lambda$, this becomes Xi's rank formula

$$b_k = \operatorname{rank}(T - \lambda I)^{k-1} - 2\operatorname{rank}(T - \lambda I)^k + \operatorname{rank}(T - \lambda I)^{k+1}.$$

Other primary components contribute equal ranks at all three stages and cancel. Therefore three full uniqueness proofs are unnecessary for theorem coverage; one proof of the residue-field layer count, followed by this translation, supplies the common invariant. For an irreducible polynomial of degree greater than one, K-dimensions of the layers must be divided by $\deg p$; pretending that every irreducible is an eigenvalue loses the rational case.

15.2 The replacement map has a genuine boundary

Grillet VIII.7, pp. 342–346/PDF 354–358, P, constructs the $K[X]$-action, identifies invariant subspaces as submodules, obtains a nonzero annihilator from finite dimensionality of $\operatorname{End}_K(V)$, and then applies the elementary-divisor theorem. Polynomial division proves the cyclic basis; replacing T by $T - \lambda I$ converts it into a Jordan chain. His displayed main Jordan theorem assumes an algebraically closed field. Its proof also works when the relevant minimal polynomial splits, but that weaker hypothesis is a mathematical sharpening of the displayed statement.

Hungerford VII.4, pp. 355–365/PDF 374–384, P, carries this translation through both invariant factors and elementary divisors. It proves rational canonical form, primary rational form, and the split Jordan specialization. Proposition 4.9, pp. 363–364/PDF 382–383, is an additional computational bridge: it presents the operator module by $tI - A$, proves exactness by reducing polynomial vectors to constant vectors, and identifies the nonunit Smith entries as the invariant factors. This is more than merely renaming Jordan blocks.

Learning goal	Coverage that may be compressed after mastery	Material still needed
Primary decomposition when an annihilator is known	Xi 2.51 and the finite $K[t]$-specializations of the two PID primary-decomposition proofs	One explicit Bézout projector calculation; the distinction between coprime and noncoprime factors; the elementwise formulation for arbitrary torsion modules if that scope is wanted
Jordan existence over a splitting field	One of the two existence routes suffices for the theorem	Xi's lift-correction argument is a valuable alternate proof; his filtration method should remain a basis-construction exercise
Jordan uniqueness and block recovery	The three filtration proofs can be replaced by one common layer argument and translation	Actual rank/nullity calculations; constructing a conjugating basis; field and splitting hypotheses
Rational form over an arbitrary field	Xi's split Jordan proof is insufficient	A PID cyclic-decomposition theorem plus polynomial division; a nonsplit example; the $tI - A$ presentation if effective invariant-factor computation is required

Learning goal	Coverage that may be compressed after mastery	Material still needed
Finitely generated modules over a PID	Reading both complete existence proofs is unnecessary for the same classification theorem	Choose one proof; retain Hungerford's bounded-primary splitting lemma as an optional stronger result and Grillet's relative-basis theorem when integral presentations are needed
Cayley–Hamilton	Once independently proved, repeated derivations can be shortened	Preserve a coherent dependency order: Xi proves it **before** Jordan; Hungerford derives it from rational form in VII.5, p. 367/PDF 386; Grillet derives it after Jordan and passes to an algebraic closure, pp. 345–346/PDF 357–358

In particular, a route that uses Xi's characteristic-polynomial splitting argument to prove Jordan and then uses Jordan as its only proof of Cayley–Hamilton would be circular. One may instead follow Xi's preceding quotient proof, or follow the PID route and establish torsion through linear dependence among $I, T, T^2, \ldots$.

The exercises are not interchangeable. Xi §2.5, pp. 83–84/PDF 89–90, P, includes explicit Jordan bases, powers, matrix-root equations, the polynomial semisimple/nilpotent decomposition, a commutator condition forcing nilpotence, and the finite exponential/logarithm correspondence. Hungerford VII.4 exercises, pp. 365–366/PDF 384–385, and VII.5 exercises, pp. 369–370/PDF 388–389, P, add invariance of invariant factors under field extension, descent of similarity, rational-form classification with nonsplit polynomials, a commutative-ring Cayley–Hamilton exercise, and norm/trace connections. These are separate destinations, even where some standard tasks overlap. This reading does not claim that every exercise in the chapters has been solved or that every application occurs in only one book.

15.3 Tensor products: the universal property is already proved in Xi

Xi II §6.2, Definition 6.11 and Theorem 6.15, pp. 273–275/PDF 279–281, P, defines a tensor product by a bilinear map whose pairs of basis vectors form a basis. He constructs such a space, proves uniqueness by the basis map, and proves the universal property by assigning the images of those basis tensors. This applies to arbitrary vector spaces with bases: the relevant expansions have finite support. The finite-dimensional qualifications on the adjacent Hom-and-dual identifications should not be transferred to the universal property itself.

Hungerford IV.5, pp. 207–211/PDF 226–230, P, and Grillet XI.5, pp. 434–438/PDF 446–450, P, build the tensor product of a right and a left module over an arbitrary ring. Both start with the free abelian group on pairs and quotient by additivity and balancing relations. A balanced map extends to the free group, kills the relation subgroup, and therefore descends uniquely. Grillet motivates this with the same vector-space basis calculation that Xi uses.

The additional content is the applicability to modules without bases, noncommutative sidedness, and the subsequent exactness theory. It is inaccurate to say that one needs a graduate text merely to obtain tensor universality. Once the quotient construction and its factorization proof have been learned in either graduate source, the other construction proof is logically redundant. Xi's coordinate realization is still useful as the explicit free case.

Hungerford's right-exactness proof, p. 210/PDF 229, P, deserves separate retention. Given a surjection $B \to C$, it constructs the inverse map from $D \otimes C$ into the quotient of $D \otimes B$ by the preceding image; independence of a chosen lift is exactly what the quotient enforces. This teaches a use of universality that is not supplied merely by the statement that basis tensors form a basis.

15.4 Exterior algebra: related bases, different constructions and scope

Xi II §6.4, pp. 292–299/PDF 298–305, P, assumes characteristic zero and finite dimension at the section opening. He realizes exterior powers inside tensor powers as alternating tensors and defines the product by normalized antisymmetrization. The associativity proof, pp. 293–295/PDF 299–301, proves that antisymmetrizing within a block and then globally agrees with global antisymmetrization. The ordered-wedge basis proof, pp. 296–297/PDF 302–303, first eliminates repetitions and sorts indices for spanning, then isolates the coefficient of each increasing tensor word for independence. This is a complete basis proof, not just a dimension assertion.

Grillet XIII.4, pp. 524–527/PDF 536–539, P, instead takes

$$\Lambda(M) = T(M)/(v \otimes v : v \in M)$$

over a commutative ring with identity. Universality follows by killing the defining ideal. He identifies the degree-n relation submodule with the span of tensors having repeated entries and proves the universal property for alternating n-linear maps. For free M, pp. 525–526/PDF 537–538 prove the ordered-wedge basis by describing relations among tensor basis words, grouping the terms caused by repeated general vectors, and splitting off a free complement spanned by increasing words.

Thus both sources prove spanning **and** independence. Xi's independence comes from coefficients in an alternating subspace; Grillet's comes from an explicit quotient and a complement to its relation submodule. Grillet does not need division by $n!$, characteristic zero, or finite rank for the ordered-basis statement.

A concise bridge between them is an original deduction: when $n!$ is invertible, normalized antisymmetrization descends from the quotient exterior power to the alternating subspace. Ordered wedge classes map to normalized alternating basis tensors, so the map is an isomorphism in the free finite-dimensional case. Xi's product is normalized accordingly. This does not justify identifying the quotient with an alternating subspace in arbitrary characteristic. In characteristic two, the equation $u \otimes v = -v \otimes u$ alone does not impose $u \otimes u = 0$; the quotient construction must retain the latter relation.

For a general algebra route, Grillet's quotient construction can be the single main proof. Retain Xi's alternating realization as a characteristic-zero model and as practice in coefficient calculations. Both sources connect the top exterior power with determinants; the proof of the ordered basis does not by itself replace exercises with minors, decomposability, or geometric interpretation. No absence claim about exterior algebra in Hungerford is made: a literal search found no match, which is not proof of noncoverage.

15.5 A smaller coherent route and stopping tests

For the intended bridge from undergraduate linear algebra to modules, use this order:

1. Learn invariant subspaces, quotients, polynomial evaluation, and Bézout. Prove the two-factor kernel decomposition and compute its projectors. If using Xi's route, include the pp. 68–69 Cayley–Hamilton proof before applying the characteristic polynomial.

2. Work through Xi's quotient-lifting Jordan construction once, then prove the common layer-count uniqueness formula. Construct at least one Jordan basis by the W_k filtration. Further repetitions of Jordan existence and uniqueness may then become short translations.
3. Learn the operator/$K[t]$-module dictionary. For broader classification, use **one** finite PID structure proof: Grillet's relative-basis proof is the natural choice when integer lattices, presentations, and rational form are downstream goals. Translate the result to operators with Hungerford VII.4, including the $tI - A$ presentation. Hungerford's complete alternative IV.6 proof becomes optional proof comparison; its stronger bounded-primary lemma remains available when that generality is needed.
4. Use one balanced-tensor quotient construction and its universal property. Derive the free-basis model, induced maps, and a right-exactness example. Then use the exterior quotient and ordered-basis proof, followed by the characteristic-zero alternation comparison.

These are mastery gates, not estimated reading times:

- **Splitting gate:** for coprime f, g, construct both projectors from a Bézout identity and check their images, kernels, sum, and idempotence. Explain why a vector-space complement need not be T-invariant.
- **Chain gate:** reconstruct the correction e_j' without assuming a quotient basis automatically lifts to chains. From nullities distinguish nilpotent block sizes $(4, 2, 2)$ and $(4, 3, 1)$, although dimension, characteristic polynomial, minimal polynomial, and eigenspace dimension agree. Their second nullities are respectively 6 and 5.
- **Rational gate:** treat $K[t]/((t^2 + 1)^2)$ over $K = \mathbb{R}$, then after extending to $\mathbb{C}$. Explain which invariant factors remain the same and which elementary divisors split. Produce a companion matrix and a valid module presentation; merely writing a complex Jordan matrix does not pass.
- **Tensor gate:** define a map out of a tensor product by checking balancing and uniqueness; compute $(R/I) \otimes_R M \simeq M/IM$. Explain the failure of injectivity after tensoring $2\mathbb{Z} \hookrightarrow \mathbb{Z}$ with $\mathbb{Z}/2\mathbb{Z}$.
- **Exterior gate:** prove ordered wedges independent, not just spanning. Explain the role of $n!$ in Xi's model, and construct $\Lambda^2(K^2)$ in characteristic two using the repeated-vector relation.

A reader who passes these gates can compress repeated theorem coverage in the inspected portions. The route does not replace metric/spectral geometry, affine or projective geometry, tensor contraction practice, the remaining exercise collections, general ring and module theory, or homological algebra. Its economy comes from choosing one proof per common structural theorem and keeping the alternate constructions that teach a distinct method.

15.6 Evidence boundary

The newly inspected operator passages are Xi II PDF 74–90, Grillet PDF 348–358, Hungerford PDF 237–245 and 374–390, with the referenced Hungerford uniqueness passage PDF 97–100. The new multilinear passages are Xi II PDF 279–281 and 298–305; Grillet PDF 446–450 and 536–539; Hungerford PDF 226–230. The comparison does not promote these ranges into an audit of the surrounding books. Source files and retrieval identifiers are private and are not included here.

16 A selective reconstruction of Xi I and III: proof routes, prerequisites, and stopping tests

This comparison asks a local question: which already familiar parts of Xi can a returning undergraduate reconstruct from a smaller collection of proofs, and what would that reconstruction omit? The answer is conditional on competence. It is not a replacement claim about any complete book.

The broadened source reading changes two tempting conclusions. Xi III does **not** stop at the complex inner-product proof of complete reducibility: it also gives the general averaged-projection proof of Maschke. Xi III does **not** introduce modules only as terminology: its module section proves the submodule theorem over a PID and uses a determinant argument to characterize integral elements. A graduate text is useful where it supplies a missing proof or a reusable generalization; its higher placement in a curriculum does not make every local proof shorter.

Evidence follows the existing ledger: **P** = actual statement/proof text read; **V/P** = rendered source page and proof inspected; **T** = contents/location only. Physical PDF pages are one-based. The checked body offsets remain Xi I/III +6, Grillet +12, Hungerford +19, and Lang +14. The supplied Xi III scan has 361 physical pages, although the Library text reader reports only 150. Local rendering recovered the relevant pages; source bytes and rendered evidence remain private.

16.1 The resultant: Xi's matrix construction already supplies an organizing principle

Xi I §7.6, printed 190–193 / PDF 196–199, V/P. Definition 7.76 starts with the Sylvester determinant and permits zero leading coefficients in the displayed, fixed-degree coefficient lists. Proposition 7.77 does not merely assert that a resultant detects a common factor. Its proof first converts the question into a bounded-degree relation

$$fu + gv = 0, \qquad \deg u < m, \quad \deg v < n,$$

where the displayed bounds for f, g are n, m. Equating coefficients gives a square linear system whose determinant is the resultant, up to the transpose convention visible in the proof. Euclid's lemma supplies the reverse implication when f, g are coprime. Thus the proof already connects polynomial divisibility, a kernel, and determinant singularity.

There is a real hypothesis issue here, rather than a cosmetic convention. With the nominal coefficient lengths fixed, Xi's criterion includes the alternative that **both displayed leading coefficients vanish**. For instance, if $f = x + 1$ and $g = x + 2$ are both padded to nominal degree two over $\mathbb{Q}$, the resulting determinant is zero even though they are coprime. A reconstruction that silently replaces nominal degrees by actual degrees changes the statement.

Proposition 7.78, printed 191–192 / PDF 197–198, then obtains the root-product formula by replacing the roots of f by independent indeterminates and studying the polynomial $\mathrm{Res}(F, g - y)$ in y. Its roots are the values $g(x_i)$; specialization gives the desired identity. The proof uses polynomial arithmetic in a rational-function field, not a prior theory of norms or resultants. Proposition 7.79 derives the discriminant identity. The exercises at printed 193 / PDF 199 include multiplicativity, the discriminant of a product, cyclotomic-looking examples, and elimination in two variables. Reading only the opening matrix definition misses these connections.

Grillet IV.7, printed 176–179 / PDF 188–191, P. The direction is reversed. Grillet defines the resultant by the product of differences between roots in an algebraic closure. Proposition 7.2

then proves that it equals the Sylvester determinant, assuming the leading coefficients are nonzero. The proof introduces universal coefficient and root variables over $\mathbb{Z}$, shows that each root difference divides the specialized determinant, controls the total degree, and determines the remaining integer constant. The root-product definition makes the vanishing criterion immediate; the longer work is proving that the invariant is a coefficient polynomial.

These are overlapping theorems with different constructions. Xi offers the more immediate route from a concrete linear system to elimination. Grillet offers a universal polynomial identity that survives specialization. Neither route supplies the other perspective merely by stating the same resultant formula. Hungerford was searched for this topic but no exact counterpart was established; that is an access/coverage limit, not evidence that the book contains no relevant material.

A compact reconstruction. Define

$$S_{f,g} : k[x]_{<m} \oplus k[x]_{<n} \longrightarrow k[x]_{<m+n}, \qquad (u, v) \longmapsto fu + gv$$

for nonzero polynomials of actual degrees n, m. Identify its matrix with the Sylvester matrix, with a recorded ordering/sign convention. Prove $\ker S_{f,g} \neq 0$ exactly when $\gcd(f, g) \neq 1$. Then derive the product formula using either Xi's generic-root argument or Grillet's universal identity. This is a reorganization of inspected proofs, not a newly discovered structure absent from Xi.

The following are useful stopping tests before skipping these pages:

1. Derive the matrix from the map rather than memorizing its rows, and explain why the domain and codomain have the same dimension.
2. Produce the padded-degree counterexample above and state the corrected criterion.
3. Explain why $x-1$ and $x+1$ are comaximal in $\mathbb{Q}[x]$ but not in $\mathbb{Z}[x]$. Their resultant is a nonzero nonunit over $\mathbb{Z}$; evaluation at 1 and -1 sends $\mathbb{Z}[x]/(x^2-1)$ to the pairs of integers with equal parity, not to all of $\mathbb{Z}^2$.
4. Recover the discriminant identity, including its sign, and compute one cubic directly.

Narrow, visually verified erratum. Xi I Example 7.80, printed 193 / PDF 199, displays $-2p^3 - 27q^2$ for the discriminant of x^3+px+q. The correct expression is $-4p^3-27q^2$. The special case x^3+x, with roots $0, i, -i$, gives discriminant -4, whereas the displayed expression gives -2. The general cubic exercise on printed 190 / PDF 196 has the correct coefficient $-4a_0a_2^3$. This is a local example error; it does not invalidate the inspected determinant criterion or product-formula argument.

16.2 Finite fields: the short route must include construction and subfields

16.2.1 What the inspected books actually prove

Xi III §1.4 Theorem 1.21, printed 34 / PDF 40, V/P. The finite multiplicative subgroup theorem uses the maximum order of an element and the polynomial root bound. The step asserting that every element is a root of $X^r - 1$, with r the maximum order, uses the preceding finite-abelian-group structure, or can be supplied independently by the elementary least-common-multiple lemma below. It is not valid for arbitrary finite groups merely because r is a maximum.

Xi III §5.6, printed 260–266 / PDF 266–272, V/P. Theorem 5.40 recalls multiplicative cyclicity. Theorems 5.41–5.43 establish the prime-power order, uniqueness, and existence. Existence takes a splitting field of $X^{p^n} - X$, uses derivative -1 for distinct roots, and verifies that the roots form a subfield. Proposition 5.44 at printed 261 / PDF 267 proves the subfield criterion directly: necessity

uses the tower dimension; sufficiency uses divisibility of $X^{p^r} - X$ into $X^{p^n} - X$ when $r \mid n$. Theorem 5.45 gives extensions of any finite degree.

At printed 262 / PDF 268, Xi identifies the automorphism group using Frobenius and the already established Galois theorem. It then obtains irreducible polynomials in every degree. Printed 262–266 / PDF 268–272 develops Möbius inversion, cyclotomic polynomials, and the count of monic irreducible polynomials over $\mathbb{F}_q$. Thus "replace §5.6 by a finite-field existence theorem" would omit much of the section. The exercises beginning on printed 266 / PDF 272 require minimal polynomials and explicit factorizations, not just theorem recall. The remaining exercise pages 267–269 were not audited.

Grillet IV.1.6, printed 157 / PDF 169, P. Cyclicity is proved by decomposing a finite abelian group into its prime-primary subgroups. In each subgroup, a maximal-order element and the root bound force the whole subgroup to be cyclic; the relatively prime orders then combine. This avoids needing the full invariant-factor classification at that point, but it still relies on the earlier primary decomposition.

Grillet V.1.3, printed 192–193 / PDF 204–205, P. Existence and uniqueness again use $X^{p^n} - X$, derivative -1, root-set closure, and uniqueness of splitting fields. Subfield existence/uniqueness, the divisibility criterion, and the intersection formula are **exercises** on printed 193, not locally supplied proofs. A "Grillet replaces Xi §5.6" claim must either solve these exercises or admit that it has traded Xi's written proof for work assigned to the reader.

Hungerford V.5, printed 279–282 / PDF 298–301, P. Theorem 5.3 proves cyclicity using the invariant-factor theorem for finite abelian groups: all elements satisfy the polynomial determined by the largest invariant factor, and the root bound rules out more than one nontrivial factor. Proposition 5.6 and Corollary 5.7 provide the same splitting-field construction and uniqueness. Frobenius is introduced in Lemma 5.5; Proposition 5.10 identifies the cyclic Galois group using the fundamental theorem of Galois theory. The construction of fields of orders 9 and 25 is assigned as Exercise 5. Irreducible-polynomial existence is also assigned as a short proof. A local shortcut is possible, but reading these paragraphs as written imports both finite-abelian classification and general Galois theory.

16.2.2 A smaller undergraduate sequence, with its imports made explicit

For applications needing only finite fields, their cyclic multiplicative groups, Frobenius, and subfields, the following sequence is sufficient:

1. **Polynomial arithmetic and one-root adjunction.** Prove the field property of $k[x]/(h)$ for irreducible h, the degree/root bound, and the derivative criterion. Construct a splitting field of a fixed polynomial by finitely many successive adjunctions. Prove uniqueness of this finite splitting field by extending an isomorphism one root at a time. An algebraic closure, Zorn's lemma, and the full Galois correspondence are unnecessary for this finite task.
2. **Root-set field construction.** Carry out the proof for $X^{p^n} - X$ completely. Do not start by choosing an irreducible polynomial of degree n unless its existence has already been proved; that would conceal the main existence problem.
3. **Cyclicity from element orders.** In a finite abelian group, elements of orders a, b yield an element of order $\operatorname{lcm}(a, b)$: for each prime, retain from one of them a power with the largest required prime-power order, then multiply these commuting components of pairwise coprime orders. Iteration shows that the exponent is attained by an element. Applied to a finite multiplicative subgroup of a field, the root bound for $X^e - 1$ gives $|G| \leq e$, while the element

of order e gives $e \le |G|$. No classification theorem is needed.

4. **Subfields by polynomial divisibility.** Use Xi's direct proof of Proposition 5.44. Uniqueness follows because a subfield of order p^r must be exactly the roots of $X^{p^r} - X$ inside the larger field. Intersections follow by the same root-set description or the Euclidean algorithm for these polynomials.
5. **Frobenius without full Galois theory.** In $\mathbb{F}_{p^n}$, Frobenius has order n: its n-th power is the identity and a smaller positive power cannot fix all p^n elements by the root bound. Choose a generator α of the multiplicative group, so the field is $\mathbb{F}_p(\alpha)$. Its minimal polynomial has degree n; an automorphism is determined by its image of α, with at most n choices. The n Frobenius powers therefore exhaust the automorphisms.

This sequence is genuinely smaller in prerequisites than importing a full abstract-algebra chapter. It is not much smaller than the relevant core pages of Xi once those earlier facts are available. It also does not replace the Möbius-counting and cyclotomic portions.

Stopping tests. Construct $\mathbb{F}_9 = \mathbb{F}_3[u]/(u^2 + 1)$; prove the defining polynomial irreducible; show $1 + u$ has order eight; compute Frobenius on $a + bu$. Explain why $\mathbb{F}_{p^6}$ has a unique subfield of order p^2 and none of order p^4. For the counting portion, independently derive

$$q^m = \sum_{d|m} dN_d, \qquad N_m = \frac{1}{m} \sum_{d|m} \mu(m/d) q^d,$$

where N_d counts monic irreducible degree-d polynomials. A reader who can only quote cyclicity and field uniqueness has not yet passed these construction tests.

16.3 Representations and modules: Xi already contains both averaging proofs

Xi III §3.2, printed 109–115 / PDF 115–121, V/P. Theorem 3.20 averages a positive Hermitian form to make a finite-dimensional complex representation unitary. The accompanying remarks distinguish real orthogonal representations, compact-group integration, and the practical problem of finding a useful matrix change of basis. Example 3.22, printed 111–113 / PDF 117–119, explicitly splits the permutation representation of S_3, writes the two-dimensional matrices, and changes bases to geometric and eigenvector coordinates.

The crucial additional pages are printed 113–115 / PDF 119–121. Theorem 3.24 states complete reducibility in characteristic zero or characteristic not dividing $|G|$, and gives **two proofs**:

- Over $\mathbb{C}$, the averaged Hermitian form makes the orthogonal complement of an invariant subspace invariant.
- Over any allowed field, start with an arbitrary vector-space projection θ onto the invariant subspace U, and average its conjugates:

$$P(v) = |G|^{-1} \sum_{g \in G} g\, \theta(g^{-1} v).$$

  Xi proves $P|_U = 1$, constructs $V = U \oplus \ker P$, and checks invariance of the kernel by reindexing. The text then distinguishes the nonuniqueness of individual summands from the later uniqueness of isotypic data.

Therefore a comparison based only on printed 109–111 would understate Xi's algebraic proof content.

Grillet IX.7, printed 381–386 / PDF 393–398, P. The group algebra is constructed from the group basis and its universal extension property. Propositions 7.2–7.3 prove the correspondence between representations and modules, including equivalence/isomorphism and invariant-subspace/submodule compatibility. Theorem 7.4, printed 383–384 / PDF 395–396, averages a projection just as Xi does, with an equivalent placement of inverses. Its additional organization is to conclude semisimplicity of $k[G]$, and then use the preceding theory of semisimple rings for classification consequences.

There are two different reading tasks here. The local representation/module dictionary and projection proof require only basic linear algebra, finite sums, module definitions, and invertibility of $|G|$. The subsequent matrix-algebra decomposition, counts of irreducibles, and center arguments invoke the developed semisimple-ring theory. They do not arrive for free from the one-page averaging proof.

Terminology must also be translated: Grillet's definition on printed 381 uses "irreducible" for what standard module language calls **indecomposable**. The text explicitly identifies this with indecomposability on printed 383. This does not contradict its results, but it can produce false comparisons with Xi's absence-of-proper-invariant-subspaces definition unless the semisimplicity hypothesis has already been imposed.

Hungerford IX.5, Proposition 5.8, printed 454–455 / PDF 473–474, P. This is a materially different proof, labeled a sketch. The regular representation realizes elements of the group algebra as matrices. An element of the Jacobson radical can be normalized to have identity coefficient one. The radical is nilpotent because the algebra is Artinian, so its matrix has trace zero; the regular-representation calculation instead makes that trace $|G|$, giving a contradiction when $|G| \neq 0$ in the field. For the converse, the sum of all group elements generates a nonzero square-zero ideal when the characteristic divides the group order.

This is an efficient synthesis **after** radicals and Artinian ring theory. It is not the shorter elementary proof for a reader who has not learned those theories. The following matrix-algebra decomposition also invokes preceding structure theorems.

16.3.1 What the module chapter adds—and where a gap remains

Xi III §4.4, printed 186–196 / PDF 192–202, V/P. Besides module definitions, quotients, kernels, annihilators and Schur's lemma, the section develops finite free modules and their mapping property. Proposition 4.46 discusses uniqueness of rank, including the distinction between domains, commutative rings, and general noncommutative rings. Theorem 4.47, printed 193 / PDF 199, proves that a submodule of a finite-rank free module over a PID is free, by projecting onto a rank-one quotient and lifting a basis. This is a genuine construction.

At printed 194 / PDF 200, the general cyclic decomposition with nested annihilators is stated, with references to the previously developed abelian-group and $k[x]$-operator cases. A complete local proof of the general invariant-factor theorem is not supplied there. The exercises at printed 196 / PDF 202 then explicitly ask the reader to recover Jordan form from the $\mathbb{C}[x]$-module theorem and to generalize earlier abelian-group results to arbitrary PIDs.

This is a defensible location for a focused supplement from Grillet VIII.6, whose fuller proof is documented in the existing source ledger, or from another fully proved PID treatment. It does **not** justify saying that the module viewpoint itself is missing from Xi. Nor should one advertise Xi III's brief statement plus exercise as an already completed proof of the general classification.

16.3.2 A compact representation sequence and its limits

For cyclic Fourier methods and finite-group invariant complements, a small sequence suffices:

1. Define representations, intertwiners, invariant subspaces, and the group algebra; prove the representation/module dictionary by extending the action linearly.
2. Prove the averaged-projection lemma directly—Xi printed 114–115 or Grillet printed 383–384 both work.
3. Prove Schur's lemma from kernels and images, adding the algebraically closed scalar conclusion only when required.
4. For a cyclic group, identify $k[C_n] \cong k[x]/(x^n - 1)$. Over a splitting field with characteristic not dividing n, derive the eigenspace projectors by polynomial CRT or the finite geometric-series identity. Over $\mathbb{Q}$, retain irreducible cyclotomic blocks rather than claiming all constituents are one-dimensional.
5. Work one nonabelian example, such as Xi's S_3 permutation-module split, to test what cyclic intuition fails to capture.

The first four steps need neither Jacobson radicals nor Wedderburn–Artin theory, compact integration, Lie algebras, or a full character table theory. They also do not replace those topics, Xi's rotation-group and $SU(2)$ examples, or the representation appendix. Hall's previously sampled unitary and Lie-theoretic pages remain a separate extension; no new Hall proof audit was made for this comparison.

Stopping tests. Prove that the averaged map is a projection and commutes with the action; identify exactly where $|G|^{-1}$ is used. In characteristic p, use C_p acting by $I + N$ on a two-dimensional space, with $N \neq 0$ and $N^2 = 0$, to show an invariant line need not have an invariant complement. Over $\mathbb{C}$, distinguish the existence of a common invariant Hermitian form from separate diagonalizability of each group element. Finally split the S_3 permutation representation and exhibit an actual intertwining matrix. These test construction and failure cases, not just theorem names.

16.4 Integrality and cyclotomic fields: retain the elementary bridge before Lang

Xi III §4.4, Lemma 4.49 and Theorem 4.50, printed 194–195 / PDF 200–201, V/P. The text proves that an element t of a domain is integral over $\mathbb{Z}$ exactly when $\mathbb{Z}[t]$ is finitely generated as a $\mathbb{Z}$-module. The reverse direction writes multiplication by t on a finite generating family and uses the vanishing of the determinant of $tI - A$, whose polynomial is monic. It then deduces closure of integral elements under addition and multiplication using finite generation and the PID submodule result. Character values, being sums of roots of unity, are connected back to algebraic integers.

This already gives a concrete bridge from matrix algebra and modules to integrality. A reconstructed version can phrase the determinant step as the adjugate identity; no Dedekind domain or valuation is needed. It is especially useful before passing from rational spectral data back to integer coefficients.

Xi III §§5.6–5.7, printed 264–265 and 270–272 / PDF 270–271 and 276–278, V/P. Example 5.47 constructs cyclotomic polynomials through the factorization of $x^n - 1$, proves integral coefficients by monic division and induction, and postpones general irreducibility. Theorem 5.50 later proves irreducibility over $\mathbb{Q}$ by Gauss's lemma and reduction modulo a prime $p \nmid n$: if a primitive root's p-th power belonged to a different factor, the reductions of the factors would share a factor, contradicting squarefreeness of $x^n - 1$ modulo p. Iterating over prime factors of an exponent reaches all primitive roots. This is an undergraduate polynomial argument with explicit prerequisites.

Lang, Algebraic Number Theory IV.1, printed 71–75 / PDF 85–89, P. The inspected

route is different. It first studies prime-power roots of unity through $\pi = 1 - \zeta$, units of the form $(1 - \zeta^i)/(1 - \zeta)$, and the product expressing p. Valuations and ramification-index bounds then establish the degree and total ramification. The proof that $\mathbb{Z}[\zeta]$ is the full ring of integers uses localization and discriminant comparisons from earlier chapters; the general case uses composita and a discriminant criterion.

Lang therefore supplies arithmetic structure that the elementary cyclotomic irreducibility theorem does not establish. But sending an undergraduate to these pages merely to obtain $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$ imports unnecessary valuation, ramification, and discriminant theory. Xi's inspected proof is the smaller route for that purpose.

Neither inspected passage was verified as a ready-made proof of the full integral CRT image for a chosen factorization of $x^n - 1$. Such gluing conditions require their own argument over $\mathbb{Z}$. Rational factorization and the fact that each component is an algebraic integer do not prove that an arbitrary tuple of component values comes from one integer polynomial.

Stopping tests. Reconstruct the determinant proof of the integrality criterion; derive the elementary cyclotomic irreducibility argument and pinpoint why $p \nmid n$ matters; then give the parity obstruction for $x^2 - 1$ over $\mathbb{Z}$. Enter Lang when the actual next question asks about the ring of integers, prime decomposition, ramification, or discriminants—not just because cyclotomic notation has appeared.

16.5 The local substitution decision

Selected goal	Smaller sequence justified by the inspected evidence	What it does not replace
Common-factor detection and elimination	Xi's bounded-degree kernel argument, followed by one product-formula proof	Degree-degeneration care, computational elimination practice, root-location material
Finite-field existence, cyclicity, subfields, Frobenius	Finite splitting-field construction; root-set field; lcm/root-bound cyclicity; direct subfields; automorphism counting	Möbius inversion, irreducible counts, explicit arithmetic, full Galois correspondence
Finite-group invariant complements	Basic representation/module dictionary plus Xi's or Grillet's averaged projection	Character theory, matrix-algebra classification, compact/Lie theory, explicit examples not worked
General PID module structure	Xi's definitions and submodule theorem plus a complete invariant-factor proof from the chosen graduate source	A proof cannot be replaced by Xi's later exercise requesting it
Elementary integrality and cyclotomic degree	Xi's finite-module determinant criterion plus Gauss/reduction-mod-p irreducibility	Integral CRT gluing, ring-of-integers theorems, ramification, valuations and discriminants

The practical conclusion is to keep one proof of each reusable mechanism, solve at least one construction and one failure-case test, and use Xi's omitted computations as a diagnostic bank. Grillet or Hungerford can organize the next layer, but only after their local prerequisites are distinguished from the much larger theories surrounding the desired result.

16.6 Inspected-page ledger for this supplement

Source	New/current inspection	Scope and limitation
Xi I	printed 190–193 / PDF 196–199, V/P	Entire resultant proof sequence 7.76–7.80 and visible exercises; narrow cubic-example error confirmed on image
Xi III	printed 34–35 / PDF 40–41, V/P	Cyclicity theorem and surrounding finite-abelian context; earlier classification proof not re-audited here
Xi III	printed 109–115 / PDF 115–121, V/P	Hermitian averaging, explicit S_3 example, both Maschke proofs, following remarks
Xi III	printed 186–196 / PDF 192–202, V/P	Module definitions through PID submodules, structure statement, integrality proofs, exercises
Xi III	printed 260–266 / PDF 266–272, V/P	Finite fields, subfields, Frobenius, Möbius inversion, cyclotomic construction and irreducible counts; later §5.6 exercise pages not audited
Xi III	printed 270–272 / PDF 276–278, V/P	Roots-of-unity theorem, cyclotomic irreducibility proof including its final lines, exercises
Grillet	printed 157 / PDF 169, P	Cyclic finite multiplicative subgroup proof
Grillet	printed 176–179 / PDF 188–191, P	Resultant definition, universal determinant proof, discriminant identity
Grillet	printed 192–193 / PDF 204–205, P	Finite-field proof and exact exercise status of subfield results
Grillet	printed 197–200 / PDF 209–212, P	General Galois framework sampled to identify prerequisites; not used as a shorter proof recommendation
Grillet	printed 381–386 / PDF 393–398, P	Group algebra, representation/module dictionary, Maschke, adjacent classification and cyclic-group exercises
Hungerford	printed 279–282 / PDF 298–301, P	Finite-field proofs, explicit imports and exercises
Hungerford	printed 453–456 / PDF 472–475, P	Maschke radical/trace proof and surrounding algebra-structure context
Lang	printed 71–75 / PDF 85–89, P	Cyclotomic degree, ramification and integer-ring proof dependencies; formulas whose extraction was uncertain were not transcribed

Public prose contains original paraphrases and small mathematical reconstructions only. Source scans, raw excerpts, private Library identifiers and access metadata are outside the repository. No commit or push was performed.

17 Source ledger and page conventions

17.1 Evidence levels

The expanded edition adds contiguous proof readings documented in the operator supplement and the arithmetic supplement. Their ending ledgers give the new page ranges and distinguish extracted proof reading from rendered-page verification. The original selected readings below remain part of the baseline evidence, not a claim to have read every intervening page.

P means the specified text, theorem, construction, or proof was inspected in this task. **V** means a rendered source page was inspected, necessary for scans or a suspected extraction problem. **T** means a contents or introductory description was inspected; it establishes a location and stated scope, not that every theorem or exercise was audited. **C** means a theorem is cited as a standard result and is not fully reproved or source-audited here. The comparison uses these distinctions rather than treating an index hit as proof coverage.

PDF page numbers are **one-based physical page positions in the supplied file**; printed pages are the numbers visible on the book pages. The offsets below apply to the ordinary Arabic-numbered body at the checked locations, not to Roman frontmatter. A section range is a reading destination; a list of inspected pages is the actual evidence, and should not be expanded into a full-book audit.

17.2 Editions and physical files

Key	Supplied source	File pages	Body page conversion
Xi I	席南华, 基础代数 第一卷, Science Press, 2016	207, local metadata	PDF = printed + 6
Xi II	席南华, 基础代数 第二卷, Science Press, 2018	334, local metadata	PDF = printed + 6
Xi III	席南华, 基础代数 第三卷, Science Press, 2021, second printing visible	361, local metadata	PDF = printed + 6
Hungerford	Thomas W. Hungerford, Algebra, Springer edition/reprint of the 1974 text; 1980 Springer preface and 2003 printing note	523, local metadata	PDF = printed + 19
Grillet	Pierre Antoine Grillet, Abstract Algebra, second edition, 2007	684, local metadata	PDF = printed + 12
Mac Lane	Saunders Mac Lane, Categories for the Working Mathematician, second edition	331, local metadata	PDF = printed + 13
Hilton–Stammbach	Peter J. Hilton and Urs Stammbach, A Course in Homological Algebra, second edition	376, local metadata	PDF = printed + 11
Hall	Brian C. Hall, Lie Groups, Lie Algebras, and Representations, second edition, 2015	452, local metadata	PDF = printed + 13

Key	Supplied source	File pages	Body page conversion
Hartshorne	Robin Hartshorne, Algebraic Geometry, Springer	511, local metadata	PDF = printed + 15
Lang	Serge Lang, Algebraic Number Theory, second edition; supplied reprint identifies 1986 edition with later copyright information	372, local metadata	PDF = printed + 14
Hirsch	Morris W. Hirsch, Differential Topology, 1976/reprint	230 reported by Library reader; no local byte audit	PDF = printed + 10 at checked body pages

Local SHA-256 fingerprints and byte sizes for the ten materialized files are saved in `evidence/source-fingerprints.json`. They identify which supplied editions underlie the references; they are not a distribution of source bytes. The original Library filenames, which sometimes have OCR-like punctuation, are not treated as authoritative edition statements.

17.3 Xi I: concrete foundations and determinant characterizations

- **P/T:** title/edition and preface, PDF pp. 3–4; contents, pp. 5–6. The preface connects the course to Kostrikin's framework and names several other references, including Hungerford. This supports pedagogical context, not a claim that the books have identical proof coverage.
- **P:** Chapter 4 opening, printed pp. 75–76 (PDF pp. 81–82): oriented volume, multilinearity, and vanishing on dependent vectors.
- **P:** §4.4, printed p. 93 (PDF p. 99): several characterizations of the determinant, with the preceding exercises including Cauchy–Binet/Laplace identities. This is actual structural content, not merely a chapter title.
- **P/T:** §6.3 opening, printed p. 133 (PDF p. 139): factorization and examples whose irreducibility changes with the coefficient ring.
- **P/T:** §7.4 opening, printed p. 177 (PDF p. 183): elementary symmetric polynomials and invariance under ordering roots.
- **P:** §7.6 opening, printed p. 190 (PDF p. 196): resultant matrix and criterion, with the leading-coefficient degeneracy explicitly visible. This was an opening-page sample in the baseline. The continuation inspected the full proof sequence on PDF pp. 196–199 and its visible exercises; see the arithmetic supplement, including the narrow cubic-example erratum.
- Quotients of sets (§2.4, printed p. 30, PDF p. 36) and matrix-map organization (Chapter 3, starting p. 40, PDF p. 46) were located and sampled.

17.4 Xi II: the principal source comparison

- **P:** §1.4–1.6 samples, printed pp. 16, 19–24 (PDF pp. 22, 25–30): independent subspaces, complements versus quotients, duals and evaluation.
- **P:** §2.4, printed pp. 67–71 (PDF pp. 73–77), selected text: quotient operator and induced-map diagram; characteristic-polynomial factorization; Cayley–Hamilton statement and dual operator discussion.
- **P:** §2.5, printed pp. 72–77 (PDF pp. 78–83), consecutive proof reading: Theorem 2.51 uses Bézout to split annihilated spaces; 2.52–2.53 gives generalized eigenspaces; 2.56–2.57 gives Jordan

form; the proof constructs chains by quotient induction and proves uniqueness by dimensions of images of powers.
- **P:** §3.1 opening, printed p. 85 (PDF p. 91), and §3.3 opening, printed p. 109 (PDF p. 115), with examples/exercises from metric linear algebra. **P:** Theorem 3.62 and its normal-operator argument, printed pp. 126–128 (PDF pp. 132–134).
- **T:** Chapters 4–5, printed pp. 161–262 (PDF pp. 167–268): affine and Euclidean geometry, transformations, convexity, indefinite spaces, quadrics, projective geometry. Their complete proofs were not audited; no blanket replacement claim about them is made.
- **P/V:** tensor Definition 6.1, printed p. 263 (PDF p. 269), defines all multilinear scalar functions on the indicated product spaces.
- **P:** tensor Definition 6.11 and Theorem 6.15, printed pp. 273–275 (PDF pp. 279–281): basis construction, uniqueness, and universal property **with a proof**. Subsequent maps, associativity, distributivity, scalar extension, and Kronecker formulas were read on PDF pp. 280–284.
- **P:** §6.3 examples of contraction as trace/composition and structural tensors, printed pp. 280–283 (PDF pp. 286–289).
- **P/V:** Theorem 6.19 and following remark, printed p. 278 (PDF p. 284). The finite-dimensional theorem is correct; the unrestricted infinite-dimensional extension fails for the definition on p. 263. See `VERIFICATION.md` for the explicit counterexample and its scope.
- **P:** §6.4 opening and definition, printed pp. 292–293 (PDF pp. 298–299): characteristic-zero assumption and exterior product via alternation.

17.5 Xi III: selected scanned pages, not unreliable extraction

The Library extraction's earlier 150-page report is not the physical file length. PyMuPDF counted **361 pages**. The scan's extracted text is extremely sparse. The following evidence comes from rendered pages, not OCR guesses.

- **V/T:** contents, PDF pp. 5–6: group actions and structure, one-parameter groups (§2.5, p. 79), matrix Lie groups (§2.6, p. 85), representations (Chapter 3, p. 97), modules (§4.4, p. 186), algebras (§4.5, p. 197), $\mathfrak{sl}_2$ (§4.6, p. 215), Galois theory (Chapter 5, p. 224), and the representations appendix (p. 317).
- **V/P:** §3.2, printed pp. 109–111 (PDF pp. 115–117): complex Hermitian-form convention and Theorem 3.20's explicit finite-group averaging proof, with the compact-group integration comment distinguished from it.
- **V/P:** §3.7, printed pp. 152–153 (PDF pp. 158–159): dual action with inverse, transpose formula, and tensor action.
- **V/P:** §4.4, printed pp. 186–188 (PDF pp. 192–194): module motivation, unital left-module definition, examples, submodules, quotients and maps.
- **V/P:** §4.6, printed pp. 215–216 (PDF pp. 221–222): Lie module conditions and discussion of the enveloping algebra. The page explicitly sends the reader elsewhere for the PBW theory; the notes do not claim that this section proves PBW.
- **V/P:** §5.6, printed pp. 260–262 (PDF pp. 266–268): cyclic multiplicative group, finite-field existence and uniqueness, subfields and Frobenius. This supplies the finite-field prerequisites for the Paley bridge.
- **V/P:** §5.8, printed pp. 272–273, 275–277 (PDF pp. 278–279, 281–283): Noether/Hilbert 90 constructions, norm/trace examples and trace-pairing and coefficient-formula exercises. The trace pairing on p. 277 is an exercise, not a full proof at that location.
- **V/P:** §5.11 opening, printed p. 286 (PDF p. 292): the normal-basis theorem and beginning of its interpolation construction. The rest of that proof, full Galois computations/inverse

questions, and the representation appendix remain selective reading destinations, not audited proofs here.

17.6 Graduate sources: where the extra structure is actually built

Grillet. Title/preface and contents were inspected. The detailed samples include VIII.6–7 (printed pp. 336–344, PDF pp. 348–350, 354–356), X.1 and X.3 (pp. 393–395, 401–402; PDF pp. 405–407, 413–414), XI.5–7 (pp. 434–437, 441–443, 447–448; PDF pp. 446–449, 453–455, 459–460), XII.4–5 (pp. 487–489, 493–494; PDF pp. 499–501, 505–506), and XIII.2–4 (pp. 518–526; PDF pp. 530–531, 533–538). These support the module, splitting, balanced tensor, Ext/Tor, and free-algebra comparisons. The general PID proof's key basis and invariant-factor arguments were read; not every ancillary exercise was checked. IX.7, printed pp. 380–385 (PDF pp. 392–397), was inspected for the group-algebra correspondence, Maschke proof, and its nonstandard initial use of "irreducible" for indecomposable. The p. 381 definition was checked as an image. IV.1.6 (printed p. 157, PDF p. 169) was read for cyclic multiplicative subgroups, and V.1.3 (printed pp. 192–193, PDF pp. 204–205) for the finite-field existence/uniqueness proof and field-table exercises. The other IV–VI destinations for norm/trace, orders/valuations and completions are **T**, not a proof audit.

Hungerford. Prefaces, contents, and course suggestions were read, especially printed p. xviii (PDF p. 15), which explicitly places torsion PID modules before VII.1–5 in the proposed linear-algebra course. IV.6 begins at printed p. 218 (PDF p. 237); selected text on torsion/cyclic modules and the structure statements was read at PDF pp. 239–241. VII.4 starts at p. 355 (PDF p. 374), VII.5 at p. 366 (PDF p. 385). The route and theorem locations are verified, but a line-by-line comparison of all proofs in VII was not completed. IV.4 (p. 199, PDF p. 218), IV.5 (p. 207, PDF p. 226), are selective destinations grounded in contents. V.5's finite-field proofs were then read at printed pp. 279–281 (PDF pp. 298–300), including Theorem 5.3, Proposition 5.6, Corollary 5.7, Frobenius, and the explicit exercise constructing fields of orders 9 and 25.

Mac Lane. III.1 (printed pp. 55–58, PDF pp. 68–71), III.2 (pp. 60–62, PDF pp. 73–75), IV.1 (pp. 79–80, PDF pp. 92–93), and IV.2's duality example (pp. 88–89, PDF pp. 101–102) were read. These are actual definitions, examples, and proofs of universality, representability/Yoneda, and adjunction. VIII's abelian-category treatment and later monads/Kan extensions are **T/C** in this study.

Hilton–Stammbach. Edition, contents, and introduction were read; III.1–2 and III.4 were sampled in detail (printed pp. 84–86, 89–90, 97–98; PDF pp. 95–97, 100–101, 108–109). IV.1's definitions and homology construction were read at pp. 117–118 (PDF pp. 128–129). The introduction explicitly motivates the extension-first route before general derived functors. The complete comparison theorem for resolutions and the full general Ext classification are **C** here; our one-step cyclic cases are proved directly.

Hall. The supplied book is the second edition. §§4.3–4.7 were sampled at printed pp. 90–101 (PDF pp. 103–114), including the unipotent nonsemisimple example, unitary splitting proof, Schur argument, and $\mathfrak{sl}_2$ classification proof. Later root systems, PBW, highest-weight theory, and topology are **T**, not compared proof by proof against Xi.

Hartshorne. Contents and introduction, Chapter II opening, II.5, II.8, and Chapter III opening were read at PDF pp. 9–10, 14–15, 74–75, 123–126, 187–189, 216. In printed pages these include II.5 at pp. 108–111, II.8 at pp. 172–174, and III's opening at p. 201. The algebraic differential universal property and the rational-residue-field cotangent calculation are visible at these locations. The "A" notation distinguishes externally referenced theorems. Affine finite projectivity and general cohomological theorems are cited rather than re-audited in full.

Lang. Title/edition and contents were read. I.1 opening (printed p. 3, PDF p. 17), I.9 Proposition 26 and its local-to-global proof (p. 29, PDF p. 43), and III.1's complementary-module definition/proposition (p. 57, PDF p. 71) were inspected. Full valuation, completion, ramification and class field arguments remain **T/C**. A route to them is identified without claiming that the current notes replace them.

Hirsch. Located by title in Library. Preface/contents, PDF pp. 8–10; Chapter IV.1–2 samples, PDF pp. 96–105 (printed pp. 86–95). The preface's declared avoidance of tensors/forms and IV.2.2's partition-of-unity splitting proof are **P**. Local download via the Library helper returned HTTP 401; no local physical-file metadata or fingerprint is claimed. The read-only Library content route succeeded. The full transversality/Morse proofs are not audited here.

17.7 Existing Hadamard/Williamson research

The reference edition is pinned to repository commit `2834fea31fb4ca7203531ca220e55a2fb7640119`, not a moving branch tip:

- Study and original scope.
- Original constructions and exact sign conventions.
- Second-round structural and spectral proofs.
- Third-round proofs, especially §§1–4, 9, 11–13.
- Third-round limitations.
- Prior independent adversarial review.

Those documents and the relevant exact fixture descriptions were read from Git objects. The new foundation chapter gives its own derivations, identifies where it extends the scope of the earlier prime Paley proof, and retains the distinction between previous audit evidence and newly executed finite checks. Numerical historical/status claims and whole prior benchmark campaigns were not rerun for this foundational task.

18 Verification record, corrections, and limits

18.1 What the evidence means

The expanded edition adds four chapters and two proof comparisons to the corrected baseline. Separate AI agents reviewed the new chapter proofs, and independent exact calculations checked their most delicate examples. The continuation review records the findings and corrections. The additional runner `scripts/check_phase2.py` records 28 chapter check groups and one symbolic cubic-discriminant check in `evidence/phase2-checks.json`, with hashes of the executed scripts. These supplement the baseline's 18 groups; they are not formal verification.

The notes contain ordinary written proofs and explicitly cited structural theorems. They have undergone a separate self-review for hypotheses, notation, and failure cases, plus exact computational spot checks written independently of the earlier repository search implementations. There is no claim of a separate human reviewer, Lean/kernel verification, a full-book audit, or a new result about universal Hadamard existence.

`evidence/exact-checks.json` records the actually executed finite checks, software versions and their scope. `evidence/pdf-build.json` and the LaTeX logs record compilation; compilation verifies neither mathematical correctness nor source fidelity. Source fidelity is supported by the specific inspected pages listed in the source ledger, including image inspection where required.

18.2 Central statements checked by written derivation

Statement	Hypotheses and check
Quotient factorization and first isomorphism theorem	Maps, domains, kernels and representative independence are explicit; no complement is assumed
Dual exactness and annihilator identities	Algebraic dual; finite dimension for the double-dual isomorphism and dimension arguments; basis extension for restriction surjectivity
Direct sums and splitting	Idempotent equations proved for modules; vector-space splitting distinguished from module/equivariant splitting; sections form an affine Hom space
Operator/module correspondence	Intertwiner equation checked; finite-dimensional operator modules distinguished from all finitely generated $k[t]$-modules
Canonical forms	Nonzero annihilator obtained without Cayley–Hamilton; Euclidean Smith mechanism explained; arbitrary PID version cited; splitting-field hypothesis retained for Jordan form
Polynomial projectors	Congruences, image/kernel identification, and independence from Bézout representative checked; canonical primary spaces distinguished from chosen cyclic summands
Cyclic Hom and commutant dimensions	Computed from the image of 1 and divisibility; no unjustified extension to two-variable polynomial rings
Hom–tensor formulas	Currying needs no finite dimension; the canonical tensor map identifies its domain with finite-rank maps, and finite dimension of either source or target suffices for all Hom; noncommutative module sides specified
Symmetric/exterior constructions	Quotient universal properties; alternation includes repeated-vector vanishing in characteristic 2; averaging needs invertible factorial
Determinants and minors	Exterior basis proof precedes determinant definition; rank test, Cauchy–Binet and characteristic coefficients derived; determinant of a map between different spaces is a line map
Forms and spectral theorem	Congruence versus similarity; Hermitian convention explicit; positive-definite complex finite-dimensional hypotheses retained; real and indefinite failures exhibited
Ext/Tor and homology	One-step resolution requires a nonzero element of a domain; explicit nonsplit $k[t]$ extension; field Ext/Tor vanishing separated from arbitrary complex homology
Representations	Maschke needs finite group and invertible order; Schur's scalar result needs algebraic closure and finite dimension; dual action has inverse; Lie tensor action has a sum
Trace pairing	Finite separable extension for nondegeneracy; scalar-extension/CRT proof and purely inseparable counterexample included

Statement	Hypotheses and check
Bundles and sheaves	Smooth paracompact bundle splitting distinguished from arbitrary pointwise choices and algebraic sheaf extensions; no category-independent Ext vanishing claim
Hadamard bridge	Integer/real Gram equations; odd finite fields for character sum; exact Paley signs; cyclic Fourier splitting/characteristic hypotheses; compression and phase relaxation retain only necessary implications

The proofs are inspectable in the chapters. The table is a review index, not an independent formal certificate.

18.3 Independently computed finite checks

The script `scripts/check_examples.py` reconstructs fixtures directly and does not import the previous Hadamard project's search or verification code. Its 18 check groups include:

- Polynomial CRT idempotents over $\mathbb{Q}$ and characteristics $2, 3, 5, 7$; a conjugated mixed Jordan fixture checks the actual operator projections.
- Kernel-growth block reconstruction, and the commutant dimension independently computed as the nullity of a 100×100 commutator equation matrix. A real nonsplitting $t^2 + 1$ summand preserves the kernel data while increasing the dimension from ten to twelve, exhibiting the missing-hypothesis failure excluded by the corrected exercise.
- Sixteen cyclic-Hom dimension comparisons using direct intertwiner equations rather than the gcd formula being checked.
- Exterior minors, rectangular Cauchy–Binet, and characteristic coefficients; Kronecker determinant/trace; adjoints for a nonorthonormal Hermitian form.
- Explicit module-section failure over four small prime fields; a nonzero field-complex homology example with chain-homotopy equations; the $\mathfrak{sl}_2$ polynomial commutators for highest weights zero through six.
- Quadratic-character sums and both Paley block constructions for six small primes, plus an actual nonprime-field fixture $\mathbb{F}_9 = \mathbb{F}_3[u]/(u^2 + 1)$ producing order 20.
- Williamson block signs at core three, the core-nine proper-quotient false positive, and all 27 possible normalized symmetric lifts of the **fixed** stronger compressed multiset from the prior audit. None is a Williamson lift. This is a bounded adversarial-fixture replay, not an expanded search campaign.
- The core-seven positive spectral residual with no fourth Williamson row; all 16 two-color-bit equality inputs and the exact Boolean phase identity. The explicit edge-orbit correlation formula is also compared with direct integer correlations for all 1,364 four-color labelings at cores $3, 5, 7, 9, 11$, using three sign fixtures per labeling (4,092 assignments), including the parity-forced signs. This is a bounded identity check, not a search for or classification of solutions.

Finite checks can catch sign, transpose, coefficient and characteristic errors in these examples. They do not prove the universal theorems or classify all Hadamard matrices. The infinite-dimensional tensor counterexample is proved by unbounded rank, not established by a finite truncation test.

18.4 Corrections incorporated in the local continuation

The continuation review supplied six focused corrections. Exercise 4 now specifies $\dim_k V = 10$, and the chapter's parallel example identifies only the λ-primary blocks until a total dimension is known. Exercise 6, its answer, and the Hom/tensor summary use the exact finite-rank image criterion; finite dimension of either space suffices. Maschke's projection is explicitly an endomorphism of V. The second congruence example begins "Similarly." Contraction is defined for $r \geq 1$ and set to zero on $\Lambda^0 V$. Finally, E_t is defined from the involution orbits, with the zero pair and fixed point removed before passing to opposite-index classes. The original source findings below are retained. That focused six-fix revision did not rematerialize or independently reread the source books; the later substantive expansion did inspect additional proof passages, as documented in the two new source-comparison ledgers.

18.5 Cited results, distinguished from complete proofs here

- The full finitely generated module structure theorem over an arbitrary PID (Grillet VIII.6; Hungerford IV.6). The notes give the Euclidean matrix construction and derive the operator consequences; they do not pretend every PID is Euclidean.
- General derived-functor theory, independence of resolutions, and the complete Ext/extension correspondence (Hilton–Stammbach III–IV; Grillet XII). The cyclic class calculation and its extension interpretation are proved directly in the notes.
- Exhaustion of finite-dimensional complex $\mathfrak{sl}_2$ irreducibles (Hall §4.6). The polynomial models and their irreducibility are proved here; the exhaustion proof was read but is cited rather than reproduced fully.
- Primitive-element theorem for finite separable extensions; finite-field existence; the general commutative-algebra/sheaf correspondence; finite projectivity on affine schemes; implicit-function, partition-of-unity and other geometric prerequisites. Their uses and sources are stated, without implying all are reproved in this study.
- Prior study benchmarks, search completeness records, and historical matrix witnesses remain attributed to their pinned source edition. Only the small fixtures listed above were newly executed. No current worldwide open-problem status or smallest unresolved order is certified.

18.6 Confirmed source corrections

Xi I, Example 7.80, printed p. 193 / PDF p. 199. The supplied scan prints $-2p^3 - 27q^2$ for the discriminant of $x^3 + px + q$. The correct coefficient is -4. Two independent image inspections confirmed the displayed formula; the preceding general cubic exercise on printed p. 190 / PDF p. 196 has the correct term. Direct symbolic calculation gives $\mathrm{Res}(f, f') = 4p^3 + 27q^2$, and the monic cubic discriminant is its negative. The specialization $f = x^3 + x$ gives -4 directly. This is a narrow example typo, not a defect in the resultant criterion or its inspected proof.

Xi II, Theorem 6.19's following remark, printed p. 278 / PDF p. 284. The remark extends the finite-dimensional identification $\mathcal{T}^p_q(V) \cong (V^*)^{\otimes p} \otimes V^{\otimes q}$ to infinite-dimensional spaces. Definition 6.1 on printed p. 263 / PDF p. 269 defines the left side as all multilinear functions on $V^p \times (V^*)^q$. Both pages were rendered and inspected.

For $p = q = 1$, evaluation $(v, \lambda) \mapsto \lambda(v)$ belongs to that function space. If it were a finite tensor sum, there would be $\alpha_i \in V^*$ and $w_i \in V$ such that

$$\lambda(v) = \sum_{i=1}^{r} \alpha_i(v)\lambda(w_i) \qquad \text{for every } v, \lambda.$$

Functionals separate vectors, so $v = \sum_i \alpha_i(v)w_i$ for every v. This puts an infinite-dimensional space in a finite-dimensional span, a contradiction. The corrected statement retains finite-dimensionality, or restricts the multilinear function space to the image of finite tensor sums. The finite-dimensional theorem and the tensor universal property are unaffected. This is one local confirmed overstatement, not a blanket judgment of the text.

18.7 Convention differences and missing hypotheses, not source mistakes

- Grillet IX.7's initial "irreducible" means indecomposable (printed pp. 381, 383; PDF pp. 393, 395). It differs from the simple-module convention in these notes, Xi and Hall. The apparent unconditional complete-reducibility statement must be translated using his definition, not declared false.
- Xi's Hermitian convention is linear in the first argument; these notes use conjugate-linearity in the first. Adjoint and coordinate formulas are translated explicitly.
- Xi II's alternating-tensor model explicitly assumes characteristic zero. Its factorial denominators are valid there; moving the model to arbitrary characteristic without that assumption would be our error, not its theorem.
- A Jordan form needs a splitting minimal polynomial; semisimple $k[t]$-modules need square-free irreducible factors and can lack eigenvectors over the base field. Commuting operators require individual diagonalizability before the simultaneous eigenbasis argument.
- A positive spectral residual and individually feasible compressed rows need not have one common discrete lift. No linear splitting theorem supplies it.
- Fourier eigenvalues over $\mathbb{C}$ are exact algebraic values in the proof; floating-point approximations are not used to certify sign matrices.

18.8 OCR issues and unverified comparisons

Xi III's local PDF has 361 pages, whereas its earlier Library extraction reported 150. Sparse extraction and corrupted symbols are source-access limitations, not mathematical errors. The key cited pages were read as images. Xi I–II also have corrupted extracted signs and indices, so suspected substantive claims were checked against the intended notation and, where material, images. The exact source fingerprints allow later reinspection.

The following comparisons remain deliberately limited: the entire exercises of any two books; Xi II Chapters 4–5 versus a putative replacement; the whole Xi III appendix or Galois computations against Hall/Lang; complete Hungerford VII proof alternatives; and the destination books' full advanced prerequisite coverage. They are not promoted from contents evidence into proven replacement claims. No source book is called filler or wholly replaceable here.

18.9 Remaining access limitation

The exact failed action was local materialization of the Library item titled `Differential Topology (Morris W. Hirsch).pdf` using the provided download helper. It returned `hosted apps tools/list request failed with HTTP status 401`. This is a failed local-download action, not a successful or uncertain transfer and not an automatic sandbox-approval denial. The same source's read-only

Library pages were successfully read afterward. Consequently the Hirsch content needed here is sourced, but no physical local PDF page count, byte hash, or full local-PDF audit is claimed.

There is no pending mathematical/source blocker to using the present notes. Completing the optional unverified comparisons would be a new, larger audit; recovering a local Hirsch copy is only necessary for that sort of byte-level or full visual audit.